

Online Appendix

*Estimating the Number of Components in Panel Data Finite Mixture
Regression Models with an Application to Production Function Heterogeneity*

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A Appendix

A.1 Likelihood ratio test for testing $H_{02} : \alpha(1 - \alpha) = 0$

We derive the asymptotic distribution of the LRT statistic for testing the hypothesis $H_{02} : \alpha(1 - \alpha) = 0$ for testing $H_0 : M_0 = 1$ against $H_1 : M_0 = 2$. Our analysis focuses on the null hypothesis $\alpha = 0$, noting that the case $\alpha = 1$ follows by symmetry.

Consider the following restricted parameter space in which θ_1 and θ_2 are bounded away from each other while α can take the value of 0 and 1:

$$\tilde{\Theta}_{\theta_2}(c) := \{\theta_2 \in \Theta_{\theta_2} : \|\theta_1 - \theta_2\| \geq c_1, \min_j \sigma_j^2 \geq c_2 \sum_{k=1}^2 \alpha_k \sigma_k^2, \sum_{k=1}^2 \alpha_k \sigma_k^2 \geq c_3\}.$$

Define the MLE $\tilde{\theta}_2$ in this restricted parameter space by

$$\ell_n^2(\tilde{\theta}_2) = \max_{\theta_2 \in \tilde{\Theta}_{\theta_2}(c)} \ell_n^2(\theta_2). \quad (1)$$

Following the proof of Proposition (3), $\tilde{\theta}_2$ can be shown to be consistent under Assumptions 1 and 3.

To simplify the asymptotic representation and regularity conditions, we use the parameter $\lambda := \theta_1 - \theta_2$ and reparameterize (θ_1, θ_2) to (λ, θ_2) so that the model parameter is $\Psi = (\theta_2, \lambda, \alpha) \in \Theta_{\Psi}$. Under $H_0 : M_0 = 1$, the two-components model replicates the true one-component model when $\theta_2 = \theta^*$ and $\alpha = 0$. Given that λ is unidentified when $\alpha = 0$, we follow Andrews (2001) by deriving the limit of the LRT statistic for each $\lambda := \theta_1 - \theta_2$ in $\Theta_{\lambda}(c_1) := \{\lambda \in \Theta_{\lambda} : \|\lambda\| \geq c_1\}$ for $c_1 > 0$.

Define the reparameterized log-density as $\log g_2(\mathbf{w}; \theta_2, \lambda, \alpha) = \log(\alpha h(\mathbf{w}; \theta_2 + \lambda) + (1 - \alpha)h(\mathbf{w}; \theta_2))$. Collect the partial derivative of $\log g_2(\mathbf{w}; \theta_2, \lambda, \alpha)$ with respect to θ_2 and its

right partial derivative with respect to α evaluated at $(\boldsymbol{\theta}_2, \boldsymbol{\lambda}, \alpha) = (\boldsymbol{\theta}^*, \boldsymbol{\lambda}, 0)$ as

$$\mathbf{s}(\mathbf{w}; \boldsymbol{\lambda}) := \begin{pmatrix} \mathbf{s}_{\boldsymbol{\theta}_2}(\mathbf{w}) \\ \mathbf{s}_{\alpha}(\mathbf{w}; \boldsymbol{\lambda}) \end{pmatrix} := \begin{pmatrix} \nabla_{\boldsymbol{\theta}_2} \log g_2(\mathbf{w}; \boldsymbol{\theta}^*, \boldsymbol{\lambda}, 0) \\ \nabla_{\alpha} \log g_2(\mathbf{w}; \boldsymbol{\theta}^*, \boldsymbol{\lambda}, 0) \end{pmatrix} = \begin{pmatrix} \frac{\nabla_{\boldsymbol{\theta}} h(\mathbf{w}; \boldsymbol{\theta}^*)}{h(\mathbf{w}; \boldsymbol{\theta}^*)} \\ \frac{h(\mathbf{w}; \boldsymbol{\theta}^* + \boldsymbol{\lambda}) - h(\mathbf{w}; \boldsymbol{\theta}^*)}{h^*(\mathbf{w}; \boldsymbol{\theta}^*)} \end{pmatrix}. \quad (2)$$

Define $\mathcal{I}(\boldsymbol{\lambda}) := E[\mathbf{s}(\mathbf{W}; \boldsymbol{\lambda}) \mathbf{s}(\mathbf{W}; \boldsymbol{\lambda})^\top]$. Analogously to (21), define

$$\mathcal{I}(\boldsymbol{\lambda}) = \begin{pmatrix} \mathcal{I}_{\boldsymbol{\theta}_2} & \mathcal{I}_{\boldsymbol{\theta}_2 \alpha}(\boldsymbol{\lambda}) \\ \mathcal{I}_{\alpha \boldsymbol{\theta}_2}(\boldsymbol{\lambda}) & \mathcal{I}_{\alpha}(\boldsymbol{\lambda}) \end{pmatrix}, \quad \mathcal{I}_{\boldsymbol{\theta}_2} = \mathbb{E}[\mathbf{s}_{\boldsymbol{\theta}_2}(\mathbf{W}) \mathbf{s}_{\boldsymbol{\theta}_2}(\mathbf{W})^\top], \quad \mathcal{I}_{\alpha \boldsymbol{\theta}_2}(\boldsymbol{\lambda}) = \mathbb{E}[\mathbf{s}_{\alpha}(\mathbf{W}; \boldsymbol{\lambda}) \mathbf{s}_{\boldsymbol{\theta}_2}(\mathbf{W})^\top],$$

$$\mathcal{I}_{\boldsymbol{\theta}_2 \alpha}(\boldsymbol{\lambda}) = \mathcal{I}_{\alpha \boldsymbol{\theta}_2}(\boldsymbol{\lambda})^\top, \quad \mathcal{I}_{\alpha}(\boldsymbol{\lambda}) = \mathbb{E}[\mathbf{s}_{\alpha}(\mathbf{W})^2], \text{ and } \mathcal{I}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda}) = \mathcal{I}_{\alpha}(\boldsymbol{\lambda}) - \mathcal{I}_{\alpha \boldsymbol{\theta}_2}(\boldsymbol{\lambda}) \mathcal{I}_{\boldsymbol{\theta}_2}^{-1} \mathcal{I}_{\boldsymbol{\theta}_2 \alpha}(\boldsymbol{\lambda}).$$

Let $\{\mathbf{S}(\boldsymbol{\lambda}) = (\mathbf{S}_{\boldsymbol{\theta}_2}, \mathbf{S}_{\alpha}(\boldsymbol{\lambda})) : \boldsymbol{\lambda} \in \tilde{\Theta}_{\boldsymbol{\lambda}}(c_1)\}$ be a mean zero vector-valued Gaussian process such that $\mathbb{E}[\mathbf{S}(\boldsymbol{\lambda}) \mathbf{S}(\boldsymbol{\lambda})^\top] = \mathcal{I}(\boldsymbol{\lambda})$, where $\mathbf{S}_{\boldsymbol{\theta}_2}$ is independent of $\boldsymbol{\lambda}$, $\mathbf{S}_{\alpha}(\boldsymbol{\lambda})$ is 1×1 , and $\mathcal{I}(\boldsymbol{\lambda})$ is defined in (3). Let $\mathbf{S}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda}) := \mathbf{S}_{\alpha}(\boldsymbol{\lambda}) - \mathcal{I}_{\alpha \boldsymbol{\theta}_2}(\boldsymbol{\lambda}) \mathcal{I}_{\boldsymbol{\theta}_2}^{-1} \mathbf{S}_{\boldsymbol{\theta}_2}$.

Define the LRT statistics for testing $H_{02} : \alpha(1 - \alpha) = 0$ by

$$\widetilde{LR}_n^2(\mathbf{c}) := 2 \left\{ \ell_n^2(\tilde{\boldsymbol{\theta}}_2) - \ell_n^1(\hat{\boldsymbol{\theta}}_0) \right\} = 2 \left\{ \max_{(\boldsymbol{\theta}_2, \boldsymbol{\lambda}, \alpha) \in \Theta_{\boldsymbol{\theta}} \times \tilde{\Theta}_{\boldsymbol{\lambda}}(c_1) \times [0, 1/2]} \ell_n^2(\boldsymbol{\theta}_2 + \boldsymbol{\lambda}, \boldsymbol{\theta}_2, \alpha) - \ell_n^1(\hat{\boldsymbol{\theta}}_0) \right\}. \quad (4)$$

Assumption A.1. (a) $\boldsymbol{\theta}^*$ is in the interior of $\Theta_{\boldsymbol{\theta}}$. (b) $h(\mathbf{w}; \boldsymbol{\theta})$ is twice continuously differentiable on $\Theta_{\boldsymbol{\theta}}$. (c) $\mathcal{I}(\boldsymbol{\lambda})$ defined in (3) is finite and positive definite uniformly in $\Theta_{\boldsymbol{\lambda}}(c_1)$. (d) Assumption 3 holds when we replace $\tilde{\Theta}_{\boldsymbol{\theta}_2}(\mathbf{c})$ with $\tilde{\Theta}_{\boldsymbol{\theta}_2}(\mathbf{c})$.

Proposition A.1. Suppose Assumptions 1 and A.1 hold. Then, (a) $\inf_{\boldsymbol{\theta}_2 \in \Theta_{\boldsymbol{\theta}_2}^*} \|\tilde{\boldsymbol{\theta}}_2 - \boldsymbol{\theta}_2\| \rightarrow 0$ almost surely, and (b) $\widetilde{LR}_n^2(\mathbf{c}) \xrightarrow{d} \sup_{\boldsymbol{\lambda} \in \Theta_{\boldsymbol{\lambda}}(c_1)} (\max\{0, \mathcal{I}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda})^{-1/2} \mathbf{S}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda})\})^2$.

Assumption A.1(b) can be verified for a class of mixture models we consider in this paper.

A necessary condition for Assumption A.1(c) is $\sup_{\boldsymbol{\lambda} \in \tilde{\Theta}_{\boldsymbol{\lambda}}(c_1)} \mathbb{E}[\nabla_{\alpha} \log g_2(\mathbf{w}; \boldsymbol{\theta}^*, \boldsymbol{\lambda}, 0)^2] < \infty$, which is violated for the finite mixture normal regression models when $\sigma_j^2 > 2\sigma^{*2}$.

One approach is to impose a restriction that $\sigma_1^2 \leq 2\sigma^{*2}$. However, a difficulty arises in imposing the constraint $\sigma_1^2 \leq 2\sigma^{*2}$ when estimating a two-component model, given that σ^{*2} is unknown. A possible solution is to use the estimated variance from the one-component model, setting:

$$\sigma_j^2 \in [0, 2\hat{\sigma}_0^2 - c] \text{ for } j = 1, 2,$$

where $\hat{\sigma}_0^2$ is the estimated variance from the one-component model and c is a small posi-

tive constant. This ensures the asymptotic validity of the condition $\sigma_j^2 < 2\sigma^{*2}$. However, in finite samples, there remains a positive probability that values of σ_j^2 within this random interval may violate the constraint.

A.2 Asymptotic analysis of the penalised maximum likelihood estimator in Section 5.3

In this appendix, we extend the previous asymptotic analysis for testing $H_0 : M = M_0$ against $H_1 : M = M_0 + 1$ to the more general setting of testing $H_0 : M = M_0$ against $H_1 : M = M_1$, where $M_1 > M_0 + 1$ to derive conditions for $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$.

There are two primary challenges in extending our earlier analysis. First, while there are exactly M_0 ways for the $(M_0 + 1)$ -component model to replicate the M_0 -component model, the number of ways for an M_1 -component model to replicate an M_0 -component model can be substantially larger when $M_1 > M_0 + 1$. Second, the degree of singularity in the Fisher Information Matrix may become more severe, necessitating the higher-order expansions beyond the second-order for an asymptotic analysis.

To deal with the first challenge, given any integer M that is greater than M_0 , we extend the definition of the restricted parameter space (16) by partitioning the M component-specific parameters into M_0 ordered, non-empty subsets. Explicitly, we consider a partition $\mathcal{T} = (T_1, T_2, \dots, T_{M_0})$ of the index set $\{1, 2, \dots, M\}$, subject to the following conditions: (i) $T_h \cap T_{h'} = \emptyset$ for $h \neq h'$, and $|T_h| \geq 1$, (ii) $\bigcup_{h=1}^{M_0} T_h = \{1, 2, \dots, M\}$, (iii) If $j \in T_h$ and $k \in T_{h'}$ with $h < h'$, then $j < k$. The total number of such ordered, non-empty, consecutive partitions is $J_M := \binom{M-1}{M_0-1}$. Let $\{\mathcal{T}^j\}_{j=1}^{J_M}$ denote the set of all these partitions, where each $\mathcal{T}^j = (T_1^j, T_2^j, \dots, T_{M_0}^j)$.

Additionally, consider a partition $\{\Theta_{\boldsymbol{\theta},h}^*\}_{h=1}^{M_0}$ of the parameter space $\Theta_{\boldsymbol{\theta}}$, defined by (15), such that each $\Theta_{\boldsymbol{\theta},h}^*$ forms a neighborhood containing $\boldsymbol{\theta}_h^*$ but excluding $\boldsymbol{\theta}_j^*$ for all $j \neq h$. Then, for each partition $\mathcal{T}^j$, $j = 1, 2, \dots, J_M$, the restricted parameter space is defined as:

$$\Psi_{\mathcal{T}^j}^* = \left\{ \boldsymbol{\vartheta}_M \in \bar{\Theta}_{\boldsymbol{\vartheta}_M}(\mathbf{c}) : \sum_{j=1}^M \alpha_j = 1, \quad \boldsymbol{\theta}_j \in \Theta_{\boldsymbol{\theta},h}^* \text{ for all } j \in T_h^j, \quad h = 1, \dots, M_0 \right\}.$$

Note that each $\Psi_{\mathcal{T}^j}^*$ uniquely represents a distinct configuration in which the components from an M -component model are distributed across M_0 components. Furthermore, the union of $\Psi_{\mathcal{T}^j}^*$ over $j = 1, 2, \dots, J_M$ covers the parameter space $\bar{\Theta}_{\boldsymbol{\vartheta}_M}(\mathbf{c})$.

Define the *local* MLE that maximises the log-likelihood function of the M -component

model under the constraint that $\boldsymbol{\vartheta}_M \in \Psi_{\mathcal{T}^j}^*$ by

$$\ell_n^M(\hat{\boldsymbol{\vartheta}}_M^j) := \arg \sup_{\boldsymbol{\vartheta}_M \in \Psi_{\mathcal{T}^j}^*} \ell_n^M(\boldsymbol{\vartheta}_M).$$

Because $\cup_{j=1}^{J_M} \Psi_{\mathcal{T}^j}^* = \bar{\Theta}_{\boldsymbol{\vartheta}_M}(\mathbf{c})$,

$$\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = \max_{j=1,2,\dots,J_M} \{\ell_n^M(\hat{\boldsymbol{\vartheta}}_M^j) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*)\}. \quad (5)$$

In view of (5), $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$ follows if we are able to show that $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M^j) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$ for any $j = 1, \dots, J_M$.

To deal with the singularity of the Fisher Information Matrix for the M -component density function $g_M(\mathbf{w}; \boldsymbol{\vartheta}_M)$, we consider the following one-to-one reparameterization of $\boldsymbol{\vartheta}_M$. Specifically, for $j = 1, 2, \dots, J_M$, we reparameterize $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_{M-1})$ by $\beta_h^j = \sum_{k \in T_h^j} \alpha_k$, $\boldsymbol{\tau}_h^j = (\tau_{h,1}^j, \dots, \tau_{h,|T_h^j|-1}^j)^\top$, and $\tau_{h,k}^j = \alpha_k / \beta_h^j$ for $k \in T_h^j$ and $h = 1, \dots, M_0$. Given the index set $\mathcal{T}^j = (T_1^j, \dots, T_{M_0}^j)$ with $T_h^j = (k_h, k_h + 1, \dots, k_h + |T_h^j| - 1)$, and applying an additional reparameterization of $\{\boldsymbol{\theta}_k\}_{k=1}^M$ defined by (7), the reparameterized density function of the M -component model is expressed as:

$$g_M(\mathbf{w}; \boldsymbol{\Psi}_M^j, \boldsymbol{\tau}^j) = \sum_{h=1}^{M_0} \beta_h^j \left(\sum_{k \in T_h^j} \tau_k^j h(\mathbf{w}; \tilde{\boldsymbol{\theta}}_k^j) \right), \quad (6)$$

where

$$\begin{pmatrix} \tilde{\boldsymbol{\theta}}_{k_h}^j \\ \tilde{\boldsymbol{\theta}}_{k_h+1}^j \\ \vdots \\ \tilde{\boldsymbol{\theta}}_{k_h+|T_h^j|-1}^j \end{pmatrix} = \begin{pmatrix} \boldsymbol{\nu}_h^j - \sum_{\ell=2}^{|T_h^j|} \tau_{h,\ell}^j \boldsymbol{\lambda}_{h,\ell}^j \\ \boldsymbol{\nu}_h^j + \tau_{h,1}^j \boldsymbol{\lambda}_{h,2}^j \\ \vdots \\ \boldsymbol{\nu}_h^j + \tau_{h,1}^j \boldsymbol{\lambda}_{h,|T_h^j|-1}^j \end{pmatrix}, \quad h = 1, \dots, M_0. \quad (7)$$

Collecting the reparameterized parameters, we set $\boldsymbol{\Psi}_M^j := (\boldsymbol{\eta}^j, \boldsymbol{\lambda}^j)^\top$, $\boldsymbol{\tau}^j := (\boldsymbol{\tau}_1^j, \dots, \boldsymbol{\tau}_{M_0}^j)^\top$, $\boldsymbol{\eta}^j := (\boldsymbol{\beta}^j, \boldsymbol{\nu}^j)$, $\boldsymbol{\beta}^j := (\beta_1^j, \dots, \beta_{M_0-1}^j)^\top$, $\boldsymbol{\nu}^j := (\boldsymbol{\nu}_1^j, \dots, \boldsymbol{\nu}_{M_0}^j)^\top$, and $\boldsymbol{\lambda}^j := (\boldsymbol{\lambda}_1^j, \dots, \boldsymbol{\lambda}_{M_0}^j)^\top$, where $\boldsymbol{\nu}_h^j := (\nu_{h,2}^j, \dots, \nu_{h,|T_h^j|}^j)^\top$, $\boldsymbol{\lambda}_h^j := (\lambda_{h,2}^j, \dots, \lambda_{h,|T_h^j|}^j)^\top$, and $\boldsymbol{\tau}_h^j := (\tau_{h,1}^j, \dots, \tau_{h,|T_h^j|-1}^j)$ for $h = 1, \dots, M_0$. When the data is generated from the M_0 -components model density $g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*) := \sum_{h=1}^{M_0} \alpha_h^* h(\mathbf{w}; \boldsymbol{\theta}_h^*)$, we have $\tilde{\boldsymbol{\theta}}_k^{j*} = \boldsymbol{\theta}_h^*$ for all $k \in T_h^j$, hence $\boldsymbol{\nu}_h^{j*} = \boldsymbol{\theta}_h^*$ and $\boldsymbol{\lambda}_h^{j*} = \mathbf{0}$ while $\boldsymbol{\tau}^j$ is not identified. In this scenario, since $\beta_h^{j*} = \alpha_h^*$ and $\sum_{k \in T_h^j} \tau_k^j h(\mathbf{w}; \tilde{\boldsymbol{\theta}}_k^{j*}) = h(\mathbf{w}; \boldsymbol{\theta}_h^*)$ in (6), the density function $g_M(\mathbf{w}; \boldsymbol{\Psi}_M^{j*}, \boldsymbol{\tau}^j)$ coincides with the true M_0 -component model density $g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*)$.

With this reparameterization, the first-order derivatives satisfy:

$$\nabla_{\boldsymbol{\nu}_h^j} \log g_M(\mathbf{w}; \boldsymbol{\psi}_M^{j*}, \boldsymbol{\tau}^j) = \frac{\nabla_{\boldsymbol{\theta}_h} h(\mathbf{w}; \boldsymbol{\theta}_h^*)}{g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*)}, \quad \nabla_{\boldsymbol{\lambda}_h^j} \log g_M(\mathbf{w}; \boldsymbol{\psi}_M^{j*}, \boldsymbol{\tau}^j) = \mathbf{0}, \quad h = 1, \dots, M_0. \quad (8)$$

Because $\nabla_{\boldsymbol{\lambda}_h^j} \log g_M(\mathbf{w}; \boldsymbol{\psi}_M^{j*}, \boldsymbol{\tau}^j) = \mathbf{0}$, the Fisher information matrix is singular, and the standard quadratic approximation fails. As analyzed in Section 5.1, when $M = M_0 + 1$, the unique elements of $\nabla_{\boldsymbol{\lambda}_h^j \boldsymbol{\lambda}_h^j} \log g_M(\mathbf{w}; \boldsymbol{\psi}_M^{j*}, \boldsymbol{\tau}^j)$ plays the role of score function in (2) to identify $\boldsymbol{\lambda}_h^j$. However, when M is much larger than M_0 , we need higher order derivatives beyond the second order derivatives to identify $\boldsymbol{\lambda}_h^j$.

Denote the density ratio by

$$l_{\boldsymbol{\Psi}_M^j \boldsymbol{\tau}^j, i} := \frac{g_M(\mathbf{W}_i; \boldsymbol{\Psi}_M^j, \boldsymbol{\tau}^j)}{g_{M_0}(\mathbf{W}_i; \boldsymbol{\vartheta}_{M_0}^*)}$$

so that $\ell_n(\boldsymbol{\Psi}_M^j, \boldsymbol{\tau}^j) - \ell_n(\boldsymbol{\Psi}_M^{j*}, \boldsymbol{\tau}^j) = \sum_{i=1}^n \log l_{\boldsymbol{\Psi}_M^j \boldsymbol{\tau}^j, i}$.

Let $\boldsymbol{\lambda}_{h, \ell_1}^j \otimes \boldsymbol{\lambda}_{h, \ell_2}^j \otimes \dots \otimes \boldsymbol{\lambda}_{h, \ell_p}^j$ denote the tensor containing all interactions among the elements of the vectors $\boldsymbol{\lambda}_{h, \ell_1}^j, \boldsymbol{\lambda}_{h, \ell_2}^j, \dots, \boldsymbol{\lambda}_{h, \ell_p}^j$, where $\otimes$ denotes the Kronecker product. We use the higher-order derivatives of $l_{\boldsymbol{\Psi}_M^j \boldsymbol{\tau}^j, i}$ with respect to $\boldsymbol{\lambda}_h^j$ evaluated at $\boldsymbol{\Psi}_M = \boldsymbol{\Psi}_M^*$ to identify $\boldsymbol{\lambda}_h^j$: for $h = 1, 2, \dots, M_0$ and for $p \geq 2$,

$$\nabla_{\boldsymbol{\lambda}_{h, \ell_1}^j \otimes \boldsymbol{\lambda}_{h, \ell_2}^j \otimes \dots \otimes \boldsymbol{\lambda}_{h, \ell_p}^j} l_{\boldsymbol{\Psi}_M^j \boldsymbol{\tau}^j, i} = \gamma_{h, \ell_1 \ell_2 \dots \ell_p}^j(\boldsymbol{\tau}_h^j) \frac{\alpha_h^{*j} \nabla_{\boldsymbol{\theta}^{\otimes p}} h_{h, i}^*}{g_{M_0, i}^*} \quad \text{for } (\ell_1, \dots, \ell_p) \in \{2, \dots, |T_h^j|\}^p,$$

where $\boldsymbol{\theta}^{\otimes p}$ represents $\boldsymbol{\theta} \otimes \boldsymbol{\theta} \dots \otimes \boldsymbol{\theta}$ (p times), $h_{h, i}^* := h(W_i; \boldsymbol{\theta}_h^*)$, and $g_{M_0, i}^* := g_{M_0}(\mathbf{W}_i; \boldsymbol{\vartheta}_{M_0}^*)$. For $p = 2, 3$, we have $\gamma_{h, \ell_1 \ell_2}^j(\boldsymbol{\tau}_h^j) = \tau_{h, 1}^j \tau_{h, \ell_1}^j (\tau_{h, \ell_2}^j + \delta_{\ell_1 \ell_2})$ and $\gamma_{h, \ell_1 \ell_2 \ell_3}^j(\boldsymbol{\tau}_h^j) = \tau_{h, 1}^j \tau_{h, \ell_1}^j (\delta_{\ell_1 \ell_2} \delta_{\ell_2 \ell_3} (\tau_{h, 1}^j)^2 - \tau_{h, \ell_2}^j \tau_{h, \ell_3}^j)$ with δ_{ij} being Kronecker delta. The elements of $\nabla_{\boldsymbol{\lambda}_{h, \ell_1}^j \otimes \boldsymbol{\lambda}_{h, \ell_2}^j \otimes \dots \otimes \boldsymbol{\lambda}_{h, \ell_p}^j} l_{\boldsymbol{\Psi}_M^j \boldsymbol{\tau}^j, i}$ are ave-zero random variables.

Let $q := \dim(\boldsymbol{\theta})$. Let $\widetilde{\text{vech}}_p(\nabla_{\boldsymbol{\theta}^{\otimes p}} h^*/g^*)$ extract all *unique* elements of q -dimensional symmetric array $\nabla_{\boldsymbol{\theta}^{\otimes p}} h^*/g^*$, multiply by its frequency, and stacks them into a vector. For example, for $p = 3$ and $q \geq 3$, $\widetilde{\text{vech}}_3(\nabla_{\boldsymbol{\theta}^{\otimes 3}} h^*/g^*)$ with $\boldsymbol{\theta} = (\theta_1, \dots, \theta_q)^\top$ will contain (i) q elements of the form $\nabla_{\theta_i^3} h^*/g^*$, (ii) $q(q-1)$ elements of the form $3\nabla_{\theta_i^2 \theta_j} h^*/g^*$ for $i \neq j$, and (iii) $\binom{q}{3}$ of the form $6\nabla_{\theta_i \theta_j \theta_k} h^*/g^*$ for $i \neq j \neq k$.

Choose p_h^j so that the dimension of the unique elements of $\frac{\nabla_{\boldsymbol{\theta}^{\otimes p_h^j}} h_{h, i}^*}{g_{M_0, i}^*}, \frac{\nabla_{\boldsymbol{\theta}^{\otimes p_h^j}} h_{h, i}^*}{g_{M_0, i}^*}, \dots, \frac{\nabla_{\boldsymbol{\theta}^{\otimes p_h^j}} h_{h, i}^*}{g_{M_0, i}^*}$ is at least as large as the number of elements in $\boldsymbol{\lambda}_h^j$ but no larger than neces-

sary:

$$\begin{aligned} & \dim \left(\widetilde{\text{vech}}_2 \left(\frac{\alpha_h^* \nabla_{\theta \otimes \theta} h_{h,i}^*}{g_{M_0,i}^*} \right) \right) + \dots + \dim \left(\widetilde{\text{vech}}_{p_h^j} \left(\frac{\alpha_h^* \nabla_{\theta^{\otimes p_h^j}} h_{h,i}^*}{g_{M_0,i}^*} \right) \right) \geq \dim(\lambda_h^j) \\ & > \dim \left(\widetilde{\text{vech}}_2 \left(\frac{\alpha_h^* \nabla_{\theta \otimes \theta} h_{h,i}^*}{g_{M_0,i}^*} \right) \right) + \dots + \dim \left(\widetilde{\text{vech}}_{p_h^j-1} \left(\frac{\alpha_h^* \nabla_{\theta^{\otimes (p_h^j-1)}} h_{h,i}^*}{g_{M_0,i}^*} \right) \right). \end{aligned} \quad (9)$$

The value of p_h^j indicates a necessary order of local expansions for identifying λ_h^j . For example, consider the case of testing $H_0 : M_0 = 1$ against $H_1 : M_0 = M_1$ for any $M_1 > 1$. Then, $g_{M_0,i}^* = h_i^*$ and $J_{M_1} = 1$. In this case, for $p = 2$, because $\dim \left(\widetilde{\text{vech}}_2 \left(\frac{\nabla_{\theta \otimes \theta} h_i^*}{h_i^*} \right) \right) = \frac{q(q+1)}{2}$ and $\dim(\lambda) = q(M_1 - 1)$, (9) is satisfied if $\frac{q(q+1)}{2} \geq M_1 - 1$. For $p = 3$ and $q \geq 3$, the condition is $\frac{q(q+1)}{2} + \left(q + q(q-1) + \frac{q(q-1)(q-2)}{6} \right) \geq q(M_1 - 1) > \frac{q(q+1)}{2}$, which simplifies to $\frac{(q+5)(q+1)}{6} + 1 \geq M_1 > \frac{q+3}{2}$. For $p = 4$ and $q \geq 4$, the condition is given by $\frac{(q+5)(q+1)}{6} + \frac{q^3-5q^2+10q-4}{2} + 1 \geq M_1 > \frac{(q+5)(q+1)}{6} + 1$, where λ is identified for up to $M_1 = 18$ if $q = 4$.

For $h = 1, \dots, M_0$, let

$$\mathbf{v}_{\lambda_h^j \tau_h^j} := \begin{pmatrix} \text{vech}_2 \left(\sum_{\ell_1=2}^{|T_h^j|} \sum_{\ell_2=2}^{|T_h^j|} \gamma_{h,\ell_1\ell_2}^j (\tau_h^j) \lambda_{h,\ell_1}^j \otimes \lambda_{h,\ell_2}^j \right) \\ \vdots \\ \text{vech}_{p_h^j} \left(\sum_{\ell_1=2}^{|T_h^j|} \dots \sum_{\ell_{p_h^j}=2}^{|T_h^j|} \gamma_{h,\ell_1\ell_2\dots\ell_{p_h^j}}^j (\tau_h^j) \lambda_{h,\ell_1}^j \otimes \lambda_{h,\ell_2}^j \otimes \dots \otimes \lambda_{h,\ell_{p_h^j}}^j \right) \end{pmatrix},$$

where $\text{vech}_p(\cdot)$ is defined similarly to $\widetilde{\text{vech}}_p(\cdot)$ as an operator that extracts all unique elements of q -dimensional symmetric array, but without multiplying each element by its frequency, so that the elements of $\text{vech}_p(\lambda_{h,\ell_1} \otimes \dots \otimes \lambda_{h,\ell_p})$ are conformable to those of $\widetilde{\text{vech}}_p(\nabla_{\theta^{\otimes p}} h_h^*/g_{M_0}^*)$.

Collect the relevant normalized reparameterized parameters and define $\mathbf{t}_{\psi_M^j \tau^j}$ as

$$\mathbf{t}_{\psi_M^j \tau^j} := \begin{pmatrix} \boldsymbol{\eta} - \boldsymbol{\eta}^* \\ \mathbf{v}_{\lambda_1^j \tau_1^j} \\ \vdots \\ \mathbf{v}_{\lambda_{M_0}^j \tau_{M_0}^j} \end{pmatrix}. \quad (10)$$

For $j = 1, \dots, J_M$, define the vector $\mathbf{s}^j(\mathbf{W})$ as

$$\mathbf{s}^j(\mathbf{W}) = \begin{pmatrix} \mathbf{s}_\eta^j(\mathbf{W}) \\ \mathbf{s}_\lambda^j(\mathbf{W}) \end{pmatrix} \quad \text{with} \quad \mathbf{s}_\eta^j(\mathbf{W}) := \begin{pmatrix} \mathbf{s}_\alpha^j(\mathbf{W}) \\ \mathbf{s}_\nu^j(\mathbf{W}) \end{pmatrix} \quad \text{and} \quad \mathbf{s}_\lambda^j(\mathbf{W}) := \begin{pmatrix} \mathbf{s}_{\lambda_1^j}^j(\mathbf{W}) \\ \vdots \\ \mathbf{s}_{\lambda_{M_0}^j}^j(\mathbf{W}) \end{pmatrix}, \quad (11)$$

where

$$\mathbf{s}_\alpha^j(\mathbf{W}) = \begin{pmatrix} \frac{h_1(\mathbf{W}; \boldsymbol{\theta}_1^*) - h_{M_0}(\mathbf{W}; \boldsymbol{\theta}_{M_0}^*)}{g_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)} \\ \vdots \\ \frac{h_{M_0-1}(\mathbf{W}; \boldsymbol{\theta}_{M_0-1}^*) - h_{M_0}(\mathbf{W}; \boldsymbol{\theta}_{M_0}^*)}{g_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)} \end{pmatrix}, \quad \mathbf{s}_\nu^j(\mathbf{W}) = \begin{pmatrix} \frac{\alpha_1^* \nabla_{\boldsymbol{\theta}} h(\mathbf{W}; \boldsymbol{\theta}_1^*)}{g_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)} \\ \vdots \\ \frac{\alpha_{M_0}^* \nabla_{\boldsymbol{\theta}} h(\mathbf{W}; \boldsymbol{\theta}_{M_0}^*)}{g_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)} \end{pmatrix}, \quad \text{and} \quad (12)$$

$$\mathbf{s}_{\lambda_h^j}^j(\mathbf{W}) = \begin{pmatrix} \widetilde{\text{vech}}_2 \left(\frac{\alpha_h^* \nabla_{\boldsymbol{\theta} \otimes \boldsymbol{\theta}} h(\mathbf{W}; \boldsymbol{\theta}_h^*)}{g_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)} \right) \\ \vdots \\ \widetilde{\text{vech}}_{p_h^j} \left(\frac{\alpha_h^* \nabla_{\boldsymbol{\theta}^{\otimes p_h^j}} h(\mathbf{W}; \boldsymbol{\theta}_h^*)}{g_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)} \right) \end{pmatrix} \quad \text{for } h = 1, \dots, M_0. \quad (13)$$

We denote $\mathbf{s}_i^j := \mathbf{s}^j(\mathbf{W}_i)$. We assume that $l_{\Psi_M^j \boldsymbol{\tau}^j, i}$ can be expanded around $l_{\Psi_M^j \boldsymbol{\tau}^j, i} = 1$ as follows. Let $P_n(\mathbf{s}_i^j (\mathbf{s}_i^j)^\top) := n^{-1} \sum_{i=1}^n \mathbf{s}_i^j (\mathbf{s}_i^j)^\top$ and $\nu_n(\mathbf{s}_i^j) := n^{-1/2} \sum_{i=1}^n [\mathbf{s}_i^j - \mathbb{E}_{\boldsymbol{\theta}^*}[\mathbf{s}_i^j]]$.

For $\epsilon > 0$, define the neighborhood of the set Ψ_M^j corresponding to the null hypothesis $H_0 : M_0 = M$ as follows:

$$\mathcal{N}_\epsilon = \{ \Psi_M^j \in \Theta_{\Psi_M^j} : \sup_{\boldsymbol{\tau}^j \in \Theta_{\boldsymbol{\tau}^j, c_1}} \|\mathbf{t}_{\Psi_M^j \boldsymbol{\tau}^j}\| < \epsilon \},$$

where $\Theta_{\boldsymbol{\tau}^j, c_1}$ denotes the parameter space of $\boldsymbol{\tau}^j$, as implied by the parameter space Θ_{α, c_1} of α , which restricts the values of τ_h^j to be bounded away from 0 and 1.

Assumption A.2. For $M = M_0 + 1, \dots, \bar{M}$, the following assumption holds when the data is generated from the M_0 -component model. For all $j = 1, \dots, J_M$ and $i = 1, \dots, n$, $l_{\Psi_M^j \boldsymbol{\tau}^j, i} - 1$ admits an expansion

$$l_{\Psi_M^j \boldsymbol{\tau}^j, i} - 1 = \left(\mathbf{t}_{\Psi_M^j \boldsymbol{\tau}^j} \right)^\top \mathbf{s}_i^j + r_{\Psi_M^j \boldsymbol{\tau}^j, i}, \quad (14)$$

where $\mathbf{t}_{\Psi_M^j \boldsymbol{\tau}^j}$ satisfies $\Psi_M^j \rightarrow \Psi_M^{j*}$ if $\sup_{\boldsymbol{\tau}^j \in \Theta_{\boldsymbol{\tau}^j, c_1}} \|\mathbf{t}_{\Psi_M^j \boldsymbol{\tau}^j}\| \rightarrow 0$, and $(\mathbf{s}_i^j, r_{\Psi_M^j \boldsymbol{\tau}^j, i})$ satisfy, for some $C \in (0, \infty)$, $\delta > 0$, $\epsilon > 0$, and $\rho \in (0, 1)$, (a) $\mathbb{E}_{\boldsymbol{\theta}^*} \|\mathbf{s}_i^j\|^{2+\delta} < C$, (b) $\|P_n(\mathbf{s}_i^j (\mathbf{s}_i^j)^\top) - \boldsymbol{\mathcal{I}}^j\| = o_p(1)$, where $\boldsymbol{\mathcal{I}}^j := \mathbb{E}[\mathbf{s}_i^j (\mathbf{s}_i^j)^\top]$ is finite and non-singular, (c) $\mathbb{E}_{\boldsymbol{\theta}^*} [\sup_{\Psi_M^j \in \mathcal{N}_\epsilon} |r_{\Psi_M^j \boldsymbol{\tau}^j, i}| / (\|\mathbf{t}_{\Psi_M^j \boldsymbol{\tau}^j}\| \|\Psi_M^j - \Psi_M^{j*}\|)] < \epsilon$.

$\Psi_M^{j*}|||)^2] < \infty$, (d) $\sup_{\Psi_M^j \in \mathcal{N}_\varepsilon} [\nu_n(r_{\Psi_M^j, i}) / (||\mathbf{t}_{\Psi_M^j}|| ||\Psi_M^j - \Psi_M^{j*}||)] = O_p(1)$, and (e) $\sup_{\Psi_M^j \in \mathcal{N}_\varepsilon} ||\nu_n(\mathbf{s}_i^j)|| = O_p(1)$.

Assumption A.2(b) is a key regularity condition that requires the unique elements of $\nabla_{\theta} h_h^*/g_{M_0}^*$, $\nabla_{\theta \otimes \theta} h_h^*/g_{M_0}^*$, ..., $\nabla_{\theta \otimes p_h} h_h^*/g_{M_0}^*$ are linearly independent and their expectation is finite.

Proposition A.2. *Suppose that Assumption A.2 holds. Then, for any $c > 0$,*

$$\sup_{\Psi_M^j \in \mathcal{N}_{c/\sqrt{n}}} \left| \ell_n(\Psi_M^j, \boldsymbol{\tau}^j) - \ell_n(\Psi_M^{j*}, \boldsymbol{\tau}^j) - \sqrt{n} \mathbf{t}_{\Psi_M^j, \boldsymbol{\tau}^j}^\top \nu_n(\mathbf{s}_i^j) + n \mathbf{t}_{\Psi_M^j, \boldsymbol{\tau}^j}^\top \mathcal{I}^j \mathbf{t}_{\Psi_M^j, \boldsymbol{\tau}^j} / 2 \right| = o_p(1).$$

The following proposition expands $\ell_n(\Psi_M^j, \boldsymbol{\tau}^j)$ in $A_{\varepsilon_n}(\eta) := \{\Psi_M \in \mathcal{N}_{\varepsilon_n} : \ell_n(\Psi_M^j, \boldsymbol{\tau}^j) - \ell_n(\Psi_M^{j*}, \boldsymbol{\tau}^j) \geq -\eta\}$ for some $\eta \in [0, \infty)$ and for $\varepsilon_n \rightarrow 0$ slowly to ensure $\Pr(\hat{\Psi}_M \in \mathcal{N}_{\varepsilon_n}) = 1$.

Proposition A.3. *Suppose that Assumption A.2 holds. Then, for all $j = 1, \dots, J_M$, for any $\eta > 0$, and for any $\{\varepsilon_n : n = 1, 2, \dots\}$ such that $\varepsilon_n \rightarrow 0$ but $\Pr(\hat{\Psi}_M \in \mathcal{N}_{\varepsilon_n}) = 1$,*

(a) $\sup_{\Psi_M \in A_{\varepsilon_n}(\eta)} |\mathbf{t}_{\Psi_M^j, \boldsymbol{\tau}^j}| = O_p(n^{-1/2})$, (b) for any $c > 0$,

$$\sup_{\Psi_M \in A_{\varepsilon_n}(\eta) \cup \mathcal{N}_{c/\sqrt{n}}} \left| \ell_n(\Psi_M^j, \boldsymbol{\tau}^j) - \ell_n(\Psi_M^{j*}, \boldsymbol{\tau}^j) - \sqrt{n} \mathbf{t}_{\Psi_M^j, \boldsymbol{\tau}^j}^\top \nu_n(\mathbf{s}_i^j) + n \mathbf{t}_{\Psi_M^j, \boldsymbol{\tau}^j}^\top \mathcal{I}^j \mathbf{t}_{\Psi_M^j, \boldsymbol{\tau}^j} / 2 \right| = o_p(1),$$

and (c) $\sup_{\Psi_M \in A_{\varepsilon_n}(\eta)} |\ell_n(\Psi_M^j, \boldsymbol{\tau}^j) - \ell_n(\Psi_M^{j*}, \boldsymbol{\tau}^j)| = O_p(1)$.

Because a consistent MLE is in $A_{\varepsilon_n}(\eta)$ by definition, Proposition A.3 implies that $\ell_n(\hat{\Psi}_M^j, \boldsymbol{\tau}^j) - \ell_n(\Psi_M^{j*}, \boldsymbol{\tau}^j) = O_p(1)$ for a consistent MLE.

A.3 Technical Details for LRT

A.3.1 Parameter Space Partition and Local Restrictions

Recall that $\mu_1^* < \mu_2^* < \dots < \mu_{M_0}^*$. Let $\underline{\Theta}_\mu$ and $\overline{\Theta}_\mu$ denote the lower and upper bounds of Θ_μ . Define

$$\Theta_{\theta, h}^* = \begin{cases} [\underline{\Theta}_\mu, \frac{\mu_1^* + \mu_2^*}{2}] \times \Theta_\beta \times \Theta_{\sigma^2}, & h = 1 \\ [\frac{\mu_{h-1}^* + \mu_h^*}{2}, \frac{\mu_h^* + \mu_{h+1}^*}{2}] \times \Theta_\beta \times \Theta_{\sigma^2}, & 2 \leq h \leq M_0 - 1 \\ [\frac{\mu_{M_0-1}^* + \mu_{M_0}^*}{2}, \overline{\Theta}_\mu] \times \Theta_\beta \times \Theta_{\sigma^2}, & h = M_0. \end{cases} \quad (15)$$

Then $\{\Theta_{\theta, h}^*\}_{h=1}^{M_0}$ partitions Θ_θ such that $\Theta_{\theta, h}^*$ contains $\boldsymbol{\theta}_h^*$ but not $\boldsymbol{\theta}_j^*$ for $j \neq h$. Define

the restricted parameter space $\Psi_h^* \subset \bar{\Theta}_{\vartheta_{M_0+1}}(\mathbf{c})$ as

$$\Psi_h^* = \left\{ \begin{array}{l} \vartheta_{M_0+1} \in \bar{\Theta}_{\vartheta_{M_0+1}}(\mathbf{c}) : \sum_{j=1}^{M_0+1} \alpha_j = 1; \\ \boldsymbol{\theta}_j \in \Theta_{\boldsymbol{\theta},j}^* \text{ for } j \leq h-1; \quad \boldsymbol{\theta}_h, \boldsymbol{\theta}_{h+1} \in \Theta_{\boldsymbol{\theta},h}^*; \\ \boldsymbol{\theta}_j \in \Theta_{\boldsymbol{\theta},j-1}^* \text{ for } j \geq h+2 \end{array} \right\}. \quad (16)$$

Note that $\Psi_h^* \cap \Theta_{\vartheta_{M_0+1},1h}^* \neq \emptyset$, $\Psi_h^* \cap \Theta_{\vartheta_{M_0+1},1l}^* = \emptyset$ for $h \neq l$, and $\cup_{h=1}^{M_0} \Psi_h^* = \bar{\Theta}_{\vartheta_{M_0+1}}(\mathbf{c})$.

The subset $\Theta_{\vartheta_{M_0+1},1h}^*$ corresponding to $H_{0,1h}$ is defined as

$$\Theta_{\vartheta_{M_0+1},1h}^* := \left\{ \begin{array}{l} \vartheta_{M_0+1} \in \Theta_{\vartheta_{M_0+1}} : \alpha_h, \alpha_{h+1} > 0; \quad \alpha_h + \alpha_{h+1} = \alpha_h^*; \\ \boldsymbol{\theta}_h = \boldsymbol{\theta}_{h+1} = \boldsymbol{\theta}_h^*; \quad \alpha_j = \alpha_j^*, \boldsymbol{\theta}_j = \boldsymbol{\theta}_j^* \text{ for } j < h; \\ \alpha_j = \alpha_{j-1}^*, \boldsymbol{\theta}_j = \boldsymbol{\theta}_{j-1}^* \text{ for } j > h+1 \end{array} \right\} \quad (17)$$

for $h = 1, \dots, M_0$.

A.3.2 Explicit Score Function via Hermite Polynomials

$H^j(\cdot)$ is defined as the j -th order Hermite polynomial. $H^1(t) = t$, $H^2(t) = t^2 - 1$, $H^3(t) = t^3 - 3t$, and $H^4(t) = t^4 - 6t^2 + 3$. As shown in the supplementary material of Kasahara and Shimotsu (2015), the derivative of $\{\frac{1}{\sigma}\phi(\frac{t}{\sigma})\}$ is

$$\frac{\nabla_{\mu^m} \nabla_{(\sigma^2)^\ell} \{\frac{1}{\sigma}\phi(\frac{t}{\sigma})\}}{\{\frac{1}{\sigma}\phi(\frac{t}{\sigma})\}} = \left(\frac{1}{2}\right)^\ell \left(\frac{1}{\sigma}\right)^{m+2\ell} H^{m+2\ell} \left(\frac{t}{\sigma}\right).$$

Let $f^* = f(\mathbf{W}; \theta^*) = \prod_{t=1}^T f_t^*(\mathbf{W}; \theta^*)$, where f_t^* denotes the component density for observation t , which may be either a normal or a mixture density. Let $\nabla f^* = \nabla f(\mathbf{W}; \theta^*)$ denote its gradient.

Define the Hermite polynomials evaluated at $Y_t - \mathbf{X}_t^\top \boldsymbol{\beta}^* - \mu^*$ as follows:

$$H_t^{j*} = \frac{1}{(\sigma^*)^j} H^j \left(\frac{Y_t - \mathbf{X}_t^\top \boldsymbol{\beta}^* - \mu^*}{\sigma^*} \right). \quad (18)$$

A.3.3 Score function for the model (1) with normal density ((2))-(3))

Let $f_t^* = \frac{1}{\sigma^*} \phi\left(\frac{Y_t - \mathbf{X}_t^\top \boldsymbol{\beta}^* - \mu^*}{\sigma^*}\right)$. Then, the first-order derivatives of the normal density function ((2))-(3)) with respect to the parameters are given by:

$$\begin{aligned}\nabla_\mu f^* &= f^* \sum_{t=1}^T H_t^{1*}, & \nabla_{\sigma^2} f^* &= f^* \sum_{t=1}^T \frac{1}{2} H_t^{2*}, \\ \nabla_\beta f^* &= f^* \sum_{t=1}^T H_t^{1*} \mathbf{X}_t.\end{aligned}$$

The score function defined in (2) is then written in terms of the Hermite polynomials:

$$\mathbf{s}_\eta(\mathbf{W}) = \begin{pmatrix} s_\mu \\ s_\sigma \\ \mathbf{s}_\beta \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^T H_t^{1*} \\ \sum_{t=1}^T H_t^{2*} \\ \sum_{t=1}^T H_t^{1*} \mathbf{x}_t \end{pmatrix} \quad \text{and} \quad \mathbf{s}_{\lambda\lambda}(\mathbf{W}) = \begin{pmatrix} s_{\lambda_\mu \lambda_\mu} \\ s_{\lambda_\mu \lambda_\sigma} \\ s_{\lambda_\sigma \lambda_\sigma} \\ \mathbf{s}_{\lambda_\mu \lambda_\beta} \\ \mathbf{s}_{\lambda_\sigma \lambda_\beta} \\ \mathbf{s}_{\lambda_\beta \lambda_\beta} \end{pmatrix}, \quad (19)$$

where

$$\begin{pmatrix} s_{\lambda_\mu \lambda_\mu} \\ s_{\lambda_\mu \lambda_\sigma} \\ s_{\lambda_\sigma \lambda_\sigma} \\ \mathbf{s}_{\lambda_\mu \lambda_\beta} \\ \mathbf{s}_{\lambda_\sigma \lambda_\beta} \\ \mathbf{s}_{\lambda_\beta \lambda_\beta} \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^T [H_t^{2*} + \sum_{s \neq t} H_t^{1*} H_s^{1*}] \\ \frac{1}{2} \sum_{t=1}^T [H_t^{3*} + \sum_{s \neq t} H_t^{1*} H_s^{2*}] \\ \frac{1}{4} \sum_{t=1}^T [H_t^{4*} + \sum_{s \neq t} H_t^{2*} H_s^{2*}] \\ \sum_{t=1}^T [H_t^{2*} + \sum_{s \neq t} H_t^{1*} H_s^{1*}] \mathbf{X}_t \\ \frac{1}{2} \sum_{t=1}^T [H_t^{3*} + \sum_{s \neq t} H_t^{1*} H_s^{2*}] \mathbf{X}_t \\ \sum_{t=1}^T [H_t^{2*} \widetilde{\text{vech}}(\mathbf{X}_t \mathbf{X}_t^\top) + \sum_{s \neq t} H_t^{1*} H_s^{1*} \widetilde{\text{vech}}(\mathbf{X}_t \mathbf{X}_s^\top)] \end{pmatrix}. \quad (20)$$

A.3.4 Score function for the model (1) with normal mixture density (2)–(3) when $K = 2$

For brevity, we present the score function for the model (1) with a two-component normal mixture (2)–(3) when there are no covariates. The score function for the model with covariates can be analogously derived. Let

$$f^* = \prod_{t=1}^T f_t^* = \prod_{t=1}^T (\tau^* f_{1t}^* + (1 - \tau^*) f_{2t}^*),$$

where $f_{kt}^* = \frac{1}{\sigma^*} \phi\left(\frac{Y_t - \mu_k^*}{\sigma^*}\right)$ for $k = 1, 2$. In this model, we omit the covariates $\mathbf{X}_t$ and the parameter β^* for simplicity. The score function with covariates can be derived similarly to the previous section. Define the b -th order normalized Hermite polynomial for k -th component evaluated at $Y_t - \mu_k^*$ as follows:

$$H_{kt}^{b*} = \frac{1}{(\sigma^*)^b} H^b\left(\frac{Y_t - \mu_k^*}{\sigma^*}\right). \quad (21)$$

Let

$$\gamma_{1t} := \frac{\tau^* f_{1t}^*}{\tau^* f_{1t}^* + (1 - \tau^*) f_{2t}^*}, \quad \gamma_{2t} := 1 - \gamma_{1t}.$$

The score function defined in (2) is then written as:

$$\mathbf{s}_\eta(\mathbf{W}) = \begin{pmatrix} s_\tau \\ s_{\mu_1} \\ s_{\mu_2} \\ s_\sigma \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^T \left(\frac{\gamma_{1t}}{\tau^*} - \frac{\gamma_{2t}}{1-\tau^*} \right) \\ \sum_{t=1}^T \gamma_{1t} H_{1t}^{1*} \\ \sum_{t=1}^T \gamma_{2t} H_{2t}^{1*} \\ \sum_{t=1}^T \gamma_{1t} H_{1t}^{2*} + \gamma_{2t} H_{2t}^{2*} \end{pmatrix} \quad \text{and} \quad \mathbf{s}_{\lambda\lambda}(\mathbf{W}) = \begin{pmatrix} s_{\lambda_\tau \lambda_\tau} \\ s_{\lambda_{\mu_1} \lambda_{\mu_1}} \\ s_{\lambda_{\mu_2} \lambda_{\mu_2}} \\ s_{\lambda_{\mu_1} \lambda_{\mu_2}} \\ s_{\lambda_\tau \lambda_{\mu_1}} \\ s_{\lambda_\tau \lambda_{\mu_2}} \\ s_{\lambda_{\mu_1} \lambda_\sigma} \\ s_{\lambda_{\mu_2} \lambda_\sigma} \\ s_{\lambda_\sigma \lambda_\sigma} \\ s_{\lambda_\tau \lambda_\sigma} \end{pmatrix}, \quad (22)$$

where

$$\begin{pmatrix} s_{\lambda_\tau \lambda_\tau} \\ s_{\lambda_{\mu_1} \lambda_{\mu_1}} \\ s_{\lambda_{\mu_2} \lambda_{\mu_2}} \\ s_{\lambda_{\mu_1} \lambda_{\mu_2}} \\ s_{\lambda_\tau \lambda_{\mu_1}} \\ s_{\lambda_\tau \lambda_{\mu_2}} \\ s_{\lambda_{\mu_1} \lambda_\sigma} \\ s_{\lambda_{\mu_2} \lambda_\sigma} \\ s_{\lambda_\sigma \lambda_\sigma} \\ s_{\lambda_\tau \lambda_\sigma} \end{pmatrix} = \begin{pmatrix} \sum_{t=1}^T \sum_{s \neq t} \left(\frac{\gamma_{1t}}{\tau^*} - \frac{\gamma_{2t}}{1-\tau^*} \right) \left(\frac{\gamma_{1s}}{\tau^*} - \frac{\gamma_{2s}}{1-\tau^*} \right) \\ \sum_{t=1}^T (\gamma_{1t} H_{1t}^{2*} + \sum_{s \neq t} \gamma_{1t} \gamma_{1s} H_{1t}^{1*} H_{1s}^{1*}) \\ \sum_{t=1}^T (\gamma_{2t} H_{2t}^{2*} + \sum_{s \neq t} \gamma_{2t} \gamma_{2s} H_{2t}^{1*} H_{2s}^{1*}) \\ \sum_{t=1}^T (\sum_{s \neq t} \gamma_{1t} \gamma_{2s} H_{1t}^{1*} H_{2s}^{1*}) \\ \sum_{t=1}^T \left(\frac{\gamma_{1t}}{\tau^*} H_{1t}^{1*} + \sum_{s \neq t} \gamma_{1t} H_{1t}^{1*} \left(\frac{\gamma_{1s}}{\tau^*} - \frac{\gamma_{2s}}{1-\tau^*} \right) \right) \\ \sum_{t=1}^T \left(-\frac{\gamma_{2t}}{1-\tau^*} H_{2t}^{1*} + \sum_{s \neq t} \gamma_{2t} H_{2t}^{1*} \left(\frac{\gamma_{1s}}{\tau^*} - \frac{\gamma_{2s}}{1-\tau^*} \right) \right) \\ \frac{1}{2} \sum_{t=1}^T (\gamma_{1t} H_{1t}^{3*} + \sum_{s \neq t} \gamma_{1t} H_{1t}^{1*} (\gamma_{1s} H_{1s}^{2*} + \gamma_{2s} H_{2s}^{2*})) \\ \frac{1}{2} \sum_{t=1}^T (\gamma_{2t} H_{2t}^{3*} + \sum_{s \neq t} \gamma_{2t} H_{2t}^{1*} (\gamma_{1s} H_{1s}^{2*} + \gamma_{2s} H_{2s}^{2*})) \\ \frac{1}{4} \sum_{t=1}^T ((\gamma_{1t} H_{1t}^{4*} + \gamma_{2t} H_{2t}^{4*}) + \sum_{s \neq t} (\gamma_{1t} H_{1t}^{2*} + \gamma_{2t} H_{2t}^{2*}) (\gamma_{1s} H_{1s}^{2*} + \gamma_{2s} H_{2s}^{2*})) \\ \frac{1}{2} \sum_{t=1}^T \left(\frac{\gamma_{1t}}{\tau^*} H_{1t}^{2*} - \frac{\gamma_{2t}}{1-\tau^*} H_{2t}^{2*} + \sum_{s \neq t} (\gamma_{1t} H_{1t}^{2*} + \gamma_{2t} H_{2t}^{2*}) \left(\frac{\gamma_{1s}}{\tau^*} - \frac{\gamma_{2s}}{1-\tau^*} \right) \right) \end{pmatrix}. \quad (23)$$

A.3.5 Score function for testing $H_0 : M = M_0$ against $H_A : M = M_0 + 1$

We only present score function for the model (1) with normal density ((2))-((3)). The score functions for other models can be derived analogously.

Let $g^* = g(\mathbf{W}; \vartheta_{M_0}^*) = \sum_{j=1}^{M_0} \alpha_j^* f_j^*$ denote the true M_0 -component model as in equation (10), where $f_j^* = f(\mathbf{W}; \theta_j^*) = \prod_{t=1}^T \frac{1}{\sigma_j^*} \phi\left(\frac{Y_t - \mu_j^*}{\sigma_j^*}\right)$ is the j -th component density. In this section, we omit the covariate $\mathbf{x}_{it}$ and the parameters β_j^* for simplicity; the derivation with covariates can be done analogously to the previous section. Define $H_{j,it}^{b*}$ as a shorthand for the b -th order normalized Hermite polynomial evaluated at $\frac{Y_{it} - \mu_j^*}{\sigma_j^*}$, i.e., $H_{j,t}^{b*} = \frac{1}{(\sigma_j^*)^b} H^b\left(\frac{Y_t - \mu_j^*}{\sigma_j^*}\right)$. Define the weight w_i^{j*} as $w_j^* = \frac{\alpha_j^* f_j^*}{\sum_l \alpha_l^* f_l^*}$, $j = 1, \dots, M_0$. The score functions are

$$\begin{aligned} \mathbf{s}_\alpha(\mathbf{W}) &= \begin{pmatrix} \frac{w_1^*}{\alpha_1^*} - \frac{w_{M_0}^*}{\alpha_{M_0}^*} \\ \vdots \\ \frac{w_{M_0-1}^*}{\alpha_{M_0-1}^*} - \frac{w_{M_0}^*}{\alpha_{M_0}^*} \end{pmatrix}, & \mathbf{s}_\mu(\mathbf{W}) &= \begin{pmatrix} w_1^* \sum_{t=1}^T H_{1,t}^{1*} \\ \vdots \\ w_{M_0}^* \sum_{t=1}^T H_{M_0,t}^{1*} \end{pmatrix}, \\ \mathbf{s}_\sigma(\mathbf{W}) &= \begin{pmatrix} w_1^* \sum_{t=1}^T H_{1,t}^{2*} \\ \vdots \\ w_{M_0}^* \sum_{t=1}^T H_{M_0,t}^{2*} \end{pmatrix}, & \mathbf{s}_{\lambda\lambda}(\mathbf{W}) &= \begin{pmatrix} \mathbf{s}_{\lambda_\mu \lambda_\mu} \\ \mathbf{s}_{\lambda_\mu \lambda_\sigma} \\ \mathbf{s}_{\lambda_\sigma \lambda_\sigma} \end{pmatrix}, \end{aligned}$$

where

$$\begin{aligned} \mathbf{s}_{\lambda_\mu \lambda_\mu} &= \begin{pmatrix} w_1^* \sum_{t=1}^T [H_{1,t}^{2*} + \sum_{s \neq t} H_{1,t}^{1*} H_{1,s}^{1*}] \\ \vdots \\ w_{M_0}^* \sum_{t=1}^T [H_{M_0,t}^{2*} + \sum_{s \neq t} H_{M_0,t}^{1*} H_{M_0,s}^{1*}] \end{pmatrix}, & \mathbf{s}_{\lambda_\mu \lambda_\sigma} &= \begin{pmatrix} w_1^* \frac{1}{2} \sum_{t=1}^T [H_{1,t}^{3*} + \sum_{s \neq t} H_{1,t}^{1*} H_{1,s}^{2*}] \\ \vdots \\ w_{M_0}^* \frac{1}{2} \sum_{t=1}^T [H_{M_0,t}^{3*} + \sum_{s \neq t} H_{M_0,t}^{1*} H_{M_0,s}^{2*}] \end{pmatrix}, \\ \mathbf{s}_{\lambda_\sigma \lambda_\sigma} &= \begin{pmatrix} w_1^* \frac{1}{4} \sum_{t=1}^T [H_{1,t}^{4*} + \sum_{s \neq t} H_{1,t}^{2*} H_{1,s}^{2*}] \\ \vdots \\ w_{M_0}^* \frac{1}{4} \sum_{t=1}^T [H_{M_0,t}^{4*} + \sum_{s \neq t} H_{M_0,t}^{2*} H_{M_0,s}^{2*}] \end{pmatrix}. \end{aligned}$$

A.4 Asymptotic distribution under local alternatives

We derive the asymptotic distribution of the LRTS under local alternatives. For brevity, we focus on testing $H_0 : M_0 = 1$ against $H_A : M_0 = 2$. Consider the following local alternative to the homogeneous model $f(\mathbf{w}; \gamma^*, \boldsymbol{\theta}^*)$ with $\boldsymbol{\theta}^* = (\mu^*, \sigma^{*2}, (\boldsymbol{\beta}^*)^\top)^\top$. For brevity, we omit the common parameter γ in this section. In a reparameterized parameter, $\boldsymbol{\Psi}^* = ((\boldsymbol{\nu}^*)^\top, (\boldsymbol{\lambda}^*)^\top)^\top$. For $\alpha^* \in (0, 1)$ and a local parameter $\mathbf{h} = (\mathbf{h}_\nu^\top, \mathbf{h}_\lambda^\top)^\top$

with $\mathbf{h}_\lambda \in v(\boldsymbol{\theta}_\lambda)$, we consider a sequence of contiguous local alternatives $(\alpha_n, \boldsymbol{\psi}_n^\top)^\top = (\alpha_n, \boldsymbol{\nu}_n^\top, \boldsymbol{\lambda}_n^\top) \in \Theta_\alpha \times \Theta_\nu \times \Theta_\lambda$ such that, with $\mathbf{t}_\lambda(\boldsymbol{\lambda}, \alpha)$ given by (35),

$$\mathbf{h}_\nu = \sqrt{n}(\boldsymbol{\nu}_n - \boldsymbol{\nu}^*), \quad \mathbf{h}_\lambda = \sqrt{n}\mathbf{t}_\lambda(\boldsymbol{\lambda}_n, \alpha_n), \quad \text{and} \quad \alpha_n = \alpha^* + o(1). \quad (24)$$

Equivalently, the non-reparameterized contiguous local alternatives are given by

$$\boldsymbol{\theta}_{1,n} = \boldsymbol{\nu}_n + (1 - \alpha_n)\boldsymbol{\lambda}_n \quad \text{and} \quad \boldsymbol{\theta}_{2,n} = \boldsymbol{\nu}_n - \alpha_n\boldsymbol{\lambda}_n \quad (25)$$

for $\boldsymbol{\nu}_n = \boldsymbol{\nu}^* + n^{-1/2}\mathbf{h}_\nu$ and $\boldsymbol{\lambda}_n = (\lambda_{1,n}, \lambda_{2,n}, \dots, \lambda_{q,n})^\top$ with

$$\lambda_{j,n} = n^{-1/4}(\alpha_n(1 - \alpha_n))^{-1/2}h_{\lambda,j} \quad \text{for } j = 1, \dots, q,$$

where $\mathbf{h}_\lambda = (h_{\lambda,1}^2, 2h_{\lambda,1}h_{\lambda,2}, h_{\lambda,2}^2, \dots, 2h_{\lambda,q-1}h_{\lambda,q}, h_{\lambda,q}^2)^\top$. The local alternatives are of order $n^{1/4}$ rather than $n^{1/2}$. See the discussion following Proposition 4.

The following proposition provides the asymptotic distribution of the LRT test statistics under contiguous local alternatives.

Proposition A.4. *Suppose that the assumptions in Proposition 6 hold for $M_0 = 1$. Consider a sequence of contiguous local alternatives $\boldsymbol{\vartheta}_{2,n} = (\alpha_n, \boldsymbol{\theta}_{1,n}^\top, \boldsymbol{\theta}_{2,n}^\top)^\top$ given in (25), where α_n and $\boldsymbol{\lambda}_n$ satisfy (24). Then, under $H_{1,n} : \boldsymbol{\vartheta} = \boldsymbol{\vartheta}_{2,n}$, we have $LR_n \xrightarrow{d} (\tilde{\mathbf{t}}_\lambda)^\top \boldsymbol{\mathcal{I}}_{\lambda,\eta} \tilde{\mathbf{t}}_\lambda$, where $\tilde{\mathbf{t}}_\lambda$ has the same distribution as $\hat{\mathbf{t}}_\lambda$ in Proposition 4 but $\mathbf{G}_{\lambda,\eta}$ is replaced with $(\boldsymbol{\mathcal{I}}_{\lambda,\eta})^{-1}\mathbf{S}_{\lambda,\eta} + \mathbf{h}_\lambda$.*

The result follows from Le Cam's third lemma applied to the LAN expansion established in Proposition 4. Under the local alternatives, $\sqrt{n}\mathbf{t}_\lambda$ acquires a non-zero mean shift $\mathbf{h}_\lambda$, shifting $\mathbf{G}_{\lambda,\eta}$ from $N(\mathbf{0}, \boldsymbol{\mathcal{I}}_{\lambda,\eta}^{-1})$ to $N(\mathbf{h}_\lambda, \boldsymbol{\mathcal{I}}_{\lambda,\eta}^{-1})$.

Importantly, a set of contiguous local alternatives considered in (25) excludes a sequence such that $\alpha_n \rightarrow 0$ or 1.

A.5 Proof of Propositions

Proof of Proposition 1. We first consider a model with an intercept parameter and a variance parameter but without covariates with $\mathbf{W}_i = \{y_{it}\}_{t=1}^T$ for the model (1) with normal density ((2))-((3)).

Define

$$s_i^2 = \frac{1}{T-1} \sum_{t=1}^T (Y_{it} - \bar{Y}_i)^2 \quad \text{with} \quad \bar{Y}_i = \frac{1}{T} \sum_{t=1}^T Y_{it},$$

where $(T-1)s_i^2/\sigma^{*2}$ follows a chi-square distribution with $T-1$ degrees of freedom. Let $i^* = \arg \min_{i=1,\dots,n} \{s_i^2\}$ so that $s_{i^*}^2 = \min\{s_1^2, \dots, s_n^2\}$ is the minimum of s_i^2 across all

values of i . We consider a sequence of parameters $\boldsymbol{\vartheta}_{2,n} = (\alpha_n, \boldsymbol{\theta}_{1,n}^\top, \boldsymbol{\theta}_{2,n}^\top)^\top$ with $\alpha_n = 1/n$, $\boldsymbol{\theta}_{1,n} = (\mu_{1,n}, \sigma_{1,n}^2)^\top = (\bar{Y}_{i^*}, s_{i^*}^2)^\top$, and $\boldsymbol{\theta}_{2,n} = \boldsymbol{\theta}^* = (\mu^*, \sigma^{*2})^\top$ for all n . It suffices to show that $LR_n^*(\boldsymbol{\vartheta}_{2,n})$ is unbounded in probability.

Define

$$\ell(\mathbf{W}_i; \boldsymbol{\theta}) := \log f(\mathbf{W}_i; \boldsymbol{\theta}) = -\frac{T}{2} \log \sigma^2 - \frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left(\frac{Y_{it} - \mu}{\sigma} \right)^2.$$

Then, the LRT statistic for a two-component mixture is written as

$$\begin{aligned} LR_n^*(\boldsymbol{\vartheta}_{2,n}) &= 2 \left\{ \sum_{i=1}^n \log \left(\alpha_n \prod_{t=1}^T \frac{1}{\sigma_{1,n}} \phi \left(\frac{Y_{it} - \mu_{1,n}}{\sigma_{1,n}} \right) + (1 - \alpha_n) \prod_{t=1}^T \frac{1}{\sigma^*} \phi \left(\frac{Y_{it} - \mu^*}{\sigma^*} \right) \right) - \sum_{i=1}^n \ell(\mathbf{W}_i; \boldsymbol{\theta}^*) \right\} \\ &= 2 \sum_{i \neq i^*} \{ \log (\exp(\log \alpha_n + \ell(\mathbf{W}_i; \boldsymbol{\theta}_{1,n})) + \exp(\log(1 - \alpha_n) + \ell(\mathbf{W}_i; \boldsymbol{\theta}^*))) - \ell(\mathbf{W}_i; \boldsymbol{\theta}^*) \} \\ &\quad + 2 \{ \log (\exp(\log \alpha_n + \ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}_{1,n})) + \exp(\log(1 - \alpha_n) + \ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}^*))) - \ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}^*) \}. \end{aligned} \quad (26)$$

The first term on the right-hand side of (26) can be rewritten as

$$= 2(n-1) \log \left(\frac{n-1}{n} \right) + 2 \sum_{i \neq i^*} \log \left(1 + \frac{1}{n-1} \exp(\ell(\mathbf{W}_i; \boldsymbol{\theta}_{1,n}) - \ell(\mathbf{W}_i; \boldsymbol{\theta}^*)) \right),$$

which is bounded from below by -2 as $n \rightarrow \infty$ because $\lim_{n \rightarrow \infty} 2(n-1) \log \left(\frac{n-1}{n} \right) = -2$ and $\log \left(1 + \frac{1}{n-1} \exp(\ell(\mathbf{W}_i; \boldsymbol{\theta}_{1,n}) - \ell(\mathbf{W}_i; \boldsymbol{\theta}^*)) \right) \geq 0$ for all n .

The second term on the right-hand side of (26) is written as

$$2\{-\log n + \ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}_{1,n})\} + 2 \log (1 + (n-1) \exp(\ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}^*) - \ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}_{1,n}))) - 2\ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}^*), \quad (27)$$

where $2\{-\log n + \ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}_{1,n})\}$ diverges to infinity as $n \rightarrow \infty$ by Lemma 2, the second term in (27) is bounded below by zero, and the third term is bounded in probability because $\ell(\mathbf{W}_{i^*}; \boldsymbol{\theta}^*) = O_p(1)$. Therefore, for any $M < \infty$, we have $\Pr(LR_n^*(\boldsymbol{\vartheta}_{2,n}) \leq M) \rightarrow 0$ as $n \rightarrow \infty$.

For a model with covariates for the model (1) with normal density ((2))-(3)), we can consider a sequence of parameters $\boldsymbol{\vartheta}_{2,n} = (\alpha_n, \boldsymbol{\theta}_{1,n}^\top, \boldsymbol{\theta}_{2,n}^\top)^\top$ with $\alpha_n = 1/n$, $\boldsymbol{\theta}_{1,n} = (\mu_{1,n}, \sigma_{1,n}^2, \boldsymbol{\beta}_{1,n}^\top)^\top = (\bar{Y}_{i^*}, s_{i^*}^2, \mathbf{0}^\top)^\top$ with $\boldsymbol{\theta}_{2,n} = \boldsymbol{\theta}^* = (\mu^*, \sigma^{*2}, (\boldsymbol{\beta}^*)^\top)^\top$. Then, repeating the above argument, the stated result follows.

For the model (1) with normal mixture density (2)-(3), suppose that $\mu_{j1} < \mu_{j2}$ for

$j = 1, 2$. Then, in view of (44), we can bound the log-likelihood function from below as

$$\begin{aligned} & \sum_{i=1}^n \log \left(\alpha \prod_{t=1}^T \sum_{k=1}^{K_\epsilon} \tau_{1k} \frac{1}{\sigma_1} \phi \left(\frac{Y_{it} - \mu_{1k}}{\sigma_1} \right) + (1 - \alpha) \prod_{t=1}^T \sum_{k=1}^{K_\epsilon} \tau_{2k} \frac{1}{\sigma_2} \phi \left(\frac{Y_{it} - \mu_{2k}}{\sigma_2} \right) \right) \\ & \geq \sum_{i=1}^n \log \left(\alpha \prod_{t=1}^T \frac{1}{\sigma_1} \phi \left(\frac{Y_{it} - \mu_{11}}{\sigma_1} \right) + (1 - \alpha) \prod_{t=1}^T \frac{1}{\sigma_2} \phi \left(\frac{Y_{it} - \mu_{21}}{\sigma_2} \right) \right). \end{aligned}$$

Therefore, the LRT statistic for a two-component mixture, $LR_n^*(\boldsymbol{\vartheta}_{2,n})$, is bounded below by the right hand side of (26). Then, repeating the argument following (26), the stated result follows from Lemma 2. This proves part (i). $\square$

Proof of Proposition 2. Note that, for $Z = \frac{Y - \mu}{\sigma}$,

$$\frac{\partial}{(\partial \mu)} \left[\frac{1}{\sigma} \phi(Z) \right] = \frac{Z}{\sigma} \frac{1}{\sigma} \phi(Z), \quad \frac{\partial^2}{(\partial \mu)^2} \left[\frac{1}{\sigma} \phi(Z) \right] = \frac{Z^2 - 1}{\sigma^2} \frac{1}{\sigma} \phi(Z),$$

$$\text{and} \quad \frac{\partial}{\partial \sigma^2} \left[\frac{1}{\sigma} \phi(Z) \right] = \frac{1}{2\sigma^2} (Z^2 - 1) \frac{1}{\sigma} \phi(Z).$$

Hence, for $j = 1$ (the case $j = 2$ follows identically with $\bar{\alpha}$ replaced by $1 - \bar{\alpha}$), a straightforward calculation gives

$$\left. \frac{\partial^2 g_2}{(\partial \mu_j)^2} \right|_{(\bar{\alpha}, \theta^*, \theta^*)} = \bar{\alpha} f^* \frac{1}{\sigma^{*2}} \left[(Z_1 + \dots + Z_T)^2 - T \right] \quad \text{and} \quad \left. \frac{\partial g_2}{\partial \sigma_j^2} \right|_{(\bar{\alpha}, \theta^*, \theta^*)} = \bar{\alpha} f^* \frac{1}{2\sigma^{*2}} \sum_{t=1}^T (Z_t^2 - 1), \quad (28)$$

where

$$Z_t := \frac{Y_t - \mu^*}{\sigma^*}, \quad t = 1, \dots, T.$$

Suppose, for contradiction, that real constants a, b exist such that

$$\left. \frac{\partial^2 g_2}{(\partial \mu_j)^2} \right|_{(\bar{\alpha}, \theta^*, \theta^*)} = a + b \left. \frac{\partial g_2}{\partial \sigma_j^2} \right|_{(\bar{\alpha}, \theta^*, \theta^*)}$$

with positive probability. Since $\bar{\alpha} f^* / \sigma^{*2} > 0$ almost surely, substituting (28) into the above equation and dividing both sides by $\bar{\alpha} f^* / \sigma^{*2}$ gives

$$(Z_1 + \dots + Z_T)^2 - T = \frac{a \sigma^{*2}}{\bar{\alpha} f^*} + \frac{b}{2} \sum_{t=1}^T (Z_t^2 - 1). \quad (29)$$

We derive a contradiction by considering two cases.

Case 1: $a = 0$. The right-hand side of (29) reduces to $\frac{b}{2} \sum_t (Z_t^2 - 1)$, so the equation becomes

$$\left(1 - \frac{b}{2}\right) \sum_{t=1}^T Z_t^2 + 2 \sum_{1 \leq s < r \leq T} Z_s Z_r = T - \frac{bT}{2},$$

after expanding $(Z_1 + \dots + Z_T)^2 = \sum_t Z_t^2 + 2 \sum_{s < r} Z_s Z_r$. The left-hand side is a polynomial in $(Z_1, \dots, Z_T)$ whose cross-terms $Z_s Z_r$ have coefficient 2, while the right-hand side is a constant. Because $T \geq 2$ the cross-terms are present, so this is a nonzero polynomial that cannot equal a constant on a set of positive Lebesgue measure. Since $(Z_1, \dots, Z_T)$ has a density, the identity fails almost surely, a contradiction.

Case 2: $a \neq 0$. Recall that $f^* = \prod_{t=1}^T \frac{1}{\sigma^*} \phi(Z_t) = (\sigma^*)^{-T} (2\pi)^{-T/2} \exp(-\frac{1}{2} \sum_t Z_t^2)$, so $1/f^* = (\sigma^*)^T (2\pi)^{T/2} \exp(\frac{1}{2} \sum_t Z_t^2)$. Rearranging (29) isolates the non-polynomial term:

$$\frac{a \sigma^{*2}}{\bar{\alpha}} (\sigma^*)^T (2\pi)^{T/2} \exp\left(\frac{1}{2} \sum_{t=1}^T Z_t^2\right) = P(Z_1, \dots, Z_T),$$

where P collects the remaining polynomial terms. Since $a \neq 0$, the left-hand side is a nonzero multiple of $\exp(\frac{1}{2} \sum_t Z_t^2)$, which grows super-polynomially along any ray $\|Z\| \rightarrow \infty$. No polynomial can match this growth, so the equality cannot hold on any set of positive Lebesgue measure. Since $(Z_1, \dots, Z_T)$ has a density, this is again a contradiction.

In both cases we reach a contradiction, so no constants a, b satisfy the supposed identity with positive probability, and

$$\Pr \left[\frac{\partial^2 g_2}{(\partial \mu_j)^2} \Big|_{(\bar{\alpha}, \theta^*, \theta^*)} = a + b \frac{\partial g_2}{\partial \sigma_j^2} \Big|_{(\bar{\alpha}, \theta^*, \theta^*)} \right] = 0.$$

□

Proof of Proposition 3. Our proof closely follows the proof of Theorem 3.3 in Hathaway (1985) by verifying Assumptions 1, 2, 3, and 5 of Kiefer and Wolfowitz (1956).

We first consider a model with the component-specific density function (1) with ((2))-((3)). Because our model has additional free parameters β_j s, as in the proof of Kasahara and Shimotsu (2015), we consider the joint density of $m_q := M(q+1)$ observations instead of $M+1$ observations in Hathaway (1985, p. 798), where $q := \dim(\beta)$. The joint density function of m_q observations is itself a mixture of M^{m_q} components, where each component is given by $\prod_{j=1}^{m_q} \prod_{t=1}^T f(Y_{jt}; \mu_{i_j} + \mathbf{X}_{jt}^\top \beta_{i_j}, \sigma_{i_j})$ for some choices $i_j \in \{1, \dots, M\}$, with the density of $N(\mu, \sigma^2)$ denoted by $f(y; \mu, \sigma) := \frac{1}{\sigma} \phi((y - \mu)/\sigma)$.

Assumptions 1, 2, and 3 of Kiefer and Wolfowitz (1956) are easily verified for the joint

density of m_q observations. We verify Assumption 5 of Kiefer and Wolfowitz (1956) for the joint density function of m_q observations by showing that

$$E \left[\log \prod_{j=1}^{m_q} \prod_{t=1}^T f(Y_{jt}; \mu_{i_j}^* + \mathbf{X}_{jt}^\top \boldsymbol{\beta}_{i_j}^*, \sigma_{i_j}^*) \right] > -\infty \quad (30)$$

for $\boldsymbol{\vartheta}_M^* \in \Theta_M^*$ and that

$$E \sup_{\boldsymbol{\vartheta}_M \in \bar{\Theta}_{\boldsymbol{\vartheta}_M}(c)} \left[\log \prod_{j=1}^{m_q} \prod_{t=1}^T f(Y_{jt}; \mu_{i_j} + \mathbf{X}_{jt}^\top \boldsymbol{\beta}_{i_j}, \sigma_{i_j}) \right] < \infty \quad (31)$$

for all component choices $i_j \in \{1, \dots, M\}$, which correspond to equations (3.1) and (3.4) in Hathaway (1985), respectively. (30) follows from the argument in the proof of Theorem 3.3 of Hathaway (1985).

For (31), because $\boldsymbol{\vartheta}_M \in \bar{\Theta}_{\boldsymbol{\vartheta}_M}(c)$, there exists $c \in (0, 1]$ such that $\min_{j,k} \sigma_j / \sigma_k > c$. Proceeding as in Hathaway (1985, pp. 798–799), we can show that

$$\sup_{\boldsymbol{\vartheta}_M \in \bar{\Theta}_{\boldsymbol{\vartheta}_M}(c)} \log \left[\prod_{j=1}^{m_q} \prod_{t=1}^T f(Y_{jt}; \mu_{i_j} + \mathbf{X}_{jt}^\top \boldsymbol{\beta}_{i_j}, \sigma_{i_j}) \right]$$

is no larger than, for some $\ell \in \{1, \dots, M\}$ and $j_1, j_2, \dots, j_{q+1} \in \{1, \dots, m_q\}$,

$$\sup_{\mu_\ell, \boldsymbol{\beta}_\ell, \sigma_\ell} \log \left[\delta(\sigma_\ell) \prod_{r=1}^{q+1} \prod_{t=1}^T f(Y_{j_r t}; \mu_\ell + \mathbf{X}_{j_r t}^\top \boldsymbol{\beta}_\ell, \sigma_\ell) \right], \quad (32)$$

where $\delta(\sigma_\ell) = (2\pi)^{-T(M-1)(q+1)/2} (c\sigma_\ell)^{-T(M-1)(q+1)}$, because $f(Y_{jt}; \mu_{i_j} + \mathbf{X}_{jt}^\top \boldsymbol{\beta}_{i_j}, \sigma_{i_j}) = (2\pi)^{-1/2} (\sigma_{i_j})^{-1} \exp(-\{Y_{jt} - (\mu_{i_j} + \mathbf{X}_{jt}^\top \boldsymbol{\beta}_{i_j})\}^2 / 2\sigma_{i_j}^2) \leq (2\pi)^{-1/2} (\sigma_{i_j})^{-1} \leq (2\pi)^{-1/2} (c\sigma_\ell)^{-1}$ for $j \notin \{j_1, j_2, \dots, j_{q+1}\}$.

Note that $\prod_{q=1}^{q+1} \prod_{t=1}^T f(Y_{j_q t}; \mu_\ell + \mathbf{x}_{j_q t}^\top \boldsymbol{\beta}_\ell, \sigma_\ell)$ is the likelihood function of a linear Gaussian model. Therefore, the maximised value of (32) equals $C_1 - C_2 \log(SSR)$, where C_1 and C_2 are a finite constant that depends only on M , k , and T while SSR is the sum of squared residuals obtained from regressing $\{\{Y_{j_q t}\}_{t=1}^T\}_{q=1}^{q+1}$ on $\{\{1, \mathbf{x}_{j_q t}\}_{t=1}^T\}_{q=1}^{q+1}$. Because there are $T(q+1)$ observations and $q+1$ regression parameters, the residual degrees of freedom are $(T-1)(q+1)$, which are positive under $T \geq 2$. Thus $SSR > 0$ a.s., and the finite moment conditions in Assumption 3(a) ensure $\mathbb{E}|\log(SSR)| < \infty$. Therefore, the expected value of (32) is finite, and (31) holds. This verifies Assumption 5 of Kiefer and Wolfowitz (1956), and the stated consistency result follows.

When the component-specific density function is given by (1) with (2)–(3), where the

number of sub-components in (3) is K , the joint density function of m_q observations can be written as a mixture of $M^{m_q} \times K^{m_q T}$ components, where each component is given by $\prod_{j=1}^{m_q} \prod_{t=1}^T f(Y_{jt}; \mu_{i_j i_k} + \mathbf{X}_{jt}^\top \boldsymbol{\beta}_{i_j}, \sigma_{i_j})$ for some choices $i_j \in \{1, \dots, M\}$ and $i_k \in \{1, \dots, K\}$. Then, repeating the same argument as above, we can show that the expected value of $\sup_{\boldsymbol{\theta}_M \in \Theta_{\boldsymbol{\theta}_M}(\mathbf{c})} \log \left[\prod_{j=1}^{m_q} \prod_{t=1}^T f(Y_{jt}; \mu_{i_j i_k} + \mathbf{X}_{jt}^\top \boldsymbol{\beta}_{i_j}, \sigma_{i_j}) \right]$ is bounded, and Assumption 5 of Kiefer and Wolfowitz (1956) holds. Because Assumptions 1, 2, and 3 of Kiefer and Wolfowitz (1956) are also easily verified, the consistency result follows. $\square$

Proof of Proposition 4. The proof is similar to that of Proposition 3 in Kasahara and Shimotsu (2015).

Given the value of $\mathbf{c}$ and $\alpha \in [c_1, 1 - c_1]$, define the space for reparameterized parameters as $\boldsymbol{\psi} := (\boldsymbol{\nu}^\top, \boldsymbol{\lambda}^\top)^\top \in \Theta_{\boldsymbol{\psi}_\alpha}$, where $\Theta_{\boldsymbol{\psi}_\alpha} = \{\boldsymbol{\psi} = (\boldsymbol{\nu}^\top, \boldsymbol{\lambda}^\top)^\top : (\alpha, \boldsymbol{\nu} + (1 - \alpha)\boldsymbol{\lambda}, \boldsymbol{\nu} - \alpha\boldsymbol{\lambda}) \in \Theta_{\boldsymbol{\theta}_2}(\mathbf{c})\}$. Under the null hypothesis $H_{01} : \boldsymbol{\theta}_1 = \boldsymbol{\theta}_2 = \boldsymbol{\theta}^*$, we have $\boldsymbol{\lambda} = \mathbf{0}$ and $\boldsymbol{\nu} = \boldsymbol{\theta}^*$. We rewrite the reparameterized parameters under the null hypothesis as $\boldsymbol{\psi}^* = ((\boldsymbol{\theta}^*)^\top, \mathbf{0}^\top)^\top$. We denote the reparameterized density function and its logarithm as

$$g(\mathbf{w}; \boldsymbol{\psi}, \alpha) := \alpha f(\mathbf{w}; \boldsymbol{\nu} + (1 - \alpha)\boldsymbol{\lambda}) + (1 - \alpha)f(\mathbf{w}; \boldsymbol{\nu} - \alpha\boldsymbol{\lambda}) \text{ and } l(\mathbf{w}; \boldsymbol{\psi}, \alpha) = \log g(\mathbf{w}; \boldsymbol{\psi}, \alpha). \quad (33)$$

Let $L_n(\boldsymbol{\psi}, \alpha) := \sum_{i=1}^n l(\mathbf{W}_i; \boldsymbol{\psi}, \alpha)$ be the reparameterized log-likelihood function. For each $\alpha \in [c_1, 1 - c_1]$, define the reparameterized MLE as

$$\hat{\boldsymbol{\psi}}_\alpha = \arg \max_{\boldsymbol{\psi} \in \Theta_{\boldsymbol{\psi}_\alpha}} L_n(\boldsymbol{\psi}, \alpha). \quad (34)$$

Collect the relevant normalized reparameterized parameters and define $\mathbf{t}(\boldsymbol{\psi}, \alpha)$ as

$$\mathbf{t}(\boldsymbol{\psi}, \alpha) = \begin{pmatrix} \mathbf{t}_\nu \\ \mathbf{t}_\lambda(\boldsymbol{\lambda}, \alpha) \end{pmatrix} = \begin{pmatrix} \boldsymbol{\nu} - \boldsymbol{\nu}^* \\ \alpha(1 - \alpha)\mathbf{v}(\boldsymbol{\lambda}) \end{pmatrix}, \quad (35)$$

where $\mathbf{v}(\boldsymbol{\lambda})$ is given by (22).

As discussed in the proof of Lemma 4, taking the fourth-order Taylor expansion of $L_n(\boldsymbol{\psi}, \alpha)$ around $(\boldsymbol{\psi}^*, \alpha)$, we may write $2\{L_n(\boldsymbol{\psi}, \alpha) - L_n(\boldsymbol{\psi}^*, \alpha)\}$ as a quadratic function of

$\sqrt{n}\mathbf{t}(\boldsymbol{\psi}, \alpha)$ as

$$2\{L_n(\boldsymbol{\psi}, \alpha) - L_n(\boldsymbol{\psi}^*, \alpha)\} = 2(\sqrt{n}\mathbf{t}(\boldsymbol{\psi}, \alpha))^\top \mathbf{S}_n - (\sqrt{n}\mathbf{t}(\boldsymbol{\psi}, \alpha))^\top \mathcal{I}_n(\sqrt{n}\mathbf{t}(\boldsymbol{\psi}, \alpha)) + R_n(\boldsymbol{\psi}, \alpha) \quad (36)$$

$$= \mathbf{G}_n^\top \mathcal{I}_n \mathbf{G}_n - [\sqrt{n}\mathbf{t}(\boldsymbol{\psi}, \alpha) - \mathbf{G}_n]^\top \mathcal{I}_n [\sqrt{n}\mathbf{t}(\boldsymbol{\psi}, \alpha) - \mathbf{G}_n] + R_n(\boldsymbol{\psi}, \alpha), \quad (37)$$

where $\mathbf{S}_n := n^{-1/2} \sum_{i=1}^n \mathbf{s}(\mathbf{W}_i)$ and $\mathbf{G}_n := \mathcal{I}_n^{-1} \mathbf{S}_n$, where $\mathcal{I}_n$ is the negative of the sample Hessian defined in the proof of Lemma 4.

Noting that $L_n(\boldsymbol{\psi}^*, \alpha) = L_{0,n}(\boldsymbol{\gamma}_0^*, \boldsymbol{\theta}_0^*)$, write

$$LR_n = \max_{\alpha \in [c_1, 1-c_1]} 2\{L_n(\hat{\boldsymbol{\psi}}_\alpha, \alpha) - L_n(\boldsymbol{\psi}^*, \alpha)\} - 2\{L_{0,n}(\hat{\boldsymbol{\gamma}}_0, \hat{\boldsymbol{\theta}}_0) - L_{0,n}(\boldsymbol{\gamma}_0^*, \boldsymbol{\theta}_0^*)\}. \quad (38)$$

Define

$$\mathbf{S}_n = \begin{pmatrix} \mathbf{S}_{\eta n} \\ \mathbf{S}_{\lambda n} \end{pmatrix} := \begin{pmatrix} n^{-1/2} \sum_{i=1}^n \mathbf{s}_\eta(\mathbf{W}_i) \\ n^{-1/2} \sum_{i=1}^n \mathbf{s}_{\lambda\lambda}(\mathbf{W}_i) \end{pmatrix}, \quad \mathbf{S}_{\lambda, \eta n} := \mathbf{S}_{\lambda n} - \mathcal{I}_{\lambda\eta} \mathcal{I}_\eta^{-1} \mathbf{S}_{\eta n}, \quad \mathbf{G}_{\lambda, \eta n} := \mathcal{I}_{\lambda, \eta}^{-1} \mathbf{S}_{\lambda, \eta n}, \\ t_{\eta, \lambda} := t_\eta + \mathcal{I}_\eta^{-1} \mathcal{I}_{\eta\lambda} t_\lambda(\boldsymbol{\lambda}, \alpha),$$

and split the quadratic form in (36) to obtain

$$2\{L_n(\boldsymbol{\psi}, \alpha) - L_n(\boldsymbol{\psi}^*, \alpha)\} = B_n(\sqrt{n}\mathbf{t}_{\eta, \lambda}) + C_n(\sqrt{n}\mathbf{t}_\lambda(\boldsymbol{\lambda}, \alpha)) + R_n(\boldsymbol{\Psi}, \alpha), \quad (39)$$

where

$$B_n(\mathbf{t}_{\eta, \lambda}) = 2\mathbf{t}_{\eta, \lambda}^\top \mathbf{S}_{\eta n} - \mathbf{t}_{\eta, \lambda}^\top \mathcal{I}_\eta \mathbf{t}_{\eta, \lambda}, \\ C_n(\mathbf{t}_\lambda) = 2\mathbf{t}_\lambda^\top \mathbf{S}_{\lambda, \eta n} - \mathbf{t}_\lambda^\top \mathcal{I}_{\lambda, \eta} \mathbf{t}_\lambda \\ = \mathbf{G}_{\lambda, \eta n}^\top \mathcal{I}_{\lambda, \eta} \mathbf{G}_{\lambda, \eta n} - (\mathbf{t}_\lambda - \mathbf{G}_{\lambda, \eta n})^\top \mathcal{I}_{\lambda, \eta} (\mathbf{t}_\lambda - \mathbf{G}_{\lambda, \eta n}), \quad (40)$$

with $\mathbf{G}_{\lambda, \eta n} \xrightarrow{d} \mathbf{G}_{\lambda, \eta} = (\mathcal{I}_{\lambda, \eta})^{-1} \mathbf{S}_{\lambda, \eta}$ and $\mathbf{S}_{\lambda, \eta n} \xrightarrow{d} \mathbf{S}_{\lambda, \eta} \sim N(0, \mathcal{I}_{\lambda, \eta})$. In addition, $R_n(\hat{\boldsymbol{\psi}}_\alpha, \alpha) = o_p(1)$ holds from Lemma 4(a) and $\sqrt{n}\mathbf{t}(\hat{\boldsymbol{\Psi}}_\alpha, \alpha) = O_p(1)$.

Because $\Delta_{(\gamma, \theta)} f(x; \hat{\boldsymbol{\gamma}}_0^*, \hat{\boldsymbol{\theta}}_0^*)$ is identical to $\Delta_\eta f(x; \boldsymbol{\Psi}^*, \alpha)$, a standard analysis gives $2[L_{0,n}(\hat{\boldsymbol{\gamma}}_0, \hat{\boldsymbol{\theta}}_0) - L_{0,n}(\boldsymbol{\gamma}_0^*, \boldsymbol{\theta}_0^*)] = \max_{\mathbf{t}_\eta} B_n(\sqrt{n}\mathbf{t}_\eta) + o_p(1)$. Note that the possible values of both $\sqrt{n}\mathbf{t}_\eta$ and $\sqrt{n}\mathbf{t}_{\eta, \lambda}$ approach $\mathbb{R}^q$. Therefore, in view of (39) and (40), we can write equation (38) as

$$LR_n = \max_{\alpha \in [c_1, 1-c_1]} C_n(\sqrt{n}\mathbf{t}_\lambda(\hat{\boldsymbol{\lambda}}_\alpha, \alpha)) + o_p(1), \quad (41)$$

where $\hat{\boldsymbol{\lambda}}_\alpha$ is as defined in (34).

The asymptotic distribution of LR_n follows from applying Theorem 3(c) of (Andrews, 1999, p. 1362) to (39) and (41). First, Assumption 2 of Andrews (1999) holds because

Assumption 2* of Andrews (1999) holds by Lemma 4(a) (the remainder bound) together with Proposition 2. Second, Assumption 3 of Andrews (1999) holds with $B_T = n^{1/2}$ and $T = n$ because $\mathbf{S}_{\lambda, \eta n} \xrightarrow{d} \mathbf{S}_{\lambda, \eta} \sim N(\mathbf{0}, \mathcal{I}_{\lambda, \eta})$ by Lemma 4(b) and $\mathcal{I}_{\lambda, \eta}$ is non-singular by Lemma 4(c). Assumption 4 of Andrews (1999) holds from part (a). Assumption 5 of Andrews (1999) follows from Assumption 5* and Lemma 3 of Andrews (1999) with $b_T = n^{1/2}$ because $\alpha(1 - \alpha)v(\Theta_\lambda)$ is locally equal to Λ_λ . Therefore, it follows from Theorem 3(c) of Andrews (1999) that $C_n(\sqrt{n}\mathbf{t}_\lambda(\hat{\lambda}, \alpha)) \xrightarrow{d} (\hat{\mathbf{t}}_\lambda)^\top \mathcal{I}_{\lambda, \eta} \hat{\mathbf{t}}_\lambda$, where $\hat{\mathbf{t}}_\lambda$ is defined by (23). $\square$

Proof of Proposition 5. Under $H_{2,0}$, we obtain $\vartheta_{M_0+1} \in \Theta_{\vartheta_{M_0+1}, 2h}^*$

$$\begin{aligned} & \mathbb{E}[\{\nabla_{\alpha_h} \log f_{M_0+1}(\mathbf{W}_i, \vartheta_{M_0+1})\}^2] \\ &= \int \frac{\{f(\mathbf{w}; \boldsymbol{\theta}_h) - f(\mathbf{w}; \boldsymbol{\theta}_{M_0}^*)\}^2}{\sum_{j=1}^{M_0} \alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)} d\mathbf{w} \\ &= \int \frac{\{f(\mathbf{w}; \boldsymbol{\theta}_h)\}^2}{\sum_{j=1}^{M_0} \alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)} d\mathbf{w} + \int \frac{\{f(\mathbf{w}; \boldsymbol{\theta}_{M_0}^*)\}^2}{\sum_{j=1}^{M_0} \alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)} d\mathbf{w} - 2 \int \frac{f(\mathbf{w}; \boldsymbol{\theta}_h) f(\mathbf{w}; \boldsymbol{\theta}_{M_0}^*)}{\sum_{j=1}^{M_0} \alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)} d\mathbf{w}. \end{aligned} \quad (42)$$

The latter two terms on the right-hand side of (42) are bounded because $f(\mathbf{w}; \boldsymbol{\theta}_{M_0}^*) / \sum_{j=1}^{M_0} \alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*) \leq (1/\alpha_{M_0}^*)$ for any $\mathbf{w}$ and $f(\mathbf{w}; \boldsymbol{\theta})$ integrates to one. Therefore, the left-hand side of (42) goes to infinity if and only if the first term on the right-hand side of (42) goes to infinity.

Because $\max_j \alpha_j \leq \sum_{j=1}^{M_0} \alpha_j \leq M_0 \max_j \alpha_j$, we obtain

$$\frac{1}{M_0 \max_j \{\alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)\}} \frac{\{f(\mathbf{w}; \boldsymbol{\theta}_h)\}^2}{\sum_{j=1}^{M_0} \alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)} \leq \frac{\{f(\mathbf{w}; \boldsymbol{\theta}_h)\}^2}{\sum_{j=1}^{M_0} \alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)} \leq \frac{\{f(\mathbf{w}; \boldsymbol{\theta}_h)\}^2}{\max_j \{\alpha_j^* f(\mathbf{w}; \boldsymbol{\theta}_j^*)\}}.$$

Without loss of generality, we assume that $\sigma_{M_0}^* = \max\{\sigma_1^*, \dots, \sigma_{M_0}^*\}$ and that the maximum is unique. We focus on models without covariates because the law of iterated expectations implies that the stated result also holds for models with covariates if it holds for models without covariates.

We first prove the case for the component-specific density function $f(\mathbf{w}; \boldsymbol{\theta}) = \prod_{t=1}^T f(y_t; \boldsymbol{\theta})$, where $f(y_t; \boldsymbol{\theta})$ is given by ((2))–((3)) with $\beta_j = \mathbf{0}$. Under the normal density function ((2))–((3)) without covariates, there exists a sufficiently large but finite positive constant B , such that $\max_j \{\alpha_j^* f(\mathbf{w}, \mu_j^*, \sigma_j^2)\} = \alpha_{M_0}^* f(\mathbf{w}, \mu_{M_0}^*, \sigma_{M_0}^2)$ when $|y_t| > B$ for all

$t = 1, \dots, T$. Note that

$$\begin{aligned} \frac{\{f(\mathbf{w}; \mu_h, \sigma_h)\}^2}{f(\mathbf{w}; \mu_{M_0}^*, \sigma_{M_0}^*)} &= \prod_{t=1}^T \frac{\sigma_{M_0}^*}{(2\pi)^{1/2} \sigma_h^2} \exp \left\{ -\frac{1}{\sigma_h^2} (y_t - \mu_h)^2 + \frac{1}{2(\sigma_{M_0}^*)^2} (y_t - \mu_{M_0}^*)^2 \right\} \\ &= \left(\frac{\sigma_{M_0}^*}{(2\pi)^{1/2} \sigma_h^2} \right)^T \exp \left\{ -\frac{1}{\sigma_h^2} \sum_{t=1}^T (y_t - \mu_h)^2 + \frac{1}{2(\sigma_{M_0}^*)^2} \sum_{t=1}^T (y_t - \mu_{M_0}^*)^2 \right\}. \end{aligned} \quad (43)$$

Then, the integral of the right-hand side of (43) over $|y_t| \geq B$ for $t = 1, \dots, T$ is infinite if $\sigma_h^2/\sigma_{M_0}^{2*} > 2$, and the stated result holds. Conversely, when $\sigma_h^2 < 2\sigma_{M_0}^{2*}$, the leading coefficient of y_t^2 in the exponent is strictly negative, so the integrand is bounded by a product of Gaussian densities and the integral is finite.

When $f(y_t; \theta)$ is given by normal mixture density (2)–(3), suppose that $\mu_{j1} < \mu_{j2} < \dots < \mu_{K_\epsilon}$ for all $j = 1, \dots, M_0$. Then, there exists a constant B such that, when $y_t > B$ for all $t = 1, \dots, T$, we have $\max_j \{\alpha_j^* f(\mathbf{w}, \theta_j^*)\} = \alpha_{M_0}^* f(\mathbf{w}, \theta_{M_0}^*)$ and

$$\frac{1}{\sigma_j} \exp \left(-\frac{1}{2} \left(\frac{y_t - \mu_{K_\epsilon}}{\sigma_j} \right)^2 \right) \geq \sum_{k=1}^{K_\epsilon} \tau_{jk} \frac{1}{\sigma_j} \exp \left(-\frac{1}{2} \left(\frac{y_t - \mu_{jk}}{\sigma_j} \right)^2 \right) \geq \frac{1}{\sigma_j} \exp \left(-\frac{1}{2} \left(\frac{y_t - \mu_{j1}}{\sigma_j} \right)^2 \right). \quad (44)$$

Then, it follows that

$$\frac{\{f(\mathbf{w}; \theta_h)\}^2}{f(\mathbf{w}; \theta_{M_0}^*)} \geq \left(\frac{\sigma_{M_0}^*}{(2\pi)^{1/2} \sigma_h^2} \right)^T \exp \left\{ -\frac{1}{\sigma_h^2} \sum_{t=1}^T (y_t - \mu_{1h})^2 + \frac{1}{2(\sigma_{M_0}^*)^2} \sum_{t=1}^T (y_t - \mu_{M_0K}^*)^2 \right\},$$

where the integral of the right-hand side over $y_t > B$ for $t = 1, \dots, T$ is infinite if $\sigma_h^2/\sigma_{M_0}^{2*} > 2$ as in (43). □

Proof of Proposition 6. Collect the score vector for testing $H_{0,1h}$ for $h = 1, \dots, M_0$ into one vector as

$$\tilde{\mathbf{s}}(\mathbf{W}) = \begin{pmatrix} \tilde{\mathbf{s}}_\eta(\mathbf{W}) \\ \tilde{\mathbf{s}}_{\lambda\lambda}(\mathbf{W}) \end{pmatrix}, \text{ where } \tilde{\mathbf{s}}_\eta(\mathbf{W})_{(M_0+p+q+1) \times 1} = \begin{pmatrix} \mathbf{s}_\alpha(\mathbf{W}) \\ \mathbf{s}_\nu(\mathbf{W}) \end{pmatrix} \text{ and } \tilde{\mathbf{s}}_{\lambda\lambda}(\mathbf{W}) = \begin{pmatrix} \mathbf{s}_{\lambda\lambda}^1(\mathbf{W}) \\ \vdots \\ \mathbf{s}_{\lambda\lambda}^{M_0}(\mathbf{W}) \end{pmatrix}, \quad (45)$$

where

$$\begin{aligned} \mathbf{s}_\alpha(\mathbf{W}) &= \begin{pmatrix} f(\mathbf{W}; \boldsymbol{\theta}_1^*) - f(\mathbf{W}; \boldsymbol{\theta}_{M_0}^*) \\ \vdots \\ f(\mathbf{W}; \boldsymbol{\theta}_{M_0-1}^*) - f(\mathbf{W}; \boldsymbol{\theta}_{M_0}^*) \end{pmatrix} \Big/ f_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*), \\ \mathbf{s}_\nu(\mathbf{W}) &= \sum_{j=1}^{M_0} \alpha_j^* \frac{\nabla_\nu f(\mathbf{W}; \boldsymbol{\theta}_j^*)}{f_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)}, \text{ and } \mathbf{s}_{\lambda\lambda}^h(\mathbf{W}) = \frac{\tilde{\nabla}_{\boldsymbol{\theta}_h \boldsymbol{\theta}_h^\top} f(\mathbf{W}; \boldsymbol{\theta}_h^*)}{f_{M_0}(\mathbf{W}; \boldsymbol{\vartheta}_{M_0}^*)} \text{ for } h = 1, 2, \dots, M_0, \end{aligned} \quad (46)$$

with $\tilde{\nabla}_{\boldsymbol{\theta}_h \boldsymbol{\theta}_h^\top} f(\mathbf{W}; \boldsymbol{\theta}_h^*) := (c_{11} \nabla_{\theta_{h1} \theta_{h1}} f^*, \dots, c_{qq} \nabla_{\theta_{hq} \theta_{hq}} f^*, c_{12} \nabla_{\theta_{h1} \theta_{h2}} f^*, \dots, c_{(q-1)q} \nabla_{\theta_{h(q-1)} \theta_{hq}} f^*)^\top$ for $\boldsymbol{\theta}_h := (\theta_{h1}, \theta_{h2}, \theta_{h3}, \dots, \theta_{hq})^\top := (\mu_h, \sigma_h^2, \beta_{h1}, \dots, \beta_{hq-2})^\top$ and $c_{jk} = 1/2$ for $j \neq k$ and $c_{jk} = 1$ for $j = k$. Define

$$\begin{aligned} \tilde{\mathcal{I}} &:= \mathbb{E}[\tilde{\mathbf{s}}(\mathbf{W}) \tilde{\mathbf{s}}(\mathbf{W})^\top], \quad \tilde{\mathcal{I}}_\eta := \mathbb{E}[\tilde{\mathbf{s}}_\eta(\mathbf{W}) \tilde{\mathbf{s}}_\eta(\mathbf{W})^\top], \quad \tilde{\mathcal{I}}_{\lambda\eta} := \mathbb{E}[\tilde{\mathbf{s}}_{\lambda\lambda}(\mathbf{W}) \tilde{\mathbf{s}}_\eta(\mathbf{W})^\top], \\ \tilde{\mathcal{I}}_{\eta\lambda} &:= \tilde{\mathcal{I}}_{\lambda\eta}^\top, \quad \tilde{\mathcal{I}}_{\lambda\lambda} := \mathbb{E}[\tilde{\mathbf{s}}_{\lambda\lambda}(\mathbf{W}) \tilde{\mathbf{s}}_{\lambda\lambda}(\mathbf{W})^\top], \quad \tilde{\mathcal{I}}_{\lambda,\eta} := \tilde{\mathcal{I}}_{\lambda\lambda} - \tilde{\mathcal{I}}_{\lambda\eta} \tilde{\mathcal{I}}_\eta^{-1} \tilde{\mathcal{I}}_{\eta\lambda}. \end{aligned} \quad (47)$$

Then, the asymptotic distribution of the normalized score function is given by

$$\tilde{\mathbf{S}}_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n \tilde{\mathbf{s}}(\mathbf{W}_i) \xrightarrow{d} \tilde{\mathbf{S}} \sim N(\mathbf{0}, \tilde{\mathcal{I}}),$$

where, in view of (45), $\tilde{\mathbf{S}}$ may be partitioned as $\tilde{\mathbf{S}} = (\tilde{\mathbf{S}}_\eta^\top, \tilde{\mathbf{S}}_{\lambda\lambda}^\top)^\top$ with $n^{-1/2} \sum_{i=1}^n \tilde{\mathbf{s}}_\eta(\mathbf{W}_i) \xrightarrow{d} \tilde{\mathbf{S}}_\eta$ and $n^{-1/2} \sum_{i=1}^n \tilde{\mathbf{s}}_{\lambda\lambda}(\mathbf{W}_i) \xrightarrow{d} \tilde{\mathbf{S}}_{\lambda\lambda}$.

Let $\tilde{\mathbf{S}}_{\lambda,\eta} := (\mathbf{S}_{\lambda,\eta}^1, \dots, \mathbf{S}_{\lambda,\eta}^{M_0})^\top := \tilde{\mathbf{S}}_{\lambda\lambda} - \tilde{\mathcal{I}}_{\lambda\eta} \tilde{\mathcal{I}}_\eta^{-1} \tilde{\mathbf{S}}_\eta \sim N(\mathbf{0}, \tilde{\mathcal{I}}_{\lambda,\eta})$ be a $\mathbb{R}^{M_0(q)(q+1)/2}$ -valued random vector. For $h = 1, 2, \dots, M_0$, define $\tilde{\mathcal{I}}_{\lambda,\eta}^h := \mathbb{E}[\mathbf{S}_{\lambda,\eta}^h (\mathbf{S}_{\lambda,\eta}^h)^\top]$ and $\mathbf{G}_{\lambda,\eta}^h := (\tilde{\mathcal{I}}_{\lambda,\eta}^h)^{-1} \mathbf{S}_{\lambda,\eta}^h$.

Define $\hat{\mathbf{t}}_\lambda^h$ analogously to $\hat{\mathbf{t}}_\lambda$ as

$$r_\lambda^h(\hat{\mathbf{t}}_\lambda^h) = \inf_{\mathbf{t}_\lambda^h \in \Lambda_\lambda} r^h(\mathbf{t}_\lambda^h); \quad r_\lambda^h(\mathbf{t}_\lambda^h) := (\mathbf{t}_\lambda^h - \mathbf{G}_{\lambda,\eta}^h)^\top \tilde{\mathcal{I}}_{\lambda,\eta}^h (\mathbf{t}_\lambda^h - \mathbf{G}_{\lambda,\eta}^h) \quad \text{for } h = 1, 2, \dots, M_0. \quad (48)$$

The local quadratic-form approximation of the log-likelihood function $LR_n^{M_0,h}$ around $\Theta_{\boldsymbol{\vartheta}_{M_0+1,1h}}^* \subset \Theta_{\boldsymbol{\vartheta}_{M_0+1}}$ has an identical structure to the approximation that we derive in Section 5.1 in testing H_{01} in the test of homogeneity. Specifically, the bounded ratios $\alpha_j^* f_j^* / g_{M_0}^*$ ensure that Lemma 4 conditions transfer to the M_0 -component case, and the Hermite polynomial structure of the score is preserved under the reparameterisation $(\boldsymbol{\lambda}_h, \boldsymbol{\nu}_h)$. Consequently, we can show that $LR_n^{M_0,h} \xrightarrow{d} (\hat{\mathbf{t}}_\lambda^h)^\top \tilde{\mathcal{I}}_{\lambda,\eta}^h \hat{\mathbf{t}}_\lambda^h$. Then, given (24), the asymptotic null distribution of the LRTS for testing H_{01} is given by the maximum over $(\hat{\mathbf{t}}_\lambda^h)^\top \tilde{\mathcal{I}}_{\lambda,\eta}^h \hat{\mathbf{t}}_\lambda^h$ for $h = 1, 2, \dots, M_0$. We now proceed with the proof.

For $h = 1, \dots, M_0$, let $\mathcal{N}_h^* \subset \Theta_{\boldsymbol{\vartheta}_{M_0+1}}(\mathbf{c})$ be a sufficiently small closed neighborhood of $\Theta_{\boldsymbol{\vartheta}_{M_0+1}, 1h}^*$ such that $\alpha_h, \alpha_{h+1} > 0$ holds and $\Theta_{\boldsymbol{\vartheta}_{M_0+1}, 1k}^* \not\subset \mathcal{N}_h^*$ if $k \neq h$. Consider the following one-to-one reparameterization from the $(M_0 + 1)$ -component model parameter $\boldsymbol{\vartheta}_{M_0+1} = (\alpha_1, \dots, \alpha_{M_0}, \boldsymbol{\theta}_1^\top, \dots, \boldsymbol{\theta}_h^\top, \boldsymbol{\theta}_{h+1}^\top, \dots, \boldsymbol{\theta}_{M_0+1}^\top)^\top$. Similar to (20), the one-to-one reparameterization for testing the null hypothesis $H_{0,1h}$ is given by

$$\begin{pmatrix} \boldsymbol{\lambda}_h \\ \boldsymbol{\nu}_h \end{pmatrix} := \begin{pmatrix} \boldsymbol{\theta}_h - \boldsymbol{\theta}_{h+1} \\ \tau \boldsymbol{\theta}_h + (1 - \tau) \boldsymbol{\theta}_{h+1} \end{pmatrix} \text{ so that } \begin{pmatrix} \boldsymbol{\theta}_h \\ \boldsymbol{\theta}_{h+1} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\nu} + (1 - \tau) \boldsymbol{\lambda} \\ \boldsymbol{\nu} - \tau \boldsymbol{\lambda} \end{pmatrix},$$

and α_j is reparameterized for $j = 1, 2, \dots, M_0$ as

$$\begin{aligned} (\pi_1, \dots, \pi_{h-1}, \pi_h, \pi_{h+1}, \dots, \pi_{M_0-1}) &= (\alpha_1, \dots, \alpha_{h-1}, (\alpha_h + \alpha_{h+1}), \alpha_{h+2}, \dots, \alpha_{M_0}) \\ \tau &= \alpha_h / (\alpha_h + \alpha_{h+1}) \end{aligned}$$

so that $\pi_h = \alpha_h + \alpha_{h+1}$ and $\pi_{M_0} = 1 - \sum_{j=1}^{M_0-1} \pi_j$.

Collect the reparameterized parameters except τ as

$$\boldsymbol{\psi}_{h,\tau} = (\boldsymbol{\eta}^\top, \boldsymbol{\lambda}_h^\top)^\top \quad \text{with} \quad \boldsymbol{\eta} = (\pi_1, \dots, \pi_{M_0-1}, \boldsymbol{\theta}_1^\top, \dots, \boldsymbol{\theta}_{h-1}^\top, \boldsymbol{\nu}_h^\top, \boldsymbol{\theta}_{h+2}^\top, \dots, \boldsymbol{\theta}_{M_0+1}^\top)^\top.$$

In the reparameterized model, the null restriction $\boldsymbol{\theta}_h = \boldsymbol{\theta}_{h+1}$ implied by $H_{0,1h}$ holds if and only if $\boldsymbol{\lambda}_h = \mathbf{0}$. Under $H_{0,1h}$, we have $\lambda_h^* = 0$ and $\boldsymbol{\eta}^* = (\alpha_1^*, \dots, \alpha_{M_0-1}^*, (\boldsymbol{\theta}_1^*)^\top, \dots, (\boldsymbol{\theta}_{M_0}^*)^\top)^\top$. Define the log-likelihood under the reparameterized parameters as

$$f_{M_0+1}^h(\mathbf{w}; \boldsymbol{\psi}_{h,\tau}, \tau) = \pi_h g^h(\mathbf{w}, \boldsymbol{\psi}_{h,\tau}, \tau) + \sum_{j=1}^{h-1} \pi_j f(\mathbf{w}; \boldsymbol{\theta}_j) + \sum_{j=h}^{M_0} \pi_{j+1} f(\mathbf{w}; \boldsymbol{\theta}_{j+1}),$$

where $g^h(\mathbf{w}, \boldsymbol{\psi}_{h,\tau}, \tau)$ is defined similarly to (33) as

$$g^h(\mathbf{w}, \boldsymbol{\psi}_{h,\tau}, \tau) = \tau f(\mathbf{w}; \boldsymbol{\nu}_h + (1 - \tau) \boldsymbol{\lambda}_h) + (1 - \tau) f(\mathbf{w}; \boldsymbol{\nu}_h - \tau \boldsymbol{\lambda}_h). \quad (49)$$

Define the local MLE of $\boldsymbol{\psi}_{h,\tau}$ by

$$\hat{\boldsymbol{\psi}}_{h,\tau} := \arg \max_{\boldsymbol{\psi}_{h,\tau} \in \mathcal{N}_h^*} L_n^h(\boldsymbol{\psi}_{h,\tau}, \tau), \quad (50)$$

where $L_n^h(\boldsymbol{\psi}_{h,\tau}, \tau) := \sum_{i=1}^n \log f_{M_0+1}^h(\mathbf{W}_i; \boldsymbol{\psi}_{h,\tau}, \tau)$. Because $\boldsymbol{\psi}_{h,\tau}^*$ is the only parameter value in $\mathcal{N}_h^*$ that generates the true density, $\hat{\boldsymbol{\psi}}_{h,\tau} - \boldsymbol{\psi}_{h,\tau}^* = o_p(1)$ follows.

Define the LRTS for testing $H_{0,1h}$ as $LR_n^{M_0,h} := \max_{\tau \in [c_1, 1-c_1]} 2(L_n^h(\hat{\boldsymbol{\psi}}_{h,\tau}, \tau) - L_{0,n}(\hat{\boldsymbol{\vartheta}}_{M_0}))$.

Then, in view of (24), the stated result holds if

$$(LR_n^{M_0,1}, \dots, LR_n^{M_0,M_0})^\top \xrightarrow{d} (\hat{\mathbf{t}}_\lambda^1)^\top \mathcal{I}_{\eta,\lambda}^1(\hat{\mathbf{t}}_\lambda^1), \dots, (\hat{\mathbf{t}}_\lambda^{M_0})^\top \mathcal{I}_{\eta,\lambda}^{M_0}(\hat{\mathbf{t}}_\lambda^{M_0})^\top. \quad (51)$$

Observe that $L_n^h(\psi_{h,\tau}, \tau) - L_n^h(\psi_{h,\tau}^*, \tau)$ admits the same expansion as $L_n(\hat{\psi}, \alpha) - L_n(\psi^*, \alpha)$ in (37) and (39) when $(\alpha, \mathbf{t}(\psi, \alpha), \mathbf{t}_\lambda(\lambda, \alpha), \mathbf{S}_n, \mathbf{G}_n, \mathcal{I}_n, R_n(\Psi, \alpha))$ is replaced with $(\tau, \mathbf{t}^h(\psi^h, \tau), \mathbf{t}_\lambda^h(\lambda^h, \tau), \mathbf{S}_n^h, \mathbf{G}_n^h, \mathcal{I}_n^h, R_n^h(\Psi^h, \tau))$, where $(\mathbf{S}_n^h, \mathcal{I}_n^h)$ is defined similarly to $(\mathbf{S}_n, \mathcal{I}_n)$ but $(s_\eta, s_{\lambda\lambda})$ is replaced with $(\tilde{s}_\eta, s_{\lambda\lambda}^h)$ and $\mathbf{G}_n^h := (\mathcal{I}_n^h)^{-1} \mathbf{S}_n^h$. Applying the proof of Lemma 4, we have $\mathbf{S}_n^h \xrightarrow{d} \mathbf{S}^h \sim N(\mathbf{0}, \mathcal{I}^h)$ and $\mathcal{I}_n^h \xrightarrow{p} \mathcal{I}^h$. Then, (51) follows from the proof of Lemma 4 and Proposition 4 for each local MLE when $(\mathbf{G}_n, \hat{\mathbf{t}}_\lambda, \mathcal{I}_{\lambda,\eta})$ is replaced with $(\mathbf{G}_n^h, \hat{\mathbf{t}}_\lambda^h, \mathcal{I}_{\lambda,\eta}^h)$ and the results are collected; note that $(\mathbf{S}_n^1, \dots, \mathbf{S}_n^{M_0}) \xrightarrow{d} (\mathbf{S}^1, \dots, \mathbf{S}^{M_0})$. $\square$

Proof of Proposition 8. We first prove that when $M < M_0$, $\Pr(LR_n^M > \hat{c}_{1-q_n}^M) \rightarrow 1$ as $n \rightarrow \infty$. Under Assumption 6, by the standard extremum estimator consistency argument (e.g., Theorem 2.1 of Newey and McFadden, 1994) gives $\hat{\boldsymbol{\vartheta}}_M \xrightarrow{p} \boldsymbol{\vartheta}_M^*$ for $M \leq M_0$, where $\boldsymbol{\vartheta}_M^*$ is the pseudo-true parameter that minimizes the Kullback–Leibler divergence within the M -component model. Because $\boldsymbol{\vartheta}_M^*$ lies in the interior of $\bar{\Theta}_{\boldsymbol{\vartheta}_M}(c)$ (the mixing proportions are bounded away from 0 and 1, and the variance ratios are bounded below), it follows from Theorem 3.2 of White (1982) that, for $M \leq M_0$,

$$\sqrt{n}(\hat{\boldsymbol{\vartheta}}_M - \boldsymbol{\vartheta}_M^*) \xrightarrow{d} N(0, A^M(\boldsymbol{\vartheta}_M^*)^{-1} B^M(\boldsymbol{\vartheta}_M^*) A^M(\boldsymbol{\vartheta}_M^*)^{-1}). \quad (52)$$

Then, from (52) and the mean value expansion, we have $Q_n^M(\hat{\boldsymbol{\vartheta}}_M) - Q^M(\boldsymbol{\vartheta}_M^*) = O_p(n^{-1/2})$ and

$$\frac{LR_n^M}{n} := 2 \left(Q_n^{M+1}(\hat{\boldsymbol{\vartheta}}_{M+1}) - Q_n^M(\hat{\boldsymbol{\vartheta}}_M) \right) = 2 \left(Q^{M+1}(\boldsymbol{\vartheta}_{M+1}^*) - Q^M(\boldsymbol{\vartheta}_M^*) \right) + o_p(1)$$

for $M = 1, 2, \dots, M_0 - 1$.

Because $Q^{M+1}(\boldsymbol{\vartheta}_{M+1}^*) - Q^M(\boldsymbol{\vartheta}_M^*) > 0$ by Assumption 6(f), LR_n^M/n converges to the positive constant $2(Q^{M+1}(\boldsymbol{\vartheta}_{M+1}^*) - Q^M(\boldsymbol{\vartheta}_M^*))$ as $n \rightarrow \infty$. By Lemma 7, $-n^{-1} \log q_n = o(1)$ and $\hat{c}_{1-q_n}^M - c_{1-q_n}^M = o_p(1)$ implies that $n^{-1} \hat{c}_{1-q_n}^M = o_p(1)$. Therefore, when $M < M_0$, we have $\Pr(LR_n^M > \hat{c}_{1-q_n}^M) = \Pr(LR_n^M/n > \hat{c}_{1-q_n}^M/n) \rightarrow 1$ as $n \rightarrow \infty$.

When $M = M_0$, because $LR_n^{M_0} = O_p(1)$ by Proposition 6 and $\hat{c}_{1-q_n}^{M_0} \rightarrow \infty$ by $q_n = o(1)$, $\Pr(LR_n^{M_0} > \hat{c}_{1-q_n}^{M_0}) \rightarrow 0$ as $n \rightarrow \infty$. $\square$

Proof of Proposition A.5. We first prove that $\lim_{n \rightarrow \infty} \Pr(\widehat{M}_{PL} < M_0) = 0$. To show

$\Pr(\hat{M}_{PL} < M_0) \rightarrow 0$, it suffices to prove that for each $M < M_0$,

$$\Pr(p\ell_n^M > p\ell_n^{M_0}) \rightarrow 0.$$

By definition,

$$\frac{p\ell_n^M - p\ell_n^{M_0}}{n} = (Q_n^M(\hat{\boldsymbol{\vartheta}}_M) - Q_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0})) - \frac{p_{n,k_M} - p_{n,k_{M_0}}}{n}. \quad (53)$$

From $p_{n,k} = o(n)$ in Assumption A.3(c),

$$\frac{p_{n,k_M} - p_{n,k_{M_0}}}{n} \rightarrow 0. \quad (54)$$

By the union bound and the triangle inequality, for any $\epsilon > 0$,

$$\Pr\left(\left|Q_n^M(\hat{\boldsymbol{\vartheta}}_M) - Q_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) - \left(Q^M(\boldsymbol{\vartheta}_M^*) - Q^{M_0}(\boldsymbol{\vartheta}_{M_0}^*)\right)\right| > 4\epsilon\right)$$

is at most

$$\Pr\left(\left|Q^M(\hat{\boldsymbol{\vartheta}}_M) - Q^M(\boldsymbol{\vartheta}_M^*)\right| > \epsilon\right) + \Pr\left(\left|Q^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) - Q^{M_0}(\boldsymbol{\vartheta}_{M_0}^*)\right| > \epsilon\right) \quad (55)$$

$$+ \Pr\left(\left|Q_n^M(\hat{\boldsymbol{\vartheta}}_M) - Q^M(\hat{\boldsymbol{\vartheta}}_M)\right| > \epsilon\right) + \Pr\left(\left|Q_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) - Q^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0})\right| > \epsilon\right). \quad (56)$$

The consistency of $\hat{\boldsymbol{\vartheta}}_M$ and $\hat{\boldsymbol{\vartheta}}_{M_0}$, together with the Continuous Mapping Theorem, implies that the two terms in (55) converge to 0 as $n \rightarrow \infty$.

For small $\delta > 0$, let $B(\delta, \boldsymbol{\vartheta}_M^*)$ and $B(\delta, \boldsymbol{\vartheta}_{M_0}^*)$ be open balls in the parameter spaces $\Theta_{\boldsymbol{\vartheta}_M}$ and $\Theta_{\boldsymbol{\vartheta}_{M_0}}$, respectively. Since $\hat{\boldsymbol{\vartheta}}_M$ and $\hat{\boldsymbol{\vartheta}}_{M_0}$ are consistent, with probability approaching one, the two terms in (56) are bounded by

$$\Pr\left(\sup_{\boldsymbol{\vartheta}_M \in B(\delta, \boldsymbol{\vartheta}_M^*)} |Q_n^M(\boldsymbol{\vartheta}_M) - Q^M(\boldsymbol{\vartheta}_M)| > \epsilon\right) \quad \text{and} \quad \Pr\left(\sup_{\boldsymbol{\vartheta}_{M_0} \in B(\delta, \boldsymbol{\vartheta}_{M_0}^*)} |Q_n^{M_0}(\boldsymbol{\vartheta}_{M_0}) - Q^{M_0}(\boldsymbol{\vartheta}_{M_0})| > \epsilon\right).$$

By applying Lemma 2.4 of Newey and McFadden (1994) under Assumption 6(b), and noting that $\log g_M(\mathbf{w}; \boldsymbol{\vartheta}_M)$ is continuous at each $\boldsymbol{\vartheta}_M$ and the envelope function $\sup_{\boldsymbol{\vartheta}_M \in B(\delta, \boldsymbol{\vartheta}_M^*)} |\log g_M(\mathbf{w}; \boldsymbol{\vartheta}_M)|$ has finite expectation for $M \leq M_0$, these two probabilities converge to 0 by the Uniform Law of Large Numbers. Thus, we obtain

$$Q_n^M(\hat{\boldsymbol{\vartheta}}_M) - Q_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) \xrightarrow{p} Q^M(\boldsymbol{\vartheta}_M^*) - Q^{M_0}(\boldsymbol{\vartheta}_{M_0}^*). \quad (57)$$

Therefore, from (53), (54), and (57), and in view of Assumptions 6(f) and A.3(c), we

have

$$\frac{p\ell_n^M - p\ell_n^{M_0}}{n} \xrightarrow{p} Q^M(\theta_M^*) - Q^{M_0}(\theta_{M_0}^*) < 0.$$

Hence, $\Pr(p\ell_n^M > p\ell_n^{M_0}) \rightarrow 0$ for $M < M_0$, which proves that

$$\lim_{n \rightarrow \infty} \Pr(\widehat{M}_{PL} < M_0) = 0. \quad (58)$$

We proceed to prove that $\lim_{n \rightarrow \infty} \Pr(\widehat{M}_{PL} > M_0) = 0$ by showing that, for $M = M_0 + 1, \dots, \bar{M}$,

$$\Pr(p\ell_n^M > p\ell_n^{M_0}) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

By definition of $p\ell_n^M$, we have

$$\Pr(p\ell_n^M > p\ell_n^{M_0}) = \Pr\left(\frac{\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0})}{p_{n,k_{M_0}}} + 1 - \frac{p_{n,k_M}}{p_{n,k_{M_0}}} > 0\right).$$

For any $\epsilon > 0$, $\lim_{n \rightarrow \infty} \Pr\left(\frac{\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0})}{p_{n,k_{M_0}}} > \epsilon\right) = 0$ because $p_{n,k_{M_0}} \rightarrow \infty$ and $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) = O_p(1)$ by Assumptions A.3(b) and A.4, respectively. Therefore,

$$\frac{\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0})}{p_{n,k_{M_0}}} + 1 - \frac{p_{n,k_M}}{p_{n,k_{M_0}}} \xrightarrow{p} 1 - \lim_{n \rightarrow \infty} \frac{p_{n,k_M}}{p_{n,k_{M_0}}},$$

which is strictly negative by Assumption A.3(d). Thus, $\Pr(p\ell_n^M > p\ell_n^{M_0}) \rightarrow 0$ as $n \rightarrow \infty$, and $\lim_{n \rightarrow \infty} \Pr(\widehat{M}_{PL} > M_0) = 0$ follows. $\square$

Proof of Proposition A.6. Because $\ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$ under standard regularity conditions in Assumption 6, Assumption A.4 holds when $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$ in view of $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) = (\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*)) - (\ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*))$. Because a consistent MLE is in $A_{\varepsilon_n}(\eta)$ defined in Appendix A.2, Propositions 3 and A.3 imply that $\ell_n(\hat{\boldsymbol{\Psi}}_M^j, \boldsymbol{\tau}^j) - \ell_n(\boldsymbol{\Psi}_M^{j*}, \boldsymbol{\tau}^j) = O_p(1)$, and the stated result holds. $\square$

Proof of Corollary 1. It is straightforward to verify that the penalty function $p_{n,k} = \frac{k}{2} \log(n)$ satisfies Assumption A.3. Thus, result (a) directly follows from Proposition A.6. For (b), Assumption A.3(c) holds for AIC penalty. Then, repeating the argument in the proof of Proposition A.5 up to (58) proves that $p\ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) > \max\{p\ell_n^1(\hat{\boldsymbol{\vartheta}}_1), \dots, p\ell_n^{M_0-1}(\hat{\boldsymbol{\vartheta}}_{M_0-1})\}$

holds with probability approaching one as $n \rightarrow \infty$. This implies that

$$\begin{aligned} \lim_{n \rightarrow \infty} \Pr \left(\widehat{M}_{PL} > M_0 \right) &\geq \lim_{n \rightarrow \infty} \Pr \left(p\ell_n^{M_0+1}(\hat{\boldsymbol{\vartheta}}_{M_0+1}) > \max \left\{ p\ell_n^1(\hat{\boldsymbol{\vartheta}}_1), \dots, p\ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) \right\} \right) \\ &= \lim_{n \rightarrow \infty} \Pr \left(p\ell_n^{M_0+1}(\hat{\boldsymbol{\vartheta}}_{M_0+1}) > p\ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) \right) \\ &= \lim_{n \rightarrow \infty} \Pr \left(2 \left[\ell_n^{M_0+1}(\hat{\boldsymbol{\vartheta}}_{M_0+1}) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) \right] > 2(k_{M_0+1} - k_{M_0}) \right), \quad (59) \end{aligned}$$

where k_{M_0} and k_{M_0+1} are the number of parameters for M_0 and $(M_0 + 1)$ components model. Since the term $2 \left[\ell_n^{M_0+1}(\hat{\boldsymbol{\vartheta}}_{M_0+1}) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) \right]$ is $O_p(1)$ and converges in distribution as specified in Proposition 6, and because k_{M_0} and k_{M_0+1} are finite, it follows from (59) that $\lim_{n \rightarrow \infty} \Pr \left(\widehat{M}_{PL} > M_0 \right) > 0$, and part (b) follows. $\square$

Proof of Proposition A.1. The part (a) can be proven similar to Proposition 3, and is thus omitted.

The proof for part (b) closely follows Theorem 2(b) in Andrews (2001). For each $\boldsymbol{\lambda} \in \Theta_{\lambda}(c_1)$, we approximate the log-likelihood function $\ell_n^2(\boldsymbol{\theta}_2, \boldsymbol{\lambda}, \alpha)$ around the true parameter values $(\boldsymbol{\theta}^*, 0)$ using the partial derivative with respect to $\boldsymbol{\theta}_2$ and the right partial derivative with respect to α to obtain

$$\begin{aligned} 2\{\ell_n^2(\boldsymbol{\theta}_2, \boldsymbol{\lambda}, \alpha) - \ell_n^2(\boldsymbol{\theta}^*, \boldsymbol{\lambda}, 0)\} &= \mathbf{G}_n(\boldsymbol{\lambda})^\top \mathcal{I}_n(\boldsymbol{\lambda}) \mathbf{G}_n(\boldsymbol{\lambda}) \\ &\quad - \left[\sqrt{n} \mathbf{t}(\boldsymbol{\theta}_2, \alpha) - \mathbf{G}_n(\boldsymbol{\lambda}) \right]^\top \mathcal{I}_n \left[\sqrt{n} \mathbf{t}(\boldsymbol{\theta}_2, \alpha) - \mathbf{G}_n(\boldsymbol{\lambda}) \right] + R_n(\boldsymbol{\theta}_2, \boldsymbol{\lambda}, \alpha), \quad (60) \end{aligned}$$

where $R_n(\boldsymbol{\theta}_2, \boldsymbol{\lambda}, \alpha)$ is a remainder term, and $\mathcal{I}_n(\boldsymbol{\lambda})$, $\mathbf{G}_n(\boldsymbol{\lambda})$, and $\mathbf{t}(\boldsymbol{\theta}_2, \alpha)$ are defined as

$$\mathcal{I}_n(\boldsymbol{\lambda}) := n^{-1} \sum_{i=1}^n \mathbf{s}(\mathbf{W}_i; \boldsymbol{\lambda}) \mathbf{s}(\mathbf{W}_i; \boldsymbol{\lambda})^\top, \quad \mathbf{G}_n(\boldsymbol{\lambda}) := \mathcal{I}_n(\boldsymbol{\lambda})^{-1} \mathbf{S}_n(\boldsymbol{\lambda}), \quad \text{and } \mathbf{t}(\boldsymbol{\theta}_2, \alpha) := \begin{pmatrix} \boldsymbol{\theta}_2 - \boldsymbol{\theta}^* \\ \alpha \end{pmatrix}$$

with $\mathbf{S}_n(\boldsymbol{\lambda}) = n^{-1/2} \sum_{i=1}^n \mathbf{s}(\mathbf{W}_i; \boldsymbol{\lambda})$ and $\mathbf{s}(\mathbf{W}_i; \boldsymbol{\lambda})$ defined in (2). $(\boldsymbol{\theta}, \pi)$ and $(B_T, D\ell_T(\boldsymbol{\theta}_0, \pi), \mathcal{J}_{T\pi}, Z_{T\pi})$ in Andrews (2001) correspond to our $((\boldsymbol{\theta}_2^\top, \alpha)^\top, \boldsymbol{\lambda})$ and $(n^{1/2}, n^{1/2} \mathbf{S}_n(\boldsymbol{\lambda}), \mathcal{I}_n(\boldsymbol{\lambda}), \mathbf{G}_n(\boldsymbol{\lambda}))$.

We prove the stated result by applying Theorem 2(b) of Andrews (2001) to (60). (β, δ, π) and $(B_T, G_\pi, \mathcal{J}_\pi, Z_\pi, Z_{\beta\pi})$ in Andrews (2001, pp. 697-699) correspond to our $(\alpha, \boldsymbol{\theta}_2, \boldsymbol{\lambda})$ and $(n^{1/2}, \mathbf{S}(\boldsymbol{\lambda}), \mathcal{I}(\boldsymbol{\lambda}), \mathbf{G}(\boldsymbol{\lambda}), \mathbf{G}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda}))$, where $\mathbf{G}(\boldsymbol{\lambda}) := \mathcal{I}(\boldsymbol{\lambda})^{-1} \mathbf{S}(\boldsymbol{\lambda})$, $\mathbf{G}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda}) := \mathcal{I}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda})^{-1} \mathbf{S}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda})$, and ψ in Andrews (2001, pp. 697-699) does not exist in our setting. The stated result then follows because $\mathbf{s}_{\boldsymbol{\theta}_2}(\mathbf{w})$ is identical to the score of the one-component model and $\hat{\lambda}'_{\beta\pi} (H \mathcal{J}_{*\pi}^{-1} H')^{-1} \hat{\lambda}_{\beta\pi}$ in Theorem 2(b) of Andrews (2001) is distributed as $(\max\{0, \mathcal{I}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda})^{-1/2} \mathbf{S}_{\alpha, \boldsymbol{\theta}_2}(\boldsymbol{\lambda})\})^2$. We proceed to verify the assumptions of Theorem 2(b) of Andrews (2001) (hereafter, A-Assumptions 2^{2*}, 3-5, 7, and 8). A-Assumption

2^{2*} (a) and (b) follow from our Assumption A.1(a)(b). A-Assumption 2^{2*} (c) holds because our Assumption A.1(c) and (d) and the uniform law of large numbers imply that $\sup_{\lambda \in \Theta_\lambda(c_1)} \|\mathcal{I}_n(\lambda) - \mathcal{I}(\lambda)\| \rightarrow_p 0$ and $\mathcal{I}(\lambda)$ is continuous.

A-Assumption 3 follows from Theorem 10.2 of Pollard (1990) if (i) $\tilde{\Theta}_\lambda$ is totally bounded, (ii) the finite dimensional distributions of $G_n(\cdot)$ converge to those of $G(\cdot)$, and (iii) $\{G_n(\cdot) : n \geq 1\}$ is stochastically equicontinuous. Condition (i) follows from the compactness of $\tilde{\Theta}_\lambda$ while conditions (ii) follow from Assumption A.1(b) and (c) and the multivariate CLT. Condition (iii) can be verified by our Assumption A.1(b) and (c) and Theorem 2 of Andrews (1994) because $\nabla_{(\theta_2^\top, \alpha)^\top} \log g_2(\cdot; \theta_2, \lambda, \alpha)$ are Lipschitz functions indexed by a finite dimensional parameter λ by Assumption A.1(b).

A-Assumption 4 follows from Lemma 1 of Andrews (2001) because, for each $\lambda \in \tilde{\Theta}_\lambda(c_1)$, $(\tilde{\theta}_2(\lambda), \tilde{\alpha}(\lambda)) = \arg \max_{(\theta_2, \alpha) \in \Theta_{\theta_2} \times [0, 1/2]} \ell_n^2(\theta_2, \lambda, \alpha)$ converges to $(\theta_2^*, 0)$ in probability from the standard consistency proof. A-Assumption 5 holds because (i) the set $[0, 1/2]$ equals a nonnegative half-line locally around 0, and (ii) $\Theta_{\theta_2} - \theta_2^*$ is locally equal to $\mathbb{R}^{\dim(\theta_2)}$. A-Assumption 7(a) is not relevant for our problem. A-Assumptions 7(b) and 8 follow from our proof of A-Assumption 5. □

Proof of Proposition A.2. For notational brevity, we drop the superscript j from Ψ_M^j, τ^j, s_i^j and $\mathcal{I}^j$. By using the Taylor expansion of $2 \log(1 + x) = 2x - x^2(1 + o(1))$ for small x , we have uniformly $\Psi_M \in \mathcal{N}_{c/\sqrt{n}}$,

$$\ell_n(\Psi_M, \tau) - \ell_n(\Psi_M^*, \tau) = 2 \sum_{k=1}^n \log(1 + h_{\Psi_M \tau, i}) = nP_n(2h_{\Psi_M \tau, i} - [1 + o_p(1)]h_{\Psi_M \tau, i}^2), \quad (61)$$

where $h_{\Psi_M \tau, i} := \sqrt{l_{\Psi_M \tau, i}} - 1$. The stated result holds if we show that

$$\sup_{\Psi_M \in \mathcal{N}_{c/\sqrt{n}}} |nP_n(h_{\Psi_M \tau, i}^2) - n\mathbf{t}_{\Psi_M \tau}^\top \mathcal{I} \mathbf{t}_{\Psi_M \tau} / 4| = o_p(1) \quad \text{and} \quad (62)$$

$$\sup_{\Psi_M \in \mathcal{N}_{c/\sqrt{n}}} |nP_n(h_{\Psi_M \tau, i}) - \sqrt{n}\mathbf{t}_{\Psi_M \tau}^\top \nu_n(s_i) / 2 + n\mathbf{t}_{\Psi_M \tau}^\top \mathcal{I} \mathbf{t}_{\Psi_M \tau} / 8| = o_p(1), \quad (63)$$

because then the right-hand side of (61) is equal to $\sqrt{n}\mathbf{t}_{\Psi_M \tau}^\top \nu_n(s_i) - \mathbf{t}_{\Psi_M \tau}^\top \mathcal{I} \mathbf{t}_{\Psi_M \tau} / 2 + o_p(1)$ uniformly in $\Psi_M \in \mathcal{N}_{c/\sqrt{n}}$.

We first show (62). Let

$$m_{\Psi_M \tau, i} := l_{\Psi_M \tau, i} - 1 = \mathbf{t}_{\Psi_M \tau}^\top \mathbf{s}_i + r_{\Psi_M \tau, i}.$$

Observe that

$$\max_{1 \leq i \leq n} \sup_{\Psi_M \in \mathcal{N}_{c/\sqrt{n}}} |m_{\Psi_M \tau i}| = \max_{1 \leq i \leq n} \sup_{\Psi_M \in \mathcal{N}_{c/\sqrt{n}}} |\mathbf{t}_{\Psi_M \tau}^\top \mathbf{s}_i + r_{\Psi_M \tau, i}| = o_p(1), \quad (64)$$

from Assumptions A.2(a) and (c) and Lemma 9. Write $4P_n(h_{\Psi_M \tau, i}^2)$ as

$$4P_n(h_{\Psi_M \tau, i}^2) = P_n \left(\frac{4(l_{\Psi_M \tau, i} - 1)^2}{(\sqrt{l_{\Psi_M \tau, i}} + 1)^2} \right) = P_n(l_{\Psi_M \tau, i} - 1)^2 - P_n \left((l_{\Psi_M \tau, i} - 1)^3 \frac{(\sqrt{l_{\Psi_M \tau, i}} + 3)}{(\sqrt{l_{\Psi_M \tau, i}} + 1)^3} \right). \quad (65)$$

From (14),

$$P_n(l_{\Psi_M \tau, i} - 1)^2 = \mathbf{t}_{\Psi_M \tau}^\top P_n(\mathbf{s}_i \mathbf{s}_i') \mathbf{t}_{\Psi_M \tau} + \zeta_{\Psi_M \tau n}, \quad (66)$$

where $\zeta_{\Psi_M \tau n} := 2\mathbf{t}_{\Psi_M \tau}^\top P_n[\mathbf{s}_i r_{\Psi_M \tau, i}] + P_n(r_{\Psi_M \tau, i})^2$. It follows from Assumptions A.2(a)(b)(c) and $(E|XY|)^2 \leq E|X|^2 E|Y|^2$ that, uniformly in $\Psi_M \in \mathcal{N}_\varepsilon$,

$$|\zeta_{\Psi_M \tau n}| = O_p(\|\mathbf{t}_{\Psi_M \tau}\|^2 \|\Psi_M - \Psi_M^*\|) + O_p(n^{-1} \|\mathbf{t}_{\Psi_M \tau}\| \|\Psi_M - \Psi_M^*\|) + O_p(n^{-1} \|\Psi_M - \Psi_M^*\|^2). \quad (67)$$

Then, (62) holds because $\|P_n(\mathbf{s}_i \mathbf{s}_i') - \mathcal{I}\| = o_p(1)$ and the second term on the right-hand side of (65) is bounded by

$$\mathcal{C} \sup_{\Psi_M \in \mathcal{N}_{c/\sqrt{n}}} P_n[|m_{\Psi_M \tau i}|^3] \leq o_p(1) \sup_{\Psi_M \in \mathcal{N}_{c/\sqrt{n}}} P_n[m_{\Psi_M \tau i}^2] = o_p(n^{-1}),$$

from $\sup_{\Psi_M \in \mathcal{N}_{c/\sqrt{n}}} \frac{(\sqrt{l_{\Psi_M \tau, i}} + 3)}{(\sqrt{l_{\Psi_M \tau, i}} + 1)^3} \leq \mathcal{C}$, (64), and $P_n(m_{\Psi_M \tau i}^2) = \mathbf{t}_{\Psi_M \tau}^\top \mathcal{I} \mathbf{t}_{\Psi_M \tau} + o_p(\|\mathbf{t}_{\Psi_M \tau}\|^2)$.

We proceed to show (63). Consider the following expansion of $h_{\Psi_M \tau, i}$:

$$h_{\Psi_M \tau, i} = (l_{\Psi_M \tau, i} - 1)/2 - h_{\Psi_M \tau, i}^2/2 = (\mathbf{t}_{\Psi_M \tau}^\top \mathbf{s}_i + r_{\Psi_M \tau, i})/2 - h_{\Psi_M \tau, i}^2/2. \quad (68)$$

Then, (63) follows from (62), (68), and Assumptions A.2(d), and the stated result follows. $\square$

Proof of Proposition A.3. For brevity, we drop the superscript M from $\mathbf{s}_i^M$ and $\mathcal{I}^M$. Define $h_{\Psi_M \alpha, i} := \sqrt{l_{\Psi_M \alpha, i}} - 1$. For part (a), it follows from $\log(1+x) \leq x$ and $h_{\Psi_M \alpha, i} = (l_{\Psi_M \alpha, i} - 1)/2 - h_{\Psi_M \alpha, i}^2/2$ (see (68)) that

$$\ell_n(\Psi_M, \alpha) - \ell_n(\Psi_M^*, \alpha) = 2 \sum_{k=1}^n \log(1 + h_{\Psi_M \alpha, i}) \leq 2nP_n(h_{\Psi_M \alpha, i}) = \sqrt{n}\nu_n(l_{\Psi_M \alpha, i} - 1) - nP_n(h_{\Psi_M \alpha, i}^2). \quad (69)$$

Observe that $h_{\Psi_M \alpha, i}^2 = (l_{\Psi_M \alpha, i} - 1)^2/(\sqrt{l_{\Psi_M \alpha, i}} + 1)^2 \geq \mathbb{I}\{l_{\Psi_M \alpha, i} \leq \kappa\} (l_{\Psi_M \alpha, i} - 1)^2/(\sqrt{\kappa} + 1)^2$

for any $\kappa > 0$. Therefore,

$$P_n(h_{\Psi_M \alpha, i}^2) \geq (\sqrt{\kappa} + 1)^{-2} P_n(\mathbb{I}\{l_{\Psi_M \alpha, i} \leq \kappa\} (l_{\Psi_M \alpha, i} - 1)^2). \quad (70)$$

In view of (66), (70) is rewritten as

$$P_n(h_{\Psi_M \alpha, i}^2) \geq (\sqrt{\kappa} + 1)^{-2} \left\{ \mathbf{t}_{\Psi_M \alpha}^\top [P_n(\mathbf{s}_i \mathbf{s}_i') - P_n(\mathbb{I}\{l_{\Psi_M \alpha, i} > \kappa\} \mathbf{s}_i \mathbf{s}_i')] \mathbf{t}_{\Psi_M \alpha} + \tilde{\zeta}_{\Psi_M \alpha n} \right\}, \quad (71)$$

where $\tilde{\zeta}_{\Psi_M \alpha n}$ satisfies the bound similar to (67). From Hölder's inequality, we have $P_n(\mathbb{I}\{l_{\Psi_M \alpha, i} > \kappa\} \|\mathbf{s}_i\|^2) \leq [P_n(\mathbb{I}\{l_{\Psi_M \alpha, i} > \kappa\})]^{\delta/(2+\delta)} [P_n(\|\mathbf{s}_i\|^{2+\delta})]^{2/(2+\delta)}$, where the right-hand side is no larger than $\kappa^{-\delta/(2+\delta)} O_p(1)$ uniformly in $\Psi_M \in \mathcal{N}_{\varepsilon_n}$ because (i) it follows from $\kappa \mathbb{I}\{l_{\Psi_M \alpha, i} > \kappa\} \leq l_{\Psi_M \alpha, i}$ that $P_n(\mathbb{I}\{l_{\Psi_M \alpha, i} > \kappa\}) \leq \kappa^{-1} P_n(l_{\Psi_M \alpha, i})$ and $\sup_{\Psi_M \in \mathcal{N}_{\varepsilon_n}} |P_n(l_{\Psi_M \alpha, i}) - 1| = o_p(1)$ from Assumptions A.2(d)(e), and (ii) $P_n(\|\mathbf{s}_i\|^{2+\delta}) = O_p(1)$ from Assumption A.2(a). Consequently, $\mathbb{P}(\sup_{\Psi_M \in \mathcal{N}_{\varepsilon_n}} P_n(\mathbb{I}\{l_{\Psi_M \alpha, i} > \kappa\} \|\mathbf{s}_i\|^2) \geq \lambda_{\min}/4) \rightarrow 0$ as $\kappa \rightarrow \infty$, where $\lambda_{\min}$ is the smallest eigenvalue of $\mathcal{I}$. Hence, we can write (71) as $P_n(h_{\Psi_M \alpha, i}^2) \geq \frac{\tau}{2}(1 + o_p(1)) \mathbf{t}_{\Psi_M \alpha}^\top \mathcal{I} \mathbf{t}_{\Psi_M \alpha} + \frac{\tau}{2} \tilde{\zeta}_{\Psi_M \alpha n}$ for $\tau := 2(\sqrt{\kappa} + 1)^{-2} > 0$ by taking κ sufficiently large. Because $\sqrt{n} \nu_n(l_{\Psi_M \alpha, i} - 1) = \sqrt{n} \mathbf{t}_{\Psi_M \alpha}^\top \nu_n(\mathbf{s}_i) + O_p(\sqrt{n} \|\mathbf{t}_{\Psi_M \alpha}\| \|\Psi_M - \Psi_M^*\|)$ from Assumptions A.2(d) and because $\tilde{\zeta}_{\Psi_M \alpha n}$ satisfies the bound analogous to (67), it follows from (69) that, uniformly in $\Psi_M \in \mathcal{N}_{\varepsilon_n}$,

$$\begin{aligned} -\eta &\leq \ell_n(\Psi_M, \alpha) - \ell_n(\Psi_M^*, \alpha) \\ &\leq \sqrt{n} \mathbf{t}_{\Psi_M \alpha}^\top \nu_n(\mathbf{s}_i) - \frac{\tau}{2} (1 + o_p(1)) n \mathbf{t}_{\Psi_M \alpha}^\top \mathcal{I} \mathbf{t}_{\Psi_M \alpha} + O_p(\sqrt{n} \|\mathbf{t}_{\Psi_M \alpha}\| \|\Psi_M - \Psi_M^*\|) \\ &\quad + O_p(n \|\mathbf{t}_{\Psi_M \alpha}\|^2 \|\Psi_M - \Psi_M^*\|) + O_p(\|\mathbf{t}_{\Psi_M \alpha}\| \|\Psi_M - \Psi_M^*\|) + O_p(\|\Psi_M - \Psi_M^*\|^2). \end{aligned} \quad (72)$$

Let $\mathbf{T}_n := \mathcal{I}^{1/2} \sqrt{n} \mathbf{t}_{\Psi_M \alpha}$. From (72), Assumptions A.2(b) and (e), and the fact $\|\Psi_M - \Psi_M^*\| \rightarrow 0$ if $\mathbf{t}_{\Psi_M \alpha} \rightarrow 0$, we obtain

$$-\eta \leq \ell_n(\Psi_M, \alpha) - \ell_n(\Psi_M^*, \alpha) \leq \|\mathbf{T}_n\| O_p(1) - \frac{\tau}{2} \|\mathbf{T}_n\|^2 + o_p(\|\mathbf{T}_n\|^2) + o_p(\|\mathbf{T}_n\|) + o_p(1),$$

uniformly in $\Psi_M \in \mathcal{N}_{\varepsilon_n}$. By rearranging the inequality and choosing a sufficiently large constant C , we have $\|\mathbf{T}_n\| C - (\tau/2) \|\mathbf{T}_n\|^2 + C \geq 0$ with an arbitrarily high probability. Namely, for any $\delta > 0$, there exist $C, n_0 < \infty$ such that

$$\mathbb{P} \left(\inf_{\Psi_M \in \mathcal{N}_{\varepsilon_n}} \|\mathbf{T}_n\| C - (\tau/2) \|\mathbf{T}_n\|^2 + C \geq 0 \right) \geq 1 - \delta, \quad \text{for all } n > n_0. \quad (73)$$

Rearranging the terms inside $\mathbb{P}(\cdot)$ gives $\sup_{\Psi_M \in \mathcal{N}_{\varepsilon_n}} (\|\mathbf{T}_n\| - (C/\tau))^2 \leq 2C/\tau + (C/\tau)^2$. Tak-

ing its square root gives $\mathbb{P}(\sup_{\Psi_{M \in \mathcal{N}_{\varepsilon_n}}} \|\mathbf{T}_n\| \leq C_1) \geq 1 - \delta$ for a constant C_1 , and part (a) follows. Part (b) follows from part (a) and Proposition A.2. $\square$

A.6 Proof of Bootstrap Validity

The key technical ingredient is the following lemma, which shows that the null distribution of the LRT is invariant under local perturbations of the null parameter. This is the panel analogue of Lemma 12 in the supplementary appendix of Kasahara and Shimotsu (2019).

Lemma 1 (Stability of the LRT under local null perturbations). *Let C_ν denote the set of sequences $\{\nu_n\}_{n \geq 1}$ satisfying $\sqrt{n}(\nu_n - \nu^*) \rightarrow \mathbf{h}_\nu$ for some finite $\mathbf{h}_\nu \in \mathbb{R}^{d_\nu}$. For each such sequence, let $P_{\nu_n}^n$ denote the product probability measure under which $\mathbf{W}_1, \dots, \mathbf{W}_n$ are i.i.d. from the M_0 -component density $g_{M_0}(\cdot; \nu_n)$.*

- (a) *Suppose Assumptions 1 and 4 hold and $M_0 = 1$. For every $\{\nu_n\} \in C_\nu$, the LRT statistic LR_n under $P_{\nu_n}^n$ converges in distribution to the same limit as under $P_{\nu^*}^n$, namely $\hat{\mathbf{t}}_\lambda^\top \mathcal{I}_{\lambda, \nu} \hat{\mathbf{t}}_\lambda$ (Proposition 4).*
- (b) *Suppose Assumptions 1, 3, and 5 hold and $M_0 = M \geq 2$. For every $\{\nu_n\} \in C_\nu$, the LRT statistic $LR_n^{M_0}$ under $P_{\nu_n}^n$ converges in distribution to the same limit as under $P_{\nu^*}^n$, namely $\max_{h=1, \dots, M_0} \{(\hat{\mathbf{t}}_\lambda^h)^\top \mathcal{I}_{\lambda, \nu}^h \hat{\mathbf{t}}_\lambda^h\}$ (Proposition 6).*

Proof of Lemma 1. We prove part (a); part (b) follows by the same argument applied to each local LRT statistic $LR_n^{M_0, h}$, $h = 1, \dots, M_0$, in the decomposition (24).

Set $\mathbf{h} = (\mathbf{h}_\nu^\top, \mathbf{0}^\top)^\top$. The proof proceeds in four steps.

Step 1 (LAN and contiguity of the null model). Under $P_{\nu^*}^n$, the correctly specified M_0 -component model is a regular parametric family: Assumption 6(c) places $\vartheta_{M_0}^*$ in the interior of $\Theta_{\vartheta_{M_0}}$, (d) ensures non-singularity of the information matrix $\mathcal{I}_\nu = B^{M_0}(\vartheta_{M_0}^*)$, and (e) provides smooth local identification. These conditions, together with the finite moment requirements in Assumption 4(a), yield the standard LAN expansion

$$\log \frac{dP_{\nu_n}^n}{dP_{\nu^*}^n} = \mathbf{h}_\nu^\top \mathbf{S}_{\nu_n} - \frac{1}{2} \mathbf{h}_\nu^\top \mathcal{I}_\nu \mathbf{h}_\nu + o_{P_{\nu^*}}(1), \quad (74)$$

where $\mathbf{S}_{\nu_n} = n^{-1/2} \sum_{i=1}^n \nabla_\nu \log g_{M_0}(\mathbf{W}_i; \nu^*)$ is the null-model score. Since $\mathbf{S}_{\nu_n} \xrightarrow{d} N(\mathbf{0}, \mathcal{I}_\nu)$ under $P_{\nu^*}^n$, mutual contiguity of $P_{\nu_n}^n$ and $P_{\nu^*}^n$ follows from Le Cam's first lemma (Lehmann and Romano, 2005, Corollary 12.3.1).

Step 2 (Score shift under contiguity). At the null point $(\boldsymbol{\nu}^*, \boldsymbol{\lambda} = \mathbf{0})$, the two-component density reduces to the one-component density: $g_2(\mathbf{W}; \boldsymbol{\nu}^*, \mathbf{0}) = g_1(\mathbf{W}; \boldsymbol{\nu}^*)$. Hence the nuisance component of the two-component score coincides with the null-model score: $\mathbf{s}_\nu(\mathbf{W}_i) = \nabla_\nu \log g_2(\mathbf{W}_i; \boldsymbol{\nu}^*, \mathbf{0}) = \nabla_\nu \log g_1(\mathbf{W}_i; \boldsymbol{\nu}^*)$. The full score vector $\mathbf{S}_n = (\mathbf{S}_{\nu n}^\top, \mathbf{S}_{\lambda n}^\top)^\top$ of the two-component model satisfies $\mathbf{S}_n \xrightarrow{d} N(\mathbf{0}, \mathcal{I})$ under $P_{\boldsymbol{\nu}^*}^n$ by Lemma 4(b)–(c). Joint convergence of $(\mathbf{S}_n, \log dP_{\nu n}^n/dP_{\boldsymbol{\nu}^*}^n)$ follows from the multivariate CLT applied to $(\mathbf{s}(\mathbf{W}_i), \nabla_\nu \log g_1(\mathbf{W}_i; \boldsymbol{\nu}^*))$ together with the LAN expansion (74). Because $\mathbf{s}_\nu(\mathbf{W}_i)$ appears in both vectors, the cross-covariance is

$$\text{Cov}(\mathbf{S}_n, \mathbf{h}_\nu^\top \mathbf{S}_{\nu n}) = \mathcal{I} \mathbf{h},$$

where the last equality uses $\mathbf{h} = (\mathbf{h}_\nu^\top, \mathbf{0}^\top)^\top$ and the block structure $\mathcal{I} = \begin{pmatrix} \mathcal{I}_\nu & \mathcal{I}_{\nu\lambda} \\ \mathcal{I}_{\lambda\nu} & \mathcal{I}_{\lambda\lambda} \end{pmatrix}$. By Le Cam’s third lemma (Lehmann and Romano, 2005, Corollary 12.3.2)—the corollary is stated for scalar statistics; the vector conclusion follows by applying it to every linear combination $\mathbf{a}^\top \mathbf{S}_n$ and invoking the Cramér–Wold device—under $P_{\nu n}^n$:

$$\mathbf{S}_n \xrightarrow{d} N(\mathcal{I} \mathbf{h}, \mathcal{I}). \quad (75)$$

Step 3 (Cancellation in the projected score). From (75), under $P_{\nu n}^n$ the component-wise means are $E[\mathbf{S}_{\nu n}] = \mathcal{I}_\nu \mathbf{h}_\nu$ and $E[\mathbf{S}_{\lambda n}] = \mathcal{I}_{\lambda\nu} \mathbf{h}_\nu$. The projected score (cf. the proof of Proposition 4) is

$$\mathbf{S}_{\lambda, \nu n} := \mathbf{S}_{\lambda n} - \mathcal{I}_{\lambda\nu} \mathcal{I}_\nu^{-1} \mathbf{S}_{\nu n}.$$

Under $P_{\nu n}^n$, its mean is

$$E[\mathbf{S}_{\lambda, \nu n}] = \mathcal{I}_{\lambda\nu} \mathbf{h}_\nu - \mathcal{I}_{\lambda\nu} \mathcal{I}_\nu^{-1} \mathcal{I}_\nu \mathbf{h}_\nu = \mathbf{0},$$

and its variance remains the Schur complement $\text{Var}[\mathbf{S}_{\lambda, \nu n}] = \mathcal{I}_{\lambda\lambda} - \mathcal{I}_{\lambda\nu} \mathcal{I}_\nu^{-1} \mathcal{I}_{\nu\lambda} =: \mathcal{I}_{\lambda, \nu}$ (using the notation of equation (21)), identical to the distribution under $P_{\boldsymbol{\nu}^*}^n$.

Step 4 (LRT distribution is unchanged). Contiguity of $P_{\nu n}^n$ and $P_{\boldsymbol{\nu}^*}^n$ preserves convergence in probability. Therefore, $\mathcal{I}_n \xrightarrow{p} \mathcal{I}$ and $R_n(\hat{\Psi}_\alpha, \alpha) = o_p(1)$ (Lemma 4(a)(c)) continue to hold under $P_{\nu n}^n$. Combined with Step 3, the proof of Proposition 4 goes through identically under $P_{\nu n}^n$. In particular, the LRT expansion (41) gives

$$LR_n = \max_{\alpha \in [c_1, 1-c_1]} C_n(\sqrt{n} \mathbf{t}_\lambda(\hat{\boldsymbol{\lambda}}_\alpha, \alpha)) + o_p(1),$$

where C_n depends on the data only through $\mathbf{G}_{\lambda, \nu n} := \mathcal{I}_{\lambda, \nu}^{-1} \mathbf{S}_{\lambda, \nu n} \xrightarrow{d} N(\mathbf{0}, \mathcal{I}_{\lambda, \nu}^{-1})$ under $P_{\nu n}^n$

(by Step 3). Hence LR_n has the same limiting distribution under $P_{\nu_n}^n$ as under $P_{\nu^*}^n$. $\square$

Proof of Proposition 7. The proof follows the argument in the proof of Theorem 15.4.2 of Lehmann and Romano (2005); see also the supplementary appendix of Kasahara and Shimotsu (2019), which establishes an analogous bootstrap validity result for multivariate normal mixture models using a similar proof strategy.

Part (a). Under $H_0: M_0 = 1$, the MLE $\hat{\vartheta}_1$ of the one-component model satisfies $\sqrt{n}(\hat{\vartheta}_1 - \vartheta_1^*) \xrightarrow{d} H$ for a P_{ν^*} -a.s. finite random variable H , by Assumption 6(a)–(d) applied with $M = 1$ (standard regular-MLE asymptotics under the correctly specified one-component model).

By the Almost Sure Representation Theorem (Lehmann and Romano, 2005, Theorem 11.2.19), there exist random variables $\tilde{\vartheta}_1$ and $\tilde{H}$ defined on a common probability space such that

- (i) $\hat{\vartheta}_1$ and $\tilde{\vartheta}_1$ have the same distribution, and
- (ii) $\sqrt{n}(\tilde{\vartheta}_1 - \vartheta_1^*) \rightarrow \tilde{H}$ almost surely.

Identifying $\nu_n = \tilde{\vartheta}_1$ in the reparameterised two-component model, property (ii) gives $\{\nu_n\} \in C_\nu$ with probability one. The bootstrap generates $\mathbf{W}_1^*, \dots, \mathbf{W}_n^*$ i.i.d. from $g_1(\cdot; \tilde{\vartheta}_1)$, and the bootstrap LRT LR_n^* is computed on this sample. By Lemma 1(a), conditionally on $\tilde{\vartheta}_1$,

$$LR_n^* \xrightarrow{d} F_1 \quad \text{under } P_{\tilde{\vartheta}_1}^n \text{ for a.e. realisation of } \tilde{\vartheta}_1.$$

Since $\hat{\vartheta}_1$ and $\tilde{\vartheta}_1$ have the same distribution (property (i)), it follows that $P^*(LR_n^* \leq x \mid \text{data}) \xrightarrow{p} F_1(x)$ at every continuity point x of F_1 .

Part (b). Under $H_0: M_0 = M$ with $M \geq 2$, the restricted MLE $\hat{\vartheta}_M$ satisfies $\sqrt{n}(\hat{\vartheta}_M - \vartheta_M^*) \xrightarrow{d} H$ by Assumption 6(a)–(d). Repeating the Almost Sure Representation argument with Lemma 1(b) in place of (a) yields the stated result.

Part (c). We verify the two conditions needed for sequential consistency.

Under-fitted models ($M < M_0$): By Assumption 6(f), $n^{-1}LR_n^M \xrightarrow{p} Q^{M+1}(\vartheta_{M+1}^*) - Q^M(\vartheta_M^*) > 0$. The bootstrap critical value $\hat{c}_{1-q_n}^{*,M}$ is the $(1 - q_n)$ -quantile of the bootstrap distribution, which converges to F_M by parts (a)–(b). Applying the Chernoff bound in the proof of Lemma 7 to the bootstrap null distribution gives $n^{-1}\hat{c}_{1-q_n}^{*,M} \xrightarrow{p} 0$. Hence $H_0: M_0 = M$ is rejected with probability approaching one.

Correctly specified model ($M = M_0$): We show $\hat{c}_{1-q_n}^{*,M_0} \xrightarrow{p} \infty$. For any $K > 0$, parts (a)–(b) give $P^*(LR_n^{*,M_0} > K \mid \text{data}) \xrightarrow{p} 1 - F_{M_0}(K) > 0$. Since $q_n \rightarrow 0$, for all large enough n we

have $q_n < 1 - F_{M_0}(K)$, which forces $\hat{c}_{1-q_n}^{*,M_0} > K$ with probability approaching one. Because K is arbitrary, $\hat{c}_{1-q_n}^{*,M_0} \xrightarrow{p} \infty$. Since $LR_n^{M_0} = O_p(1)$ under H_0 (Proposition 6),

$$P(LR_n^{M_0} > \hat{c}_{1-q_n}^{*,M_0}) \leq P(LR_n^{M_0} > K) + P(\hat{c}_{1-q_n}^{*,M_0} \leq K) \rightarrow 1 - F_{M_0}(K) + 0$$

for every $K > 0$. Taking $K \rightarrow \infty$ yields $P(LR_n^{M_0} > \hat{c}_{1-q_n}^{*,M_0}) \rightarrow 0$, so H_0 is not rejected with probability approaching one. Therefore $\widehat{M}_{\text{LRT}} = M_0 + o_p(1)$. $\square$

Remark 1 (Penalised bootstrap). *In the implementation, both estimation and bootstrap re-estimation maximise the penalised log-likelihood $\ell_n^{\text{pen}}(\boldsymbol{\vartheta}) := \ell_n(\boldsymbol{\vartheta}) + p(\boldsymbol{\vartheta})$, where the logarithmic boundary-repulsion penalty is*

$$p(\boldsymbol{\vartheta}) = (a_\alpha - 1) \sum_{j=1}^M \log \alpha_j + \sum_{j=1}^M (a_\tau - 1) \sum_{k=1}^{K_\epsilon} \log \tau_{jk} \quad (76)$$

for the static model. The penalty $p(\boldsymbol{\vartheta})$ is C^2 on the interior of $\Theta_{\boldsymbol{\vartheta}_M}$ (as a sum of log functions of parameters bounded away from zero). Variances are regularised only through the hard floor $\sigma_j^2 \geq \delta \hat{s}^2$, which does not correspond to a fixed additive term in $p(\boldsymbol{\vartheta})$. We verify that the penalised bootstrap is first-order asymptotically equivalent to the restricted bootstrap.

Null-model estimator equivalence. Under $H_0: M_0 = M$, the restricted MLE $\hat{\boldsymbol{\vartheta}}_M$ is interior to $\Theta_{\boldsymbol{\vartheta}_M}$ by Assumption 6(c). At any interior point, the penalty gradient satisfies $\nabla p(\boldsymbol{\vartheta}) = O(1)$ because the logarithmic-penalty derivatives are bounded away from zero at interior points, and the variance floor bounds variances away from zero. The penalised score equation is $\sum_{i=1}^n \mathbf{s}(\mathbf{W}_i; \boldsymbol{\vartheta}) + \nabla p(\boldsymbol{\vartheta}) = \mathbf{0}$. Expanding around $\hat{\boldsymbol{\vartheta}}_M$:

$$\hat{\boldsymbol{\vartheta}}_M^{\text{pen}} - \hat{\boldsymbol{\vartheta}}_M = -[\nabla^2 \ell_n(\hat{\boldsymbol{\vartheta}}_M)]^{-1} \nabla p(\hat{\boldsymbol{\vartheta}}_M) + o_p(n^{-1}) = O_p(n^{-1}) = o_p(n^{-1/2}), \quad (77)$$

since $\nabla^2 \ell_n = O_p(n)$ (Assumption 6(d)–(e)) and $\nabla p(\hat{\boldsymbol{\vartheta}}_M) = O_p(1)$ by interiority. Consequently, $\sqrt{n}(\hat{\boldsymbol{\vartheta}}_M^{\text{pen}} - \boldsymbol{\vartheta}_M^*) = \sqrt{n}(\hat{\boldsymbol{\vartheta}}_M - \boldsymbol{\vartheta}_M^*) + o_p(1)$, so the Almost Sure Representation argument in Proposition 7 goes through with the penalised MLE in place of the restricted MLE.

LRT equivalence. The LRT is computed using the sample log-likelihood ℓ_n evaluated at the penalised MLEs. For the null model: $|\ell_n^M(\hat{\boldsymbol{\vartheta}}_M^{\text{pen}}) - \ell_n^M(\hat{\boldsymbol{\vartheta}}_M)| = O_p(\sqrt{n}) \cdot O_p(n^{-1}) = o_p(1)$, by a Taylor expansion around $\hat{\boldsymbol{\vartheta}}_M$ and the score being $O_p(\sqrt{n})$ at the true value. For the alternative $(M+1)$ -component model under H_0 : the penalised MLE $\hat{\boldsymbol{\vartheta}}_{M+1}^{\text{pen}}$ is also interior (the logarithmic penalties bound the mixing proportions and sub-component weights away from zero, and the variance floor bounds variances away from zero), and the same Taylor argument gives $|\ell_n^{M+1}(\hat{\boldsymbol{\vartheta}}_{M+1}^{\text{pen}}) - \ell_n^{M+1}(\hat{\boldsymbol{\vartheta}}_{M+1})| = o_p(1)$. Combining both sides: $|LR_n^{\text{pen}} - LR_n| = o_p(1)$.

Bootstrap law. The same bounds hold conditionally on the data for the bootstrap penalised LRT, because the bootstrap data is generated from the penalised null-model MLE (which is interior) and the bootstrap re-estimation uses the same penalty. Therefore, the penalised bootstrap is first-order asymptotically equivalent to the restricted bootstrap, and Proposition 7 applies. The logarithmic-penalty terms in (76) diverge to $-\infty$ as $\alpha_j \rightarrow 0$ or $\tau_{jk} \rightarrow 0$, providing smooth boundary repulsion that complements the hard constraints (including the variance floor) in $\bar{\Theta}_{\Theta_M}(\mathbf{c})$.

A.7 Lemmas

Lemma 2. For any $M < \infty$, $\Pr(-\log n + \ell(\mathbf{W}_{i^*}; \bar{Y}_{i^*}, s_{i^*}^2) < M) = \exp(-C_T n^{1/T})$ as $n \rightarrow \infty$, where C_T for $T = 2, 3, \dots$ are some positive constant that depends on M and T .

Proof of Lemma 2. Because $\sum_{t=1}^T \frac{(Y_{it} - \mu)^2}{s_{i^*}^2} = T - 1$ when $i = i^*$, we have

$$\begin{aligned} -\log n + \ell(\mathbf{W}_{i^*}; \bar{Y}_{i^*}, s_{i^*}^2) &= -\log n - \frac{T}{2} \log s_{i^*}^2 - \frac{T}{2} \log(2\pi) - \frac{T-1}{2} \\ &= -\log \left(Bn (\chi_{i^*, T-1}^2)^{T/2} \right) \end{aligned} \quad (78)$$

with $B := \left(\frac{2\pi\sigma^{*2}}{T-1} \right)^{T/2} \exp(\frac{T-1}{2}) > 0$ and

$$\chi_{i^*, T-1}^2 := \frac{(T-1)s_{i^*}^2}{\sigma^{*2}}.$$

Therefore, to prove the stated result, it suffices to show that for any $\epsilon = \exp(-M)/B > 0$, $\Pr \left(n (\chi_{i^*, T-1}^2)^{T/2} > \epsilon \right) = \Pr \left(\chi_{i^*, T-1}^2 > (\epsilon/n)^{2/T} \right) \rightarrow 0$ as $n \rightarrow \infty$. Given the property of the first-order statistic, the distribution function of $\chi_{i^*, T-1}^2$ is given by $1 - [1 - F_{T-1}(t)]^n$, where $F_k(t)$ is the cumulative distribution function for chi-squared variables with k degree of freedom. It follows that

$$\Pr \left(\chi_{i^*, T-1}^2 > \left(\frac{\epsilon}{n} \right)^{2/T} \right) = \left[1 - F_{T-1} \left(\left(\frac{\epsilon}{n} \right)^{2/T} \right) \right]^n.$$

For any finite $T \geq 2$, write

$$\left[1 - F_{T-1}((\epsilon/n)^{2/T}) \right]^n = \left\{ \left[1 - F_{T-1}((\epsilon/n)^{2/T}) \right]^{\frac{1}{F_{T-1}((\epsilon/n)^{2/T})}} \right\}^{n F_{T-1}((\epsilon/n)^{2/T})}. \quad (79)$$

Then, because $(1 - F)^{\frac{1}{F}} \rightarrow \frac{1}{e}$ when $F \rightarrow 0$, the stated result follows from (79) if we can show

$$\frac{F_{T-1}((\epsilon x)^{2/T})}{x} = O(x^{-1/T}) \text{ as } x \rightarrow 0 \quad (80)$$

for $x = 1/n$, where (79)-(80) imply that $\log \left[1 - F_{T-1}((\epsilon/n)^{2/T}) \right]^n \leq -C_T n^{1/T}$ for some positive constant C_T when n is sufficiently large.

For small t , the CDF $F_k(t)$ can be expanded as $F_k(t) = \frac{t^{k/2}}{2^{k/2}\Gamma(k/2+1)} + o(t^{k/2})$ as $t \rightarrow 0$. Then, letting $k = T - 1$ and $t = (\epsilon x)^{2/T} = (\epsilon x)^{2/(k+1)}$, and simplifying the exponent $(x^{2/(k+1)})^{k/2} = x^{k/(k+1)}$, we have

$$F_{T-1}((\epsilon x)^{2/T}) = \frac{(\epsilon x)^{k/(k+1)}}{2^{k/2}\Gamma(k/2+1)} + o((\epsilon x)^{k/(k+1)}) \text{ as } x \rightarrow 0.$$

Therefore, given finite $\epsilon > 0$, noting that $x^{k/(k+1)-1} = x^{-1/(k+1)}$, we have

$$\frac{F_{T-1}((\epsilon x)^{2/T})}{x} = \frac{\epsilon^{k/(k+1)} x^{-1/(k+1)}}{2^{k/2}\Gamma(k/2+1)} + o(x^{-1/(k+1)}) = O(x^{-1/(k+1)}) \text{ as } x \rightarrow 0,$$

and the equation (80) follows with $k + 1 = T$. □

Lemma 3. For any $M < \infty$, $\Pr(-4 \log n + \ell(\mathbf{W}_{i*}; \bar{Y}_{i*}, s_{i*}^2) < M) \rightarrow 1$ as $n \rightarrow \infty$.

Proof. Following the proof of Lemma 2, because $-4 \log n + \ell(\mathbf{W}_{i*}; \bar{Y}_{i*}, s_{i*}^2) = -\log \left(B n^4 (\chi_{i*, T-1}^2)^{T/2} \right)$, it suffices to show that, for any $\epsilon > 0$, $\Pr \left(\chi_{i*, T-1}^2 > (\epsilon/n^4)^{2/T} \right) = \left[1 - F_{T-1}((\epsilon/n^4)^{2/T}) \right]^n \rightarrow 1$ as $n \rightarrow \infty$. For any finite $T \geq 2$, write

$$\left[1 - F_{T-1}((\epsilon/n^4)^{2/T}) \right]^n = \left\{ \left[1 - F_{T-1}((\epsilon/n^4)^{2/T}) \right]^{\frac{1}{F_{T-1}((\epsilon/n^4)^{2/T})}} \right\}^{n F_{T-1}((\epsilon/n^4)^{2/T})}. \quad (81)$$

Then, because $(1 - F)^{\frac{1}{F}} \rightarrow \frac{1}{e}$ when $F \rightarrow 0$, the stated result follows from (81) if we can show $\frac{F_{T-1}((\epsilon x^4)^{2/T})}{x} \rightarrow 0$ as $x \rightarrow 0$ for $x = 1/n$. By applying L'Hôpital's rule, we have

$$\lim_{x \rightarrow 0} \frac{F_{T-1}((\epsilon x^4)^{2/T})}{x} = \lim_{x \rightarrow 0} f_{T-1}((\epsilon x^4)^{2/T}) \epsilon^{2/T} (8/T) x^{8/T-1} = \lim_{x \rightarrow 0} \tilde{C}_{T, \epsilon} e^{-((\epsilon x^4)^{2/T})/2} x^{3-\frac{4}{T}} = 0,$$

where $\tilde{C}_{T, \epsilon} = \frac{\epsilon^{(T-1)/T} (8/T)}{2^{(T-1)/2} \Gamma((T-1)/2)}$ because $e^{-((\epsilon x^4)^{2/T})/2} \rightarrow 1$ and $x^{3-\frac{4}{T}} \rightarrow 0$ as $x \rightarrow 0$ for any finite $T \geq 2$. □

Lemma 4. Suppose that Assumptions 1, 3, and 4 hold. Then, under $H_0 : M_0 = 1$, for $\alpha \in [c_1, 1 - c_1]$, (a) for any $\delta > 0$, $\limsup_{n \rightarrow \infty} \Pr(\sup_{\Psi \in \Theta_{\Psi}: \|\Psi - \Psi^*\| \leq \kappa} |R_n(\Psi, \alpha)| > \delta(1 + \|\sqrt{n} \mathbf{t}(\Psi, \alpha)\|^2)) \rightarrow 0$ as $\kappa \rightarrow 0$, (b) $\mathbf{S}_n \xrightarrow{d} \mathbf{S} \sim N(0, \mathcal{I})$, and (c) $\mathcal{I}_n \xrightarrow{p} \mathcal{I}$, where $\mathcal{I}$ is finite and non-singular, and (d) $\mathbf{t}(\hat{\psi}_\alpha, \alpha) = O_p(n^{-1/2})$ for any $\alpha \in [c_1, 1 - c_1]$.

Proof of Lemma 4. The proof follows that of Proposition 2 in Kasahara and Shimotsu (2012). For a vector $\mathbf{x}$ and a function $f(\mathbf{x})$, let $\nabla_{\mathbf{x}^k} f(\mathbf{x})$ denote its k -th derivative with

respect to $\mathbf{x}$, which can be a multidimensional array. Observe that for any finite k and for a neighborhood $\mathcal{N}$ of $\boldsymbol{\psi}^*$, we obtain

$$\begin{aligned} E\|\nabla_{\boldsymbol{\psi}^k} g(\mathbf{W}_i; \boldsymbol{\psi}^*, \alpha)/g(\mathbf{W}_i; \boldsymbol{\psi}^*, \alpha)\|^2 &< \infty, \\ E\|\sup_{\boldsymbol{\psi} \in \Theta_{\boldsymbol{\psi}} \cap \mathcal{N}} \nabla_{\boldsymbol{\psi}^k} \log g(\mathbf{W}_i; \boldsymbol{\psi}, \alpha)\|^2 &< \infty \end{aligned} \quad (82)$$

because each element of $\nabla_{\boldsymbol{\psi}^k} \log g(y|\mathbf{x}, \mathbf{z}; \boldsymbol{\psi}, \alpha)$ is written as a sum of products of Hermite polynomials. Note also that the following holds:

$$\nabla_{\eta\lambda_j} L_n(\boldsymbol{\Psi}^*, \alpha) = 0, \quad \nabla_{\lambda_i\lambda_j\lambda_k} L_n(\boldsymbol{\Psi}^*, \alpha) = O_p(n^{1/2}), \quad (83)$$

$$\nabla_{\eta\eta\lambda_i} L_n(\boldsymbol{\Psi}^*, \alpha) = O_p(n), \quad \nabla_{\eta\eta\eta} L_n(\boldsymbol{\Psi}^*, \alpha) = O_p(n), \quad (84)$$

where equation (83) follows from Lemma 8(a), (c), and (d) and (82) and equation (84) is a simple consequence of (82). Furthermore, for a neighborhood $\mathcal{N}$ of $\boldsymbol{\Psi}^*$,

$$\sup_{\boldsymbol{\Psi} \in \Theta_{\boldsymbol{\Psi}} \cap \mathcal{N}} |n^{-1} \nabla^{(4)} L_n(\boldsymbol{\Psi}, \alpha) - E \nabla^{(4)} \log g(\mathbf{W}_i; \boldsymbol{\Psi}, \alpha)| = o_p(1), \quad (85)$$

$$E \nabla^{(4)} \log g(\mathbf{W}_i; \boldsymbol{\Psi}, \alpha) \text{ is continuous in } \boldsymbol{\psi} \in \Theta_{\boldsymbol{\Psi}} \cap \mathcal{N}. \quad (86)$$

Equations (85) and (86) follow from Lemma 2.4 of Newey and McFadden (1994) and the fact that $\nabla_{\boldsymbol{\psi}^k} \log g(\mathbf{w}; \boldsymbol{\psi}, \alpha)$ is written as a sum of products of Hermite polynomials.

Taking a fourth-order Taylor expansion of $L_n(\boldsymbol{\psi}, \alpha)$ around $\boldsymbol{\psi}^*$ and using (82) and (83), we can write $L_n(\boldsymbol{\psi}, \alpha) - L_n(\boldsymbol{\psi}^*, \alpha)$ as the sum of the relevant terms and the remainder term as follows:

$$\begin{aligned} L_n(\boldsymbol{\psi}, \alpha) - L_n(\boldsymbol{\psi}^*, \alpha) = \\ \nabla_{\boldsymbol{\eta}} L_n^*(\boldsymbol{\eta} - \boldsymbol{\eta}^*) + \frac{1}{2!} (\boldsymbol{\eta} - \boldsymbol{\eta}^*)^\top \nabla_{\boldsymbol{\eta}\boldsymbol{\eta}^\top} L_n^*(\boldsymbol{\eta} - \boldsymbol{\eta}^*) + \frac{1}{2!} \sum_{i=1}^q \sum_{j=1}^q \nabla_{\lambda_i\lambda_j} L_n^* \lambda_i \lambda_j \end{aligned} \quad (87)$$

$$+ \frac{3}{3!} \sum_{i=1}^q \sum_{j=1}^q (\boldsymbol{\eta} - \boldsymbol{\eta}^*)^\top \nabla_{\boldsymbol{\eta}\lambda_i\lambda_j} L_n^* \lambda_i \lambda_j \quad (88)$$

$$+ \frac{1}{4!} \sum_{i=1}^q \sum_{j=1}^q \sum_{k=1}^q \sum_{\ell=1}^q \nabla_{\lambda_i\lambda_j\lambda_k\lambda_\ell} L_n^* \lambda_i \lambda_j \lambda_k \lambda_\ell + R_n(\boldsymbol{\psi}, \alpha), \quad (89)$$

where ∇L_n^* denotes the derivative of $L_n(\boldsymbol{\psi}, \alpha)$ evaluated at $(\boldsymbol{\psi}^*, \alpha)$. In view of (83) and

(84), the remainder term is written as

$$R_n(\Psi, \alpha) = O_p(n^{1/2}) \sum_{i=1}^q \sum_{j=1}^q \sum_{k=1}^q \lambda_i \lambda_j \lambda_k + O_p(n) \left(\sum_{i=1}^q \|\boldsymbol{\eta} - \boldsymbol{\eta}^*\|^2 \lambda_i + \|\boldsymbol{\eta} - \boldsymbol{\eta}^*\|^3 \right) \quad (90)$$

$$+ O_p(n) \sum_{i=1}^q \sum_{j=1}^q \sum_{k=1}^q (\|\boldsymbol{\eta} - \boldsymbol{\eta}^*\|^4 + \|\boldsymbol{\eta} - \boldsymbol{\eta}^*\|^3 |\lambda_i| + \|\boldsymbol{\eta} - \boldsymbol{\eta}^*\|^2 |\lambda_i \lambda_j| + \|\boldsymbol{\eta} - \boldsymbol{\eta}^*\| |\lambda_i \lambda_j \lambda_k|) \quad (91)$$

$$+ \frac{1}{4!} \sum_{i=1}^q \sum_{j=1}^q \sum_{k=1}^q \sum_{\ell=1}^q \{\nabla_{\lambda_i \lambda_j \lambda_k \lambda_\ell} L_n(\boldsymbol{\psi}^\dagger, \alpha) - \nabla_{\lambda_i \lambda_j \lambda_k \lambda_\ell} L_n(\boldsymbol{\psi}^*, \alpha)\} \lambda_i \lambda_j \lambda_k \lambda_\ell \quad (92)$$

with $\boldsymbol{\psi}^\dagger$ being between $\boldsymbol{\psi}$ and $\boldsymbol{\psi}^*$. Because $\|\sqrt{nt}(\boldsymbol{\psi}, \alpha)\|^2 = n\|\boldsymbol{\eta} - \boldsymbol{\eta}^*\|^2 + n \sum_{i=1}^q \sum_{j=1}^i \alpha^2 (1-\alpha)^2 |\lambda_i \lambda_j|^2$, the right-hand side of (90) and the terms in (91) are bounded by $O_p(1)(\|\sqrt{nt}(\boldsymbol{\psi}, \alpha)\| + \|\sqrt{nt}(\boldsymbol{\psi}, \alpha)\|^2)(\|\boldsymbol{\eta} - \boldsymbol{\eta}^*\| + \|\lambda\|)$. In view of (85) and (86), (92) is bounded by $\|\sqrt{nt}(\boldsymbol{\psi}, \alpha)\|^2 [d(\boldsymbol{\psi}^\dagger) + o_p(1)]$ with $d(\boldsymbol{\psi}^\dagger) \rightarrow 0$ as $\boldsymbol{\psi}^\dagger \rightarrow \boldsymbol{\psi}^*$, where a function $d(\boldsymbol{\psi}^\dagger)$ corresponds to $n^{-1} \mathbb{E}[\nabla_{\lambda_i \lambda_j \lambda_k \lambda_\ell} L_n(\boldsymbol{\psi}^\dagger, \alpha) - \nabla_{\lambda_i \lambda_j \lambda_k \lambda_\ell} L_n(\boldsymbol{\psi}^*, \alpha)]$. Therefore, $R_n(\boldsymbol{\psi}, \alpha) = (1 + \|\sqrt{nt}(\boldsymbol{\psi}, \alpha)\|)^2 [d(\boldsymbol{\psi}^\dagger) + o_p(1) + O_p(\|\boldsymbol{\psi} - \boldsymbol{\psi}^*\|)]$, and part (a) follows.

Part (b) follows from Lemma 8(c) and (d), the Lindeberg–Levy central limit theorem, and the finiteness of $\mathcal{I}$ in part (c).

For part (c), we first provide the formula of $\mathcal{I}_n$. Partition $\mathcal{I}_n$ as

$$\mathcal{I}_n = \begin{pmatrix} \mathcal{I}_{\boldsymbol{\eta}n} & \mathcal{I}_{\boldsymbol{\eta}\boldsymbol{\lambda}n} \\ \mathcal{I}_{\boldsymbol{\eta}\boldsymbol{\lambda}n}^\top & \mathcal{I}_{\boldsymbol{\lambda}n} \end{pmatrix}, \quad \mathcal{I}_{\boldsymbol{\eta}n} : q \times q, \quad \mathcal{I}_{\boldsymbol{\eta}\boldsymbol{\lambda}n} : q \times q_\lambda, \quad \mathcal{I}_{\boldsymbol{\lambda}n} : q_\lambda \times q_\lambda,$$

where $q_\lambda = q(q+1)/2$. $\mathcal{I}_{\boldsymbol{\eta}n}$ is given by $\mathcal{I}_{\boldsymbol{\eta}n} = -n^{-1} \nabla_{\boldsymbol{\eta}\boldsymbol{\eta}^\top} L_n(\boldsymbol{\psi}^*, \alpha)$. For $\mathcal{I}_{\boldsymbol{\eta}\boldsymbol{\lambda}n}$, let $A_{ij} = n^{-1} \nabla_{\boldsymbol{\eta}\lambda_i \lambda_j} L_n(\boldsymbol{\psi}^*, \alpha)$ and write the term in (88) as $(n/2) \sum_{i=1}^q \sum_{j=1}^q (\boldsymbol{\eta} - \boldsymbol{\eta}^*)^\top A_{ij} \lambda_i \lambda_j = n \sum_{i=1}^q \sum_{j=1}^i (\boldsymbol{\eta} - \boldsymbol{\eta}^*)^\top A_{ij} c_{ij} \lambda_i \lambda_j$, where $c_{ij} = 1/2$ if $i = j$, and $c_{ij} = 1$ if $i \neq j$. Then, by defining $\mathcal{I}_{\boldsymbol{\eta}\boldsymbol{\lambda}n} = -(A_{11}, \dots, A_{qq}, A_{12}, \dots, A_{q-1,q})/\alpha(1-\alpha)$, the term in (88) equals $-n(\boldsymbol{\eta} - \boldsymbol{\eta}^*)^\top \mathcal{I}_{\boldsymbol{\eta}\boldsymbol{\lambda}n} [\alpha(1-\alpha)v(\boldsymbol{\lambda})]$. For $\mathcal{I}_{\boldsymbol{\lambda}n}$, define $B_{ijk\ell} = n^{-1}(8/4!) \nabla_{\lambda_i \lambda_j \lambda_k \lambda_\ell} L_n(\boldsymbol{\Psi}^*, \alpha)$ so that the first term in (89) is written as $(n/8) \sum_{i=1}^q \sum_{j=1}^q \sum_{k=1}^q \sum_{\ell=1}^q B_{ijk\ell} \lambda_i \lambda_j \lambda_k \lambda_\ell = (n/2) \sum_{i=1}^q \sum_{j=1}^i \sum_{k=1}^q \sum_{\ell=1}^k c_{ij} c_{k\ell} B_{ijk\ell} \lambda_i \lambda_j \lambda_k \lambda_\ell$. Define $\mathcal{I}_{\boldsymbol{\lambda}n}$ such that the $(ij, k\ell)$ element of $\mathcal{I}_{\boldsymbol{\lambda}n}$ is $-c_{ij} c_{k\ell} B_{ijk\ell} / \alpha^2 (1-\alpha)^2$, where the values of ij run over $\{(1,1), \dots, (q,q), (1,2), \dots, (q-1,q)\}$. Then, the first term in (89) equals $-(n/2)[\alpha(1-\alpha)v(\boldsymbol{\lambda})]^\top \mathcal{I}_{\boldsymbol{\lambda}n} [\alpha(1-\alpha)v(\boldsymbol{\lambda})]$. With this definition of $\mathcal{I}_n$, the expansion (87)-(89) is written as (36) in terms of $\sqrt{nt}(\boldsymbol{\psi}, \alpha)$.

We now show that $\mathcal{I}_n \rightarrow_p \mathcal{I}$. $\mathcal{I}_{\boldsymbol{\eta}n} \rightarrow_p \mathcal{I}_{\boldsymbol{\eta}}$ holds trivially. For $\mathcal{I}_{\boldsymbol{\eta}\boldsymbol{\lambda}n}$, it follows from Lemma 8(c) and the law of large numbers that $A_{ij} \rightarrow_p$

$-\mathbb{E}[\nabla_{\eta} l(\mathbf{W}; \psi^*, \alpha) \nabla_{\lambda_i \lambda_j} l(\mathbf{W}; \psi^*, \alpha)]$, giving $\mathcal{I}_{\eta \lambda_n} \rightarrow_p E[s_{\eta} s_{\lambda \lambda}^{\top} / \alpha(1 - \alpha)] = \mathcal{I}_{\eta \lambda}$. For $\mathcal{I}_{\lambda n}$, Lemma 8(d) and (e) and the law of large numbers imply that $\sum_{i=1}^q \sum_{j=1}^q \sum_{k=1}^q \sum_{\ell=1}^q B_{ijkl} \lambda_i \lambda_j \lambda_k \lambda_{\ell} \rightarrow_p$
 $-\sum_{i=1}^q \sum_{j=1}^q \sum_{k=1}^q \sum_{\ell=1}^q E[\nabla_{\lambda_i \lambda_j} l(\mathbf{W}; \psi^*, \alpha) \nabla_{\lambda_k \lambda_{\ell}} l(\mathbf{W}; \psi^*, \alpha)] \lambda_i \lambda_j \lambda_k \lambda_{\ell}$, where the factor $(8/4!) = 1/3$ in B_{ijkl} and the three derivatives on the right-hand side of Lemma 8(e) cancel each other out. Therefore, we have $\mathcal{I}_{\lambda n} \rightarrow_p E[s_{\lambda \lambda} s_{\lambda \lambda}^{\top} / \alpha^2(1 - \alpha)^2] = \mathcal{I}_{\lambda}$, and $\mathcal{I}_n \rightarrow_p \mathcal{I}$ follows.

We complete the proof of part (c) by showing that $\mathcal{I} = E[s(\mathbf{W})s(\mathbf{W})^{\top}]$ is finite and non-singular. Note that $s(\mathbf{W})$ can be expressed in Hermite polynomials as in (19). Then, the finiteness of $\mathcal{I}$ follows from Assumption 4(a) and the definition of Hermite polynomials.

To show that $\mathcal{I}$ is positive definite, it suffices to show that there exists no multicollinearity in $s(\mathbf{w})$. Suppose, to the contrary, that $s(\mathbf{w})$ is multicollinear and that there exists a non-zero vector $\mathbf{a}$ that solves the equation $\mathbf{a}^{\top} s(\mathbf{w}) = 0$ for all values of $\mathbf{w}$. Partition $s(\mathbf{w})$ as $s(\mathbf{w}) = (s_{(\mu)}^{\top}, s_{(\beta)}^{\top})^{\top}$ with $s_{(\mu)} = (s_{\mu}, s_{\sigma}, s_{\lambda_{\mu\mu}}, s_{\lambda_{\mu\sigma}}, s_{\lambda_{\sigma\sigma}})^{\top}$ and $s_{(\beta)} = (s_{\beta}^{\top}, s_{\lambda_{\mu\beta}}^{\top}, s_{\lambda_{\sigma\beta}}^{\top}, s_{\lambda_{\beta\beta}}^{\top})^{\top}$, where $s(\mathbf{w})$ is defined in (2) and (19). Similarly, partition $\mathbf{a}$ as $\mathbf{a} = (\mathbf{a}_{(\mu)}^{\top}, \mathbf{a}_{(\beta)}^{\top})^{\top}$ so that

$$\mathbf{a}^{\top} s(\mathbf{w}) = \mathbf{a}_{(\mu)}^{\top} s_{(\mu)} + \mathbf{a}_{(\beta)}^{\top} s_{(\beta)}. \quad (93)$$

By Assumption 4(b) and the property of Hermite polynomials, if $\mathbf{a}^{\top} s(\mathbf{w}) = 0$ for all $\mathbf{w}$, then $\mathbf{a}_{(\beta)} = 0$.

Then, in view of (93), the stated result follows if we can show that $\mathbf{a}_{(\mu)}^{\top} s_{(\mu)} = 0$ for all $\mathbf{w}$ implies $\mathbf{a}_{(\mu)} = 0$. Suppose that

$$\begin{aligned} \mathbf{a}_{(\mu)}^{\top} s_{(\mu)} &= a_{\mu} \sum_{t=1}^T H_t^{1*} + (a_{\sigma} + a_{\lambda_{\mu\mu}}) \sum_{t=1}^T H_t^{2*} + \frac{a_{\lambda_{\mu\mu}}}{2} \sum_{t=1}^T \sum_{s \neq t}^T H_t^{1*} H_s^{1*} \\ &\quad + a_{\lambda_{\mu\sigma}} \sum_{t=1}^T H_t^{3*} + a_{\lambda_{\mu\sigma}} \sum_{t=1}^T \sum_{s \neq t}^T H_t^{1*} H_s^{2*} + 3a_{\lambda_{\sigma\sigma}} \sum_{t=1}^T H_t^{4*} + \frac{a_{\lambda_{\sigma\sigma}}}{2} \sum_{t=1}^T \sum_{s \neq t}^T H_t^{2*} H_s^{2*} = 0 \end{aligned}$$

for all $\mathbf{w}$, where H_t^{j*} for $j = 1, 2, 3$ is defined in (18) in Appendix A.3.2.

Because the above equation holds for all values of $\mathbf{w}$, with the property of the Hermite polynomials, we have $a_{\mu} = 0$, $(a_{\sigma} + a_{\lambda_{\mu\mu}}) = 0$, $a_{\lambda_{\mu\mu}} = 0$, $a_{\lambda_{\mu\sigma}} = 0$, $a_{\lambda_{\sigma\sigma}} = 0$. This implies that $\mathbf{a}_{(\mu)} = 0$. Therefore, no multicollinearity exists in $s(\mathbf{w})$ and $\mathcal{I}$ is non-singular, proving part (c).

The proof of part (d) closely follows the proof of Theorem 1 of Andrews (1999). Let

$\mathbf{T}_n := \mathcal{I}_n^{1/2} \sqrt{nt}(\hat{\Psi}_\alpha, \alpha)$. Then, in view of (36), we have

$$\begin{aligned}
o_p(1) &\leq L_n(\hat{\psi}_\alpha, \alpha) - L_n(\psi^*, \alpha) \\
&= \mathbf{T}_n' \mathcal{I}_n^{-1/2} \mathbf{S}_n - \frac{1}{2} \|\mathbf{T}_n\|^2 + R_n(\hat{\psi}_\alpha, \alpha) \\
&= O_p(\|\mathbf{T}_n\|) - \frac{1}{2} \|\mathbf{T}_n\|^2 + (1 + \|\mathcal{I}_n^{-1/2} \mathbf{T}_n\|)^2 o_p(1) \\
&= \|\mathbf{T}_n\| O_p(1) - \frac{1}{2} \|\mathbf{T}_n\|^2 + o_p(\|\mathbf{T}_n\|) + o_p(\|\mathbf{T}_n\|^2) + o_p(1),
\end{aligned}$$

where the third equality holds because $\mathcal{I}_n^{-1/2} \mathbf{S}_n = O_p(1)$ and $R_n(\hat{\psi}_\alpha, \alpha) = o_p((1 + \|\mathcal{I}_n^{-1/2} \mathbf{T}_n\|)^2)$ from part (a). Rearranging this equation yields $\|\mathbf{T}_n\|^2 \leq 2\|\mathbf{T}_n\| O_p(1) + o_p(1)$. Denote the $O_p(1)$ term by ς_n . Then, $(\|\mathbf{T}_n\| - \varsigma_n)^2 \leq \varsigma_n^2 + o_p(1) = O_p(1)$; taking its square root gives $\|\mathbf{T}_n\| \leq O_p(1)$. In conjunction with $\mathcal{I}_n \rightarrow_p \mathcal{I}$, we obtain $\sqrt{nt}(\hat{\Psi}_\alpha, \alpha) = O_p(1)$, and part (d) follows. $\square$

Lemma 5 (Linear independence of posterior-weighted Hermite functions). *Let $\tau^* \in (0, 1)$, $\sigma^* > 0$, and $\mu_1^*, \mu_2^* \in \mathbb{R}$ with $\mu_1^* \neq \mu_2^*$. Define the densities $\phi_k(y) := (\sigma^*)^{-1} \phi((y - \mu_k^*)/\sigma^*)$ for $k = 1, 2$, the mixture density $f^*(y) := \tau^* \phi_1(y) + (1 - \tau^*) \phi_2(y)$, the posterior weights $\gamma_k(y) := \tau_k^* \phi_k(y)/f^*(y)$ with $\tau_1^* := \tau^*$ and $\tau_2^* := 1 - \tau^*$, and the Hermite functions $H_k^{a*}(y) := (\sigma^*)^{-a} H^a((y - \mu_k^*)/\sigma^*)$, where H^a is the probabilists' Hermite polynomial of degree a as defined in (18). Set*

$$\begin{aligned}
V_1(y) &:= \frac{\phi_1(y) - \phi_2(y)}{f^*(y)} = \frac{\gamma_1(y)}{\tau^*} - \frac{\gamma_2(y)}{1 - \tau^*}, \\
V_2(y) &:= \gamma_1(y) H_1^{1*}(y), \\
V_3(y) &:= \gamma_2(y) H_2^{1*}(y), \\
V_4(y) &:= \gamma_1(y) H_1^{2*}(y) + \gamma_2(y) H_2^{2*}(y).
\end{aligned} \tag{94}$$

The five functions $\{1, V_1, V_2, V_3, V_4\}$ are linearly independent on $\mathbb{R}$.

Proof. Suppose $c_0 + \sum_{j=1}^4 c_j V_j(y) = 0$ for all $y \in \mathbb{R}$. Multiplying by $f^*(y) > 0$ and using the Rodrigues identity for the probabilists' Hermite polynomials,

$$H_k^{a*}(y) \phi_k(y) = (-1)^a \partial_y^a \phi_k(y), \tag{95}$$

yields the identity

$$P_1(y) \phi_1(y) + P_2(y) \phi_2(y) = 0 \quad \forall y \in \mathbb{R},$$

where P_1, P_2 are polynomials of degree at most 2 in y depending linearly on $(c_0, \dots, c_4)$. Both sides are real-analytic, so the identity holds for all $y \in \mathbb{R}$.

Without loss of generality assume $\mu_1^* < \mu_2^*$. Dividing by $\phi_2(y) > 0$,

$$P_1(y) \frac{\phi_1(y)}{\phi_2(y)} + P_2(y) = 0, \quad \frac{\phi_1(y)}{\phi_2(y)} = \exp\left(\frac{(\mu_1^* - \mu_2^*)y}{\sigma^{*2}} + \frac{\mu_2^{*2} - \mu_1^{*2}}{2\sigma^{*2}}\right). \quad (96)$$

Since $\mu_1^* < \mu_2^*$, the exponential factor decays to 0 as $y \rightarrow +\infty$ super-polynomially, so $P_1(y)\phi_1(y)/\phi_2(y) \rightarrow 0$. Taking $y \rightarrow +\infty$ in (96) forces $P_2(y) \rightarrow 0$; but P_2 is a polynomial in y , so $P_2 \equiv 0$. Substituting back, $P_1 \equiv 0$. Reading off the coefficients of the zero polynomials P_1, P_2 in terms of $(c_0, \dots, c_4)$ yields $c_0 = c_1 = c_2 = c_3 = c_4 = 0$.

The above linear-independence statement for posterior-weighted Hermite functions at two distinct centres is the classical identifiability result for finite normal mixtures with equal variance and distinct means, due to Teicher (1963) and Yakowitz and Spragins (1968); we have given a self-contained derivation specialised to the auxiliary basis required below. $\square$

Remark 2. The proof of Lemma 5 extends mechanically to any $K_\epsilon \geq 2$ provided the within-type means $\mu_1^*, \dots, \mu_{K_\epsilon}^*$ are pairwise distinct (Assumption 1). One obtains an identity $\sum_{k=1}^{K_\epsilon} P_k(y)\phi_k(y) = 0$ with P_k polynomials of degree at most $2(K_\epsilon - 1)$; ordering $\mu_1^* < \dots < \mu_{K_\epsilon}^*$ and taking $y \rightarrow +\infty$ peels off $P_{K_\epsilon} \equiv 0$, then $P_{K_\epsilon-1} \equiv 0$, etc., by iteration.

Assumption MOD-1 (Panel-covariate joint identifying variation). Let $\mathbf{U}_t := (1, \mathbf{X}_t^\top)^\top \in \mathbb{R}^{1+q}$ for each $t \in \{1, \dots, T\}$. The $(1+q)^2 \times (1+q)^2$ matrix

$$\mathbb{E}[(\mathbf{U}_t \otimes \mathbf{U}_s)(\mathbf{U}_t \otimes \mathbf{U}_s)^\top]$$

is non-singular for every distinct pair $t \neq s$. Equivalently, no non-trivial polynomial in $(\mathbf{X}_t, \mathbf{X}_s)$ of degree at most $(1, 1)$ in $(\mathbf{X}_t, \mathbf{X}_s)$ vanishes almost surely.

Assumption MOD-1 is the natural cross-period strengthening of Assumption 4(b). It is automatically implied by the conjunction of Assumption 4(b) and Assumption 3's "identifying covariate variation" clause whenever $(\mathbf{X}_t, \mathbf{X}_s)$ admits a joint density positive on an open set of $\mathbb{R}^{2q}$ (e.g. when $\mathbf{X}$ is a continuous panel process with non-degenerate joint distribution across t), as in the empirical specifications of Section 7. It rules out degenerate panel-covariate processes such as perfect persistence ($\mathbf{X}_1 = \mathbf{X}_2$) or supports concentrated on polynomial varieties (e.g. $\{X_1 X_2 = 0\}$ a.s.) for which Assumption 4(b) by itself does not control the bilinear cross-period design rank.¹

¹An adversarial counterexample illustrates the necessity. Take $\tau^* = 1/2$, $\mu_1^* = 0$, $\mu_2^* = 1$, $\sigma^* = 1$, $T = 2$,

Lemma 6 (Companion to Lemma 4 for $K_\epsilon = 2$). *Suppose Assumptions 1 (with $K_\epsilon = 2$), 3, 4(a)(b)(d), and MOD-1 hold. Then the conclusions (a), (b), and (d) of Lemma 4 hold, and (c) holds with $\mathcal{I} = \mathbb{E}[\mathbf{s}(\mathbf{W})\mathbf{s}(\mathbf{W})^\top]$ finite and positive definite for model (1) with (2)–(3).*

Proof. Parts (a), (b), and (d) extend from the proof of Lemma 4 verbatim: each element of $\nabla_{\psi^k} \log g(\mathbf{w}; \psi, \alpha)$ for $k \leq 4$ is a finite sum of products of posterior probabilities $\gamma_{kt} \in [0, 1]$, Hermite polynomials of degree at most 4, and (when covariates are present) monomials in $\mathbf{X}_t$ of degree at most 2. The derivative bounds (82)–(86) and the Bartlett-style identities of Lemma 8 continue to apply, and the Lindeberg–Lévy CLT, the uniform LLN of Newey and McFadden (1994), and the boundary-MLE argument of Andrews (1999) proceed exactly as in the proof of Lemma 4.

It remains to prove part (c). *Finiteness* of $\mathcal{I} = \mathbb{E}[\mathbf{s}(\mathbf{W})\mathbf{s}(\mathbf{W})^\top]$ follows because each entry is a polynomial of degree at most 4 in the standardised residuals $Z_{kt} = (Y_t - \mathbf{X}_t^\top \beta^* - \mu_k^*)/\sigma^*$ and degree at most 2 in $\mathbf{X}_t$; the standardised residuals have all polynomial moments, and Assumption 4(a) supplies the required 9th moment of $\mathbf{X}$ (more than enough to cover degree 4 after squaring).

Non-singularity: suppose, toward a contradiction, that a non-zero vector $\mathbf{a}$ satisfies $\mathbf{a}^\top \mathbf{s}(\mathbf{W}) = 0$ a.s. We will deduce $\mathbf{a} = \mathbf{0}$. Partition $\mathbf{a}$ into a first-order block $\mathbf{a}_1 := (a_\tau, a_{\mu_1}, a_{\mu_2}, a_\sigma, \mathbf{a}_\beta^\top)^\top$ and a $(4 + q) \times (4 + q)$ symmetric matrix $\mathbf{A}_{\lambda\lambda}$ collecting the ten $\lambda\lambda^\top$ coefficients $(a_{\lambda\tau\tau}, a_{\lambda\mu_1\mu_1}, a_{\lambda\mu_2\mu_2}, a_{\lambda\mu_1\mu_2}, a_{\lambda\tau\mu_1}, a_{\lambda\tau\mu_2}, a_{\lambda\mu_1\sigma}, a_{\lambda\mu_2\sigma}, a_{\lambda\sigma\sigma}, a_{\lambda\tau\sigma})$ together with the $\lambda\beta$ -blocks.

Write the standardised residual $\tilde{Y}_t := Y_t - \mathbf{X}_t^\top \beta^*$. By (22)–(23) together with Lemma 8(c), the score admits the U-statistic representation

$$\mathbf{a}^\top \mathbf{s}(\mathbf{W}) = \sum_{t=1}^T u_{\mathbf{a}_1}(\tilde{Y}_t, \mathbf{X}_t) + \sum_{t \neq s} \mathbf{V}(\tilde{Y}_t)^\top \mathbf{M}(\mathbf{X}_t, \mathbf{X}_s) \mathbf{V}(\tilde{Y}_s), \quad (97)$$

where $\mathbf{V}(\tilde{Y}_t) := (V_1(\tilde{Y}_t), V_2(\tilde{Y}_t), V_3(\tilde{Y}_t), V_4(\tilde{Y}_t))^\top$ is the auxiliary basis of Lemma 5, $u_{\mathbf{a}_1}$ collects the single-period first-order and same- t second-order contributions, and $\mathbf{M}(\mathbf{X}_t, \mathbf{X}_s) = \mathbf{J}(\mathbf{X}_t)^\top \mathbf{A}_{\lambda\lambda} \mathbf{J}(\mathbf{X}_s)$ for a deterministic block matrix $\mathbf{J}(\mathbf{x}) \in \mathbb{R}^{(4+q) \times 4}$ whose first four rows are $\mathbf{I}_4$ (no $\mathbf{x}$ -factor) and whose lower q -block inserts $\mathbf{x}$ in the V_2 - and V_3 -slots (the β -block columns). The factorisation $\mathbf{J}(\mathbf{x})\mathbf{v} = \mathbf{L}(\mathbf{v})\mathbf{U}$ with $\mathbf{U} = (1, \mathbf{x}^\top)^\top$ and $\mathbf{L}(\mathbf{v})$ a deterministic linear map of $\mathbf{v} \in \mathbb{R}^4$ records the explicit dependence on $\mathbf{U}$.

Step 1 (Mixed-difference identity). Since $f^*(\mathbf{y} \mid \mathbf{x}) > 0$ everywhere, $\mathbf{a}^\top \mathbf{s}(\mathbf{W}) = 0$ a.s.

$q = 1$, and $P_{\mathbf{X}}\{(X_1, X_2) \in \{(0, 0), (1, 0), (0, 1)\}\} = 1/3$ each. Then $\mathbb{E}[\mathbf{U}\mathbf{U}^\top]$ is non-singular (LRT1(b) holds), yet $X_1 X_2 = 0$ a.s., so the matrix $\mathbf{B} = \text{diag}(0, 1) \neq \mathbf{0}$ satisfies $\mathbf{U}_1^\top \mathbf{B} \mathbf{U}_2 = X_1 X_2 = 0$ a.s. The implication “ $\mathbf{U}_1^\top \mathbf{B} \mathbf{U}_2 = 0$ a.s. $\Rightarrow \mathbf{B} = \mathbf{0}$ ” therefore requires Assumption MOD-1, not merely Assumption 4(b).

implies the identity holds for all $(\mathbf{Y}, \mathbf{X})$ in the joint support. Because $T \geq 2$ (Assumption 3), we may apply the bivariate mixed difference operator $\Delta := \Delta_{\tilde{y}_1, z_1} \Delta_{\tilde{y}_2, z_2}$ that holds $\tilde{Y}_3, \dots, \tilde{Y}_T$ and $\mathbf{X}_1, \dots, \mathbf{X}_T$ fixed and takes

$$\Delta h(\tilde{Y}_1, \tilde{Y}_2) := h(\tilde{y}_1, \tilde{y}_2) - h(z_1, \tilde{y}_2) - h(\tilde{y}_1, z_2) + h(z_1, z_2),$$

for arbitrary $\tilde{y}_1, \tilde{y}_2, z_1, z_2 \in \mathbb{R}$. All single-period terms $u_{a_1}(\tilde{Y}_t, \mathbf{X}_t)$ in (97) are additively separable in $(\tilde{y}_1, \tilde{y}_2)$ (they depend on at most one of them at a time) and so are annihilated by Δ . Using symmetry $M(\mathbf{X}_2, \mathbf{X}_1)^\top = M(\mathbf{X}_1, \mathbf{X}_2)$, the cross-period sum reduces to a single $(t, s) = (1, 2)$ pair after applying Δ :

$$2 (\mathbf{V}(\tilde{y}_1) - \mathbf{V}(z_1))^\top M(\mathbf{X}_1, \mathbf{X}_2) (\mathbf{V}(\tilde{y}_2) - \mathbf{V}(z_2)) = 0 \quad (98)$$

for almost every $(\tilde{y}_1, z_1, \tilde{y}_2, z_2) \in \mathbb{R}^4$ and for $(\mathbf{X}_1, \mathbf{X}_2)$ in the joint support.

Step 2 ($\mathbf{A}_{\lambda\lambda} = \mathbf{0}$). By Lemma 5, the five functions $\{1, V_1, V_2, V_3, V_4\}$ are linearly independent on $\mathbb{R}$, so the difference set $\{\mathbf{V}(\tilde{y}) - \mathbf{V}(z) : \tilde{y}, z \in \mathbb{R}\}$ spans $\mathbb{R}^4$. Therefore (98) forces

$$M(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{J}(\mathbf{X}_1)^\top \mathbf{A}_{\lambda\lambda} \mathbf{J}(\mathbf{X}_2) = \mathbf{0} \quad \text{a.s.}$$

For any fixed $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^4$, write $\mathbf{J}(\mathbf{X}_i) \mathbf{v}_i = \mathbf{L}(\mathbf{v}_i) \mathbf{U}_i$. Then $\mathbf{U}_1^\top \mathbf{L}(\mathbf{v}_1)^\top \mathbf{A}_{\lambda\lambda} \mathbf{L}(\mathbf{v}_2) \mathbf{U}_2 = 0$ a.s. over the joint distribution of $(\mathbf{U}_1, \mathbf{U}_2)$. Vectorising, $(\mathbf{U}_1 \otimes \mathbf{U}_2)^\top \text{vec}(\mathbf{L}(\mathbf{v}_1)^\top \mathbf{A}_{\lambda\lambda} \mathbf{L}(\mathbf{v}_2)) = 0$ a.s. over $(\mathbf{U}_1 \otimes \mathbf{U}_2)$. By Assumption MOD-1, $\mathbb{E}[(\mathbf{U}_1 \otimes \mathbf{U}_2)(\mathbf{U}_1 \otimes \mathbf{U}_2)^\top]$ is non-singular, hence $\mathbf{L}(\mathbf{v}_1)^\top \mathbf{A}_{\lambda\lambda} \mathbf{L}(\mathbf{v}_2) = \mathbf{0}$ for every fixed $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^4$. Standard basis variations of $(\mathbf{v}_1, \mathbf{v}_2)$ then force each block of $\mathbf{A}_{\lambda\lambda}$ to zero, so $\mathbf{A}_{\lambda\lambda} = \mathbf{0}$ identically.

Step 3 ($\mathbf{a}_1 = \mathbf{0}$). With $\mathbf{A}_{\lambda\lambda} = \mathbf{0}$, the identity (97) reduces to

$$\sum_{t=1}^T \left[a_\tau V_1(\tilde{Y}_t) + (a_{\mu_1} + \mathbf{X}_t^\top \mathbf{a}_{\beta,1}) V_2(\tilde{Y}_t) + (a_{\mu_2} + \mathbf{X}_t^\top \mathbf{a}_{\beta,2}) V_3(\tilde{Y}_t) + a_\sigma V_4(\tilde{Y}_t) \right] = 0 \quad \text{a.s.},$$

where $\mathbf{a}_{\beta,1}$ and $\mathbf{a}_{\beta,2}$ are the β -coefficients in the score components that multiply V_2 and V_3 respectively (they coincide if the β_j -block is not type-specific; the proof goes through with separate coefficients in either case). Varying $\tilde{Y}_1$ alone forces the bracketed function of $\tilde{Y}_1$ to equal a constant. By Lemma 5, the basis $\{1, V_1, V_2, V_3, V_4\}$ is linearly independent, so the coefficient on each V_j must equal zero, while the constant absorbs the only constant-function direction. Hence

$$a_\tau = 0, \quad a_\sigma = 0, \quad a_{\mu_1} + \mathbf{X}_1^\top \mathbf{a}_{\beta,1} = 0, \quad a_{\mu_2} + \mathbf{X}_1^\top \mathbf{a}_{\beta,2} = 0 \quad \text{a.s. } \mathbf{X}_1.$$

Writing the last two equations as $(a_{\mu_j}, \mathbf{a}_{\beta,j}^\top) \mathbf{U}_1 = 0$ a.s. for $j = 1, 2$ and invoking Assumption 4(b) (non-singularity of $\mathbb{E}[\mathbf{U}\mathbf{U}^\top]$) yields $a_{\mu_j} = 0$ and $\mathbf{a}_{\beta,j} = \mathbf{0}$. Hence $\mathbf{a}_1 = \mathbf{0}$.

Step 4 (Conclusion). Combining Steps 2 and 3, $\mathbf{a} = \mathbf{0}$, contradicting the hypothesis. Therefore $\mathcal{I}$ is positive definite.

Quantitative rank certificate. Specialised to the no-covariate sub-system in the (μ, τ, σ) -block (i.e. $\mathbf{a}_\beta = \mathbf{0}$ and $q = 0$), the proof produces a 14×14 linear system on the fourteen non- β coordinates of $\mathbf{a}$. A choice of fourteen non-redundant linear functionals from Steps 2 and 3 yields a matrix whose determinant is

$$\det = \frac{1}{16 \tau^* (1 - \tau^*)},$$

strictly positive on $\tau^* \in (0, 1)$. A symbolic verification using SymPy is provided in `memo_keps2_nonsingularity_det.sympy.txt` (run `memo_keps2_nonsingularity_det.sympy.py`); evaluations at $\tau^* \in \{1/10, 1/4, 1/2, 3/4, 9/10\}$ give determinants $\{25/36, 1/3, 1/4, 1/3, 25/36\}$. $\square$

Remark 3. Lemma 6 converts the non-singularity of $\mathcal{I}$ for $K_\epsilon \geq 2$, currently maintained as Assumption 4(c), into an analytical consequence of Assumptions 1 (with $K_\epsilon = 2$), 3, 4(a)(b)(d), and MOD-1. Propositions 4, 6, and 7, which currently rely on the maintained hypothesis for $K_\epsilon = 2$, therefore hold unconditionally for $K_\epsilon = 2$ under this strengthening, which is automatically satisfied in the empirical specifications of Section 7 because the input vector $\mathbf{x}_{it} = (k_{it}, \ell_{it}, v_{it})^\top$ admits a joint density positive on an open set across consecutive years. The proof extends to $K_\epsilon \geq 3$ mutatis mutandis via the iterated phase-peeling argument of Remark 2.

Lemma 7. Suppose that the assumptions in Proposition 8 hold. If $-n^{-1} \log q_n = o(1)$, then $n^{-1} c_{1-q_n}^M = o(1)$.

Proof. For brevity of notation, write $c_n = c_{1-q_n}^M$. By Theorem 2.1 of Foutz and Srivastava (1977), $PLR_n(M) \xrightarrow{d} \sum_{j=1}^K b_j \chi_j^2$ for $0 < b_j < \infty$ and K is finite, where $\chi_1^2, \dots, \chi_K^2$ are independent chi-square random variables with one degree of freedom. Then, we have

$$q_n = \Pr \left(\sum_{j=1}^K b_j \chi_j^2 \geq c_n \right) \leq \sum_{j=1}^K \Pr \left(\chi_j^2 \geq \frac{c_n}{K b_j} \right) \leq \frac{K}{\sqrt{1-2t}} \exp \left(-t \frac{c_n}{K b^*} \right) \quad \text{for } 0 < t < \frac{1}{2}$$

with $b^* = \max\{b_1, \dots, b_K\}$, where the last inequality follows from a Chernoff bound: $\Pr \left(\chi_j^2 \geq \frac{c_n}{b^*} \right) \leq \frac{\mathbb{E}[\exp(t(\chi_j^2 - 1))]}{\exp(t(\frac{c_n}{b^*} - 1))} = \frac{1}{\sqrt{1-2t}} \exp \left(-t \frac{c_n}{b^*} \right)$ for $0 < t < \frac{1}{2}$. Therefore, $-\frac{\log q_n}{n} \geq -\frac{1}{n} \log \left(\frac{K}{\sqrt{1-2t}} \right) + \frac{t}{K b^*} \frac{c_n}{n}$, and the stated result follows. $\square$

Lemma 8. Suppose that $g(\mathbf{w}; \boldsymbol{\psi}, \alpha)$ is defined as (33), where $\boldsymbol{\psi} = (\boldsymbol{\eta}^\top, \boldsymbol{\lambda}^\top)^\top$. Let g^* , ∇g^* , and $\nabla \log g^*$ denote $g(\mathbf{W}; \boldsymbol{\psi}, \alpha)$, $\nabla g(\mathbf{W}; \boldsymbol{\psi}, \alpha)$, and $\nabla \log g(\mathbf{W}; \boldsymbol{\psi}, \alpha)$ evaluated at $(\boldsymbol{\psi}^*, \alpha)$, respectively. Let ∇f^* denote $\nabla f(\mathbf{W}; \boldsymbol{\theta}^*)$. The following statements hold.

(a) For $l = 0, 1, \dots$, $\nabla_{(\boldsymbol{\lambda} \otimes \boldsymbol{\eta}^{\otimes l})^\top} g^* = 0$;

(b) $\nabla_{(\boldsymbol{\lambda}^{\otimes 2})^\top} g^* = \alpha(1 - \alpha) \nabla_{(\boldsymbol{\theta}^{\otimes 2})^\top} f^*$;

(c) $\nabla_{(\boldsymbol{\lambda}^{\otimes 2})^\top} \log g^* = \alpha(1 - \alpha) \nabla_{(\boldsymbol{\theta}^{\otimes 2})^\top} f^* / f^*$;

(d) $\mathbb{E}[\nabla_{\lambda_i \lambda_j} \log g^*] = 0$, $\mathbb{E}[\nabla_{\lambda_i \lambda_j \lambda_k} \log g^*] = 0$, and $\mathbb{E}[\nabla_{\eta \lambda_i \lambda_j} \log g^*] = -\mathbb{E}[\nabla_{\eta} \log g^* \nabla_{\lambda_i \lambda_j} \log g^*]$;

(e) $\mathbb{E}[\nabla_{\lambda_i \lambda_j \lambda_k \lambda_\ell} \log g^*] = -\mathbb{E}[\nabla_{\lambda_i \lambda_j} \log g^* \nabla_{\lambda_k \lambda_\ell} \log g^* + \nabla_{\lambda_i \lambda_k} \log g^* \nabla_{\lambda_j \lambda_\ell} \log g^* + \nabla_{\lambda_i \lambda_\ell} \log g^* \nabla_{\lambda_j \lambda_k} \log g^*]$.

Proof of Lemma 8. Recall that

$$g(\mathbf{w}; \boldsymbol{\psi}, \alpha) = \alpha f(\mathbf{w}; \boldsymbol{\nu} + (1 - \alpha)\boldsymbol{\lambda}) + (1 - \alpha)f(\mathbf{w}; \boldsymbol{\nu} - \alpha\boldsymbol{\lambda}).$$

First, we show that for $l = 0$ holds for (a), $\nabla_{\boldsymbol{\lambda}} g^* = \alpha(1 - \alpha) \nabla_{\boldsymbol{\theta}} f^* - \alpha(1 - \alpha) \nabla_{\boldsymbol{\theta}} f^* = 0$. For $l > 0$, by Fubini's theorem, we have

$$\begin{aligned} \nabla_{(\boldsymbol{\lambda} \otimes \boldsymbol{\eta}^{\otimes l})^\top} g &= \nabla_{\boldsymbol{\lambda}} \left(\alpha \nabla_{(\boldsymbol{\eta}^{\otimes l})^\top} f(\mathbf{w}; \boldsymbol{\nu} + (1 - \alpha)\boldsymbol{\lambda}) + (1 - \alpha) \nabla_{(\boldsymbol{\eta}^{\otimes l})^\top} f(\mathbf{w}; \boldsymbol{\nu} - \alpha\boldsymbol{\lambda}) \right) \Big|_{\boldsymbol{\nu}=\boldsymbol{\theta}^*, \boldsymbol{\lambda}=0} \\ &= \left(\alpha(1 - \alpha) \nabla_{(\boldsymbol{\lambda} \otimes \boldsymbol{\eta}^{\otimes l})^\top} f(\mathbf{w}; \boldsymbol{\nu} + (1 - \alpha)\boldsymbol{\lambda}) - \alpha(1 - \alpha) \nabla_{(\boldsymbol{\lambda} \otimes \boldsymbol{\eta}^{\otimes l})^\top} f(\mathbf{w}; \boldsymbol{\nu} - \alpha\boldsymbol{\lambda}) \right) \Big|_{\boldsymbol{\nu}=\boldsymbol{\theta}^*, \boldsymbol{\lambda}=0} \\ &= 0. \end{aligned}$$

To show part (b), note that

$$\begin{aligned} \nabla_{(\boldsymbol{\lambda}^{\otimes 2})^\top} g &= \nabla_{\boldsymbol{\lambda}} \left(\alpha(1 - \alpha) \nabla_{\boldsymbol{\lambda}^\top} f(\mathbf{w}; \boldsymbol{\nu} + (1 - \alpha)\boldsymbol{\lambda}) - \alpha(1 - \alpha) \nabla_{\boldsymbol{\lambda}^\top} f(\mathbf{w}; \boldsymbol{\nu} - \alpha\boldsymbol{\lambda}) \right) \\ &= \alpha(1 - \alpha)^2 \nabla_{(\boldsymbol{\lambda}^{\otimes 2})^\top} f(\mathbf{w}; \boldsymbol{\nu} + (1 - \alpha)\boldsymbol{\lambda}) + \alpha^2(1 - \alpha) \nabla_{(\boldsymbol{\lambda}^{\otimes 2})^\top} f(\mathbf{w}; \boldsymbol{\nu} - \alpha\boldsymbol{\lambda}) \Big|_{\boldsymbol{\nu}=\boldsymbol{\theta}^*, \boldsymbol{\lambda}=0} \\ &= \alpha(1 - \alpha) \nabla_{(\boldsymbol{\lambda}^{\otimes 2})^\top} f^*. \end{aligned}$$

For part (c), $\nabla_{\boldsymbol{\lambda} \boldsymbol{\lambda}^\top} \log g^* = \nabla_{\boldsymbol{\lambda} \boldsymbol{\lambda}^\top} g^* / g^* - (\nabla_{\boldsymbol{\lambda}} \log g^*)(\nabla_{\boldsymbol{\lambda}} \log g^*) = \alpha(1 - \alpha) \nabla_{\boldsymbol{\lambda} \boldsymbol{\lambda}^\top} f^* / f^*$ because $\nabla_{\boldsymbol{\lambda} \boldsymbol{\lambda}^\top} g^* / g^* = \alpha(1 - \alpha) \nabla_{\boldsymbol{\theta} \boldsymbol{\theta}^\top} f^* / f^*$ from part (a) and $\nabla_{\boldsymbol{\lambda}} \log g^* = \nabla_{\boldsymbol{\lambda}} g^* / g^* = 0$ from part (b).

For parts (d) and (e), observe that $\int \nabla_{\lambda_i} \log g(\mathbf{w}; \boldsymbol{\psi}, \alpha) g(\mathbf{w}; \boldsymbol{\psi}, \alpha) d\mathbf{x} = 0$ holds for any $\boldsymbol{\psi}$

in the interior of Θ_ψ , and differentiating this equation w.r.t. λ_j gives

$$\int \{\nabla_{\lambda_i \lambda_j} \log g(\mathbf{w}; \psi, \alpha) + \nabla_{\lambda_i} \log g(\mathbf{w}; \psi, \alpha) \nabla_{\lambda_j} \log g(\mathbf{w}; \psi, \alpha)\} g(\mathbf{w}; \psi, \alpha) dx = 0. \quad (99)$$

Evaluating (99) at $\psi = \psi^*$ in conjunction with part (a) gives the first equation in part (d). Differentiating (99) w.r.t. λ_k or η and evaluating at $\psi = \psi^*$ gives the latter two equations in part (d). Part (e) follows from differentiating (99) w.r.t. λ_k and λ_ℓ and evaluating at $\psi = \psi^*$ in conjunction with parts (a) and (d). $\square$

Lemma 9. [Lemma 2.1 of Liu and Shao (2003)] Suppose $X_1, \dots, X_n$ are i.i.d. random variables with $\max_{1 \leq i \leq n} \mathbb{E}|X_i|^{q+\delta} < C$ for some $\delta > 0$, $q > 0$, and $C \in (0, \infty)$. Then, $\max_{1 \leq i \leq n} |X_i| = o_p(n^{1/q})$.

Proof. For any $\varepsilon > 0$, we have

$$P\left(\max_{1 \leq i \leq n} |X_i| > \varepsilon n^{1/q}\right) \leq \sum_{1 \leq i \leq n} P(|X_i| > \varepsilon n^{1/q}) \leq \varepsilon^{-q} \mathbb{E}(|X_1|^q \mathbb{I}\{|X_1| > \varepsilon n^{1/q}\})$$

by a version of the Markov inequality and the i.i.d. assumption. As $n \rightarrow \infty$, the right-hand side tends to 0 by the dominated convergence theorem. $\square$

A.8 EM Algorithm

A.8.1 EM Algorithm for the model (1) with normal mixture density (2)–(3)

We introduce latent variables $D_i \in \{1, 2, \dots, M\}$ and $C_{it} \in \{1, 2, \dots, K\}$, where the top-level latent variable D_i indicates which of the M top-level components was chosen for observation i , and C_{it} indicates which of the K inner components was chosen for the t -th observation of unit i under the selected top-level component. Then, the complete-data log-likelihood is given as

$$\log L_c = \sum_{i=1}^n \sum_{j=1}^M \mathbb{I}(D_i = j) \left[\log \alpha_j + \sum_{t=1}^T \sum_{k=1}^{K_\epsilon} \mathbb{I}(C_{it} = k) (\log \tau_{jk} + \log \phi(Z_{jk,it}) - \log \sigma_j) \right],$$

where $Z_{jk,it} := \frac{Y_{it} - \mu_{jk} - \mathbf{x}_{it}^\top \boldsymbol{\beta}_j}{\sigma_j}$.

To obtain the penalised maximum a posteriori (MAP) estimator $\hat{\boldsymbol{\vartheta}} = \arg \max_{\boldsymbol{\vartheta} \in \bar{\Theta}(\mathbf{c})} \ell_n^{\text{pen}}(\boldsymbol{\vartheta})$, where $\ell_n^{\text{pen}} := \ell_n + p$ with penalty (76), we implement the following EM algorithm by alternating between the E-step and the M-step. The algorithm

is initialized with multiple random starting values and iterated until the relative change in log-likelihood falls below 10^{-6} or 5,000 iterations are reached, whichever comes first.

In the E-step, given current estimates $(\alpha_j^{(m)}, \theta_j^{(m)})$, compute the posterior probabilities of the latent variables as

$$\pi_{i,j}^{(m)} = P(D_i = j | \mathbf{W}_i, \{\alpha_j^{(m)}, \theta_j^{(m)}\}) = \frac{\alpha_j^{(m)} f(\mathbf{W}_i; \theta_j^{(m)})}{\sum_{r=1}^M \alpha_r^{(m)} f(\mathbf{W}_i; \theta_r^{(m)})} \text{ and}$$

$$\gamma_{jk,it}^{(m)} = P(C_{it} = k | \mathbf{W}_i, D_i = j, \theta_j^{(m)}) = \frac{\tau_{jk}^{(m)} \phi(z_{jk,it}^{(m)})}{\sum_{\ell=1}^{K_\epsilon} \tau_{j\ell}^{(m)} \phi(z_{j\ell,it}^{(m)})}$$

for each $j = 1, \dots, M$ and $k = 1, \dots, K$.

In the M-step, we maximise the expected complete-data penalised log-likelihood $E[\log L_c | \mathbf{W}, \alpha_j^{(m)}, \theta_j^{(m)}] + p(\boldsymbol{\vartheta})$ as follows: for $j = 1, \dots, M$,

1. Update α_j as

$$\alpha_j^{(m+1)} = \frac{\sum_{i=1}^n \pi_{i,j}^{(m)}}{n}, \quad \text{with } \sum_{j=1}^M \alpha_j^{(m+1)} = 1.$$

2. Update τ_{jk} as

$$\tau_{jk}^{(m+1)} = \frac{\sum_{i=1}^n \pi_{i,j}^{(m)} \sum_{t=1}^T \gamma_{jk,it}^{(m)}}{\sum_{i=1}^n \pi_{i,j}^{(m)} T} \quad \text{for } k = 1, \dots, K_\epsilon.$$

3. Update μ_j as

$$\mu_{jk}^{(m+1)} = \frac{\sum_{i,t} \pi_{i,j}^{(m)} \gamma_{jk,it}^{(m)} (Y_{it} - \mathbf{X}_{it}^\top \beta_j^{(m)})}{\sum_{i,t} \pi_{i,j}^{(m)} \gamma_{jk,it}^{(m)}} \quad \text{for } k = 1, \dots, K.$$

4. Given $\mu_j^{(m+1)}$, update β_j as

$$\beta_j^{(m+1)} = \arg \min_{\beta_j} \sum_{k=1}^{K_\epsilon} \sum_{i=1}^n \sum_{t=1}^T \pi_{i,j}^{(m)} \gamma_{jk,it}^{(m)} (Y_{it} - \mu_{jk}^{(m+1)} - \mathbf{x}_{it}^\top \beta_j)^2.$$

5. Update σ_j^2

$$\sigma_j^{(m+1)2} = \frac{\sum_{i=1}^n \pi_{i,j}^{(m)} \sum_{t=1}^T \sum_{k=1}^{K_\epsilon} \gamma_{jk,it}^{(m)} (Y_{it} - \mu_{jk}^{(m+1)} - \mathbf{x}_{it}^\top \beta_j)^2}{\sum_{i=1}^n \pi_{i,j}^{(m)} T}.$$

We iterate between the E-step and the M-step until convergence.

MAP modifications. Under the priors in Section 5.2, the M-step updates above are replaced by their MAP counterparts: (i) $\alpha_j^{(m+1)} \propto \sum_i \pi_{i,j}^{(m)} + a_\alpha - 1$, normalised to sum to one; (ii) $\tau_{jk}^{(m+1)} \propto \sum_{i,t} \pi_{i,j}^{(m)} \gamma_{jk,it}^{(m)} + a_\tau - 1$, normalised per component; (iii) $\sigma_j^{(m+1)^2} = \text{SS}_j / N_j^{\text{eff}}$, subject to $\sigma_j^2 \geq \delta \hat{s}^2$. The standard MLE formulas above correspond to $a_\alpha = a_\tau = 1$.

A.9 Information Criteria: AIC and BIC

We may also estimate the number of components by the penalised maximum likelihood estimator

$$\widehat{M}_{PL} = \arg \max_{M \in \{1, 2, \dots, \overline{M}\}} p \ell_n^M(\hat{\boldsymbol{\vartheta}}_M),$$

where

$$p \ell_n^M(\hat{\boldsymbol{\vartheta}}_M) := \ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - p_{n,k_M},$$

and $\hat{\boldsymbol{\vartheta}}_M$ is the MLE defined by (18) while p_{n,k_M} is a model-selection penalization term with $k_M := \dim(\boldsymbol{\vartheta}_M)$ representing the number of estimated parameters for mixture models (1). (The model-selection penalty p_{n,k_M} is distinct from the log-prior penalty $p(\boldsymbol{\vartheta})$ in (76) used for EM regularisation; see Section 5.2.)

The following proposition states that the penalised maximum likelihood estimator of the number of components is consistent under the regularity condition.

Assumption A.3. For all $n > 1$, the penalty function $p_{n,k}$ satisfies (a) $p_{n,k+1} \geq p_{n,k} > 0$, (b) $\lim_{n \rightarrow \infty} p_{n,k} = \infty$, (c) $p_{n,k} = o(n)$, and (d) $\lim_{n \rightarrow \infty} \frac{p_{n,k'}}{p_{n,k}} > 1$ for $k < k' \leq \overline{M}$.

Assumption A.4. For $M \in \{M_0 + 1, M_0 + 2, \dots, \overline{M}\}$, $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M) - \ell_n^{M_0}(\hat{\boldsymbol{\vartheta}}_{M_0}) = O_p(1)$.

Proposition A.5 (Consistent estimation by the maximum penalised likelihood estimator). Suppose that Assumptions 6-A.4 hold. Then, $\widehat{M}_{PL} \xrightarrow{p} M_0$.

Assumption A.3 corresponds to Assumption (C1) of Keribin (2000), specifying conditions for penalty functions. The BIC penalty function, given by $p_{n,k} = \frac{k}{2} \log(n)$, satisfies this assumption, whereas the AIC penalty function, $p_{n,k} = k$, does not satisfy Assumption A.3(b).

Assumption A.4 is a key high-level condition preventing overestimation. In Appendix A.2, we provide explicit conditions under which Assumption A.4 holds by extending the previous asymptotic analysis, originally developed for testing $H_0 : M = M_0$ against $H_1 : M = M_0 + 1$, to a more general setting of testing $H_0 : M = M_0$ against $H_1 : M = M_1$, where $M_1 > M_0 + 1$. Under this extension, we derive conditions ensuring that $\ell_n^{M_1}(\hat{\boldsymbol{\vartheta}}_{M_1}) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$ for $M > M_0 + 1$. The primary challenge encountered in

generalizing our earlier analysis is the increased severity of the singularity in the Fisher Information matrix, which necessitates expansions of order higher than second-order in the asymptotic analysis.

To address the singularity of the Fisher information matrix, we reparameterize the model and expand the density ratio $g_M(\mathbf{w}; \boldsymbol{\vartheta}_M)/g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*) - 1$ beyond the second order. This expansion yields a quadratic-form approximation of the log-likelihood difference $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M^j) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*)$ that captures higher-order terms necessary, building on a generalized version of Le Cam's differentiable in quadratic mean (DQM) expansion (Liu and Shao, 2003; Kasahara and Shimotsu, 2018). We then demonstrate that $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M^j) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$, analogously to arguments used in the proof of Proposition 6. Appendix A.2 contains the detailed discussion and Assumption A.2 presents the conditions for obtaining $\ell_n^M(\hat{\boldsymbol{\vartheta}}_M^j) - \ell_n^{M_0}(\boldsymbol{\vartheta}_{M_0}^*) = O_p(1)$. Assumption A.2 (b) is a key condition requiring that the unique elements of $\nabla_{\boldsymbol{\theta}_h} f(\mathbf{w}; \boldsymbol{\theta}_h^*)/g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*)$, $\nabla_{\boldsymbol{\theta}_h \otimes \boldsymbol{\theta}_h} f(\mathbf{w}; \boldsymbol{\theta}_h^*)/g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*)$, ..., $\nabla_{\boldsymbol{\theta}_h^{\otimes p_h}} f(\mathbf{w}; \boldsymbol{\theta}_h^*)/g_{M_0}(\mathbf{w}; \boldsymbol{\vartheta}_{M_0}^*)$ are linearly independent and their expectation is finite for $h = 1, 2, \dots, M_0$, where the value of p_h indicates a necessary order for identification.

The following proposition formalizes the results derived in Appendix A.2.

Proposition A.6. *Suppose that Assumptions 1, 3, 6, A.3, and A.2 hold. Then, Assumption A.4 holds, and $\widehat{M}_{PL} \xrightarrow{p} M_0$.*

Corollary 1 (Consistency of BIC and Inconsistency of AIC). *Suppose Assumptions 1, 3, 6, and A.2 hold. Then, (a) If $p_{n,k} = \frac{k}{2} \log(n)$ (BIC penalty), we have $\widehat{M}_{PL} \xrightarrow{p} M_0$. (b) If $p_{n,k} = k$ (AIC penalty), then $\lim_{n \rightarrow \infty} \Pr(\widehat{M}_{PL} > M_0) > 0$.*

Remark 4. *The higher-order linear independence condition in Assumption A.2(b) follows from the panel structure ($T \geq 2$) by the same argument as Proposition 2.*

Leroux (1992) established that, under conditions similar to Assumption A.3, the maximum penalised likelihood method produces an estimator that asymptotically does not underestimate the number of components. Building upon the work of Dacunha-Castelle and Gassiat (1999), Keribin (2000) derived regularity conditions necessary for the consistency of the maximum penalised likelihood estimator. Our Assumption A.2(b) corresponds to condition (P2) in Keribin (2000). However, Keribin's condition (P2) is not satisfied when the number of components M is moderately larger than the true number M_0 , as it relies on the sufficiency of a second-order expansion of the the density ratio for identification. In contrast, our Assumption A.2(b) accommodates settings where parameter identification requires a higher-order rank condition, thereby extending the applicability of the consistency result.

A.10 Nonparametric estimation of the lower bound of the number of components by the rank test

We also estimate the lower bound of the number of components without imposing the parametric assumption on error distributions by extending a method proposed by Kasahara and Shimotsu (2014) which in turn is based on the rank test of Kleibergen and Paap (2006).

We partition the support of $Y_t \in \mathcal{Y}_t$ into $|\Delta_t|$ mutually exclusive and exhaustive subsets $\Delta_t = \{\delta_1^t, \dots, \delta_{|\Delta_t|}^t\}$ so that $\mathcal{Y}_t = \cup_{i=1}^{|\Delta_t|} \delta_i$ and $\delta_i \cap \delta_j = \emptyset$ for $i \neq j$.

For each $k \in \mathcal{T} := \{1, 2, \dots, T\}$, define $\mathcal{T}_{-k} := \{s \in \mathcal{T} : s \neq t\} = \{1, \dots, t-1, t+1, \dots, T\}$ and let $\mathbf{Y}_{\mathcal{T}_{-k}} = (Y_1, \dots, Y_{k-1}, Y_{k+1}, \dots, Y_T)^\top$ be a vector of Y_t s in the group $\mathcal{T}_{-k}$. Partition the support of $\mathbf{Y}_{\mathcal{T}_{-k}}$ into $|\Delta_{\mathcal{T}_{-k}}|$ mutually exclusive and exhaustive subsets $\Delta_{\mathcal{T}_{-k}} = \{\delta_1^{\mathcal{T}_{-k}}, \dots, \delta_{|\Delta_{\mathcal{T}_{-k}}|}^{\mathcal{T}_{-k}}\}$. In particular, we construct this partition by unfolding the tensor product $\otimes_{t \neq k} \Delta_t$ using the Cartesian product denoted by $\odot$ as $\Delta_{\mathcal{T}_{-k}} = \odot_{t \neq k} \Delta_t = \Delta_1 \odot \dots \odot \Delta_{k-1} \odot \Delta_{k+1} \odot \dots \odot \Delta_T$ so that $|\Delta_{\mathcal{T}_{-k}}| = |\Delta_t|^{T-1}$.

For each $k \in \mathcal{T}$, we construct a $|\Delta_t| \times |\Delta_{\mathcal{T}_{-k}}|$ bivariate probability matrix $\mathbf{P}_k$ by arranging $\Pr(Y_k \in \delta_a, \mathbf{Y}_{-k} \in \delta_b^{\mathcal{T}_{-k}})$ for partition level $(a, b) = (1, 1), \dots, (|\Delta_t|, |\Delta_{\mathcal{T}_{-k}}|)$ as

$$\mathbf{P}_k = \begin{bmatrix} \Pr(Y_k \in \delta_1, \mathbf{Y}_{\mathcal{T}_{-k}} \in \delta_1^{\mathcal{T}_{-k}}) & \dots & \Pr(Y_k \in \delta_1, \mathbf{Y}_{\mathcal{T}_{-k}} \in \delta_{|\Delta_{\mathcal{T}_{-k}}|}^{\mathcal{T}_{-k}}) \\ \vdots & \ddots & \vdots \\ \Pr(Y_k \in \delta_{|\Delta_t|}, \mathbf{Y}_{\mathcal{T}_{-k}} \in \delta_1^{\mathcal{T}_{-k}}) & \dots & \Pr(Y_k \in \delta_{|\Delta_t|}, \mathbf{Y}_{-k} \in \delta_{|\Delta_{\mathcal{T}_{-k}}|}^{\mathcal{T}_{-k}}) \end{bmatrix}. \quad (100)$$

Collect the marginal probability distribution of Y_k and $\mathbf{Y}_{-k}$ over Δ_t and $\Delta_{\mathcal{T}_{-k}}$ conditional on being from the j -th component into a vector as

$$\begin{aligned} \mathbf{p}_k^j &:= (\Pr(Y_k \in \delta_1 | D = j), \dots, \Pr(Y_k \in \delta_{|\Delta_t|} | D = j))^\top \quad \text{and} \\ \mathbf{q}_k^j &:= (\Pr(\mathbf{Y}_{-k} \in \delta_1^{\mathcal{T}_{-k}} | D = j), \dots, \Pr(\mathbf{Y}_{-k} \in \delta_{|\Delta_{\mathcal{T}_{-k}}|}^{\mathcal{T}_{-k}} | D = j))^\top, \quad \text{respectively.} \end{aligned}$$

Then, under the conditional independence assumption as in the mixture model (1) but without imposing parametric restrictions, we may represent $\mathbf{P}_k$ as

$$\mathbf{P}_k = \sum_{j=1}^M \alpha^j \mathbf{p}_k^j (\mathbf{q}_k^j)^\top.$$

Kasahara and Shimotsu (2014) shows the rank of $\mathbf{P}_k$ identifies the lower bound of the number of components and develop a sequential hypothesis testing procedure for estimating the rank of $\mathbf{P}_k$ when the empirical quantile of the Y_t 's are used to construct the

partition. Because there are T possible ways to pick different k 's out of $\{1, \dots, T\}$, we test the maximum of the ranks of $\mathbf{P}_k$ across $k = 1, \dots, T$.

We first develop a rk-statistic of Kleibergen and Paap (2006) for testing the null hypothesis of $\text{rank}(\mathbf{P}_k) = r$. Write the singular value decomposition of $\mathbf{P}_k$ as

$$\mathbf{P}_k = \mathbf{U}^k \mathbf{S}^k (\mathbf{V}^k)^\top = \begin{bmatrix} \mathbf{U}_{11}^k & \mathbf{U}_{12}^k \\ \mathbf{U}_{21}^k & \mathbf{U}_{22}^k \end{bmatrix} \begin{bmatrix} \mathbf{S}_1^k & 0 \\ 0 & \mathbf{S}_2^k \end{bmatrix} \begin{bmatrix} \mathbf{V}_{11}^k & \mathbf{V}_{12}^k \\ \mathbf{V}_{21}^k & \mathbf{V}_{22}^k \end{bmatrix}^\top,$$

where $\mathbf{U}^k$ is a $|\Delta_t| \times |\Delta_t|$ orthonormal matrix, $\mathbf{V}^k$ is a $|\Delta_{\mathcal{T}-k}| \times |\Delta_{\mathcal{T}-k}|$ orthonormal matrix, and $\mathbf{S}^k$ is a $|\Delta_t| \times |\Delta_{\mathcal{T}-k}|$ diagonal matrix containing the singular values in decreasing order. In the partition of $\mathbf{U}^k$, $\mathbf{V}^k$, and $\mathbf{S}^k$ on the right-hand side, $\mathbf{U}_{11}^k$, $\mathbf{V}_{11}^k$, and $\mathbf{S}_1^k$ are $r \times r$, and the dimensions of the other submatrices are defined conformably. Then, the null hypothesis $\mathcal{H}_0 : \text{rank}(\mathbf{P}_k) = r$ is equivalent to $\mathcal{H}_0 : \mathbf{S}_2^k = 0$. The statistic of Kleibergen and Paap (2006) is based on an orthogonal transformation of $\mathbf{S}_2^k$ given by $\mathbf{A}_r^k = (\mathbf{A}_r^k)^\top \mathbf{P}_k \mathbf{B}_r^k$, where

$$\mathbf{A}_r^k = \begin{bmatrix} \mathbf{U}_{12}^k \\ \mathbf{U}_{22}^k \end{bmatrix} (\mathbf{U}_{22}^k)^{-1} (\mathbf{U}_{22}^k (\mathbf{U}_{22}^k)^\top)^{1/2} \quad \text{and} \quad \mathbf{B}_r^k = \begin{bmatrix} \mathbf{V}_{12}^k \\ \mathbf{V}_{22}^k \end{bmatrix} (\mathbf{V}_{22}^k)^{-1} (\mathbf{V}_{22}^k (\mathbf{V}_{22}^k)^\top)^{1/2}.$$

Let $\hat{\mathbf{P}}_k$ be a sample analogue estimator for $\mathbf{P}_k$ for which we have $\sqrt{n} \text{vec}(\hat{\mathbf{P}}_k - \mathbf{P}_k) \xrightarrow{d} N(0, \Sigma_k)$. The following proposition follows from Theorem 1 of Kleibergen and Paap (2006).

Proposition A.7. *Suppose that $\sqrt{n} \text{vec}(\hat{\mathbf{P}}_k - \mathbf{P}_k) \xrightarrow{d} N(0, \Sigma_k)$ and that $\Omega_r^k := ((\mathbf{B}_r^k)^\top \otimes (\mathbf{A}_r^k)^\top) \Sigma_k (\mathbf{B}_r^k \otimes \mathbf{A}_r^k)$ is non-singular. If $\text{rank}(\mathbf{P}_k) = r$, then $\sqrt{n} \hat{\boldsymbol{\lambda}}_r^k \rightarrow_d \mathcal{N}(0, \Omega_r^k)$ as $n \rightarrow \infty$, where $\hat{\boldsymbol{\lambda}}_r^k = \text{vec}((\hat{\mathbf{A}}_r^k)^\top \hat{\mathbf{P}}_k \hat{\mathbf{B}}_r^k)$.*

Kleibergen and Paap (2006) proposed the statistic called the rk-statistic:

$$\text{rk}^k(r) = n(\hat{\boldsymbol{\lambda}}_r^k)^\top (\hat{\Omega}_r^k)^{-1} \hat{\boldsymbol{\lambda}}_r^k,$$

where $\hat{\Omega}_r^k$ is a consistent estimator for Ω_r^k . If the assumptions of proposition A.7 hold, $\text{rk}(r)$ converges in distribution to a $\chi^2(|\Delta_t| - r)(|\Delta_{\mathcal{T}-k}| - r)$ random variable as $n \rightarrow \infty$.

When $T \geq 3$, we test the null hypothesis that $\text{rank}(\mathbf{P}_k) \leq r$ for each $k = 1, \dots, T$. By selecting partitions such that $|\Delta_t| = r + 1$, we define the following ave- and max-rk test statistics:

$$\text{ave-rk}(r) = \frac{1}{T} \sum_{k=1}^T \text{rk}^k(r) \quad \text{and} \quad \text{max-rk}(r) = \max\{\text{rk}^1(r), \dots, \text{rk}^T(r)\}.$$

See Section 3.4 of Kasahara and Shimotsu (2014) for the asymptotic distribution of these statistics.

In practice, we use the Bayesian bootstrap to obtain the bootstrap p-value for testing the rank, rejecting the null hypothesis $\text{rank}(\mathbf{P}_k) \leq r$ for all $k = 1, \dots, T$ at significance level α if the bootstrap p-value is strictly less than α ; see Appendix A.11. We estimate the lower bound for the number of components by applying the sequential hypothesis testing procedure described in Section 3.2 of Kasahara and Shimotsu (2014).

A.11 Bootstrap procedure for the ave- and max-rk statistic

We consider the following Bayesian bootstrap by drawing $\{\omega_i^{(b)}\}_{i=1}^n$ for $b = 1, \dots, B$, where $\omega_i^{(b)} = \tilde{\omega}_i^{(b)} / \sum_{j=1}^n \tilde{\omega}_j^{(b)}$ with $\tilde{\omega}_i^{(b)} \stackrel{iid}{\sim} \text{Exp}(1)$ for $i = 1, \dots, n$, and compute the bootstrap version of $\hat{\mathbf{P}}_k$ as

$$\hat{\mathbf{P}}_k^{(b)} = \begin{bmatrix} \sum_{i=1}^n \omega_i^{(b)} \mathbb{I}(Y_k \in \delta_1, \mathbf{Y}_{\mathcal{T}-k} \in \delta_1^{\mathcal{T}-k}) & \cdots & \sum_{i=1}^n \omega_i^{(b)} \mathbb{I}(Y_k \in \delta_1, \mathbf{Y}_{\mathcal{T}-k} \in \delta_{|\Delta_{\mathcal{T}-k}|}^{\mathcal{T}-k}) \\ \vdots & \ddots & \vdots \\ \sum_{i=1}^n \omega_i^{(b)} \mathbb{I}(Y_k \in \delta_{|\Delta_t|}, \mathbf{Y}_{\mathcal{T}-k} \in \delta_1^{\mathcal{T}-k}) & \cdots & \sum_{i=1}^n \omega_i^{(b)} \mathbb{I}(Y_k \in \delta_{|\Delta_t|}, \mathbf{Y}_{-k} \in \delta_{|\Delta_{\mathcal{T}-k}|}^{\mathcal{T}-k}) \end{bmatrix}$$

for $k = 1, \dots, T$. Note that we use the same weights across different k s to accomodate the dependencies across k s so that the bootstrapped correlation between $\hat{\mathbf{P}}_k^{(b)}$ and $\hat{\mathbf{P}}_\ell^{(b)}$ for $k \neq \ell$ correctly captures the corresponding sample correlation between $\hat{\mathbf{P}}_k$ and $\hat{\mathbf{P}}_\ell$.

With this bootstrapped $\{\hat{\mathbf{P}}_k^{(b)}\}_{k=1}^T$ for $b = 1, \dots, B$, we construct the bootstrapped rk-statistics as

$$\text{rk}^{k(b)}(r) = n(\hat{\boldsymbol{\lambda}}_r^{k(b)} - \hat{\boldsymbol{\lambda}}_r^k)^\top (\hat{\boldsymbol{\Omega}}_r^{k(b)})^{-1} (\hat{\boldsymbol{\lambda}}_r^{k(b)} - \hat{\boldsymbol{\lambda}}_r^k) \quad \text{for } k = 1, \dots, T, \quad (101)$$

where $\hat{\boldsymbol{\lambda}}_r^{k(b)}$ and $\hat{\boldsymbol{\Omega}}_r^{k(b)}$ are computed as described in Proposition A.7 but we compute $\hat{\mathbf{A}}_r^k$ and $\hat{\boldsymbol{\Sigma}}_k$ using $\hat{\mathbf{P}}_k^{(b)}$ in place of $\hat{\mathbf{P}}_k$.

Based on $\{\text{rk}^{k(b)}(r)\}_{k=1}^T$ derived as above, we can construct bootstrapped ave- and max-rk-statistics as:

$$\text{ave-rk}^{(b)}(r) = \frac{1}{T} \sum_{k=1}^T \text{rk}^{k(b)}(r) \quad \text{and} \quad \text{max-rk}^{(b)}(r) = \max\{\text{rk}^{1(b)}(r), \dots, \text{rk}^{T(b)}(r)\} \quad \text{for } b = 1, 2, \dots, B.$$

We then compute the bootstrap p-value as the empirical proportion of the bootstrapped test statistics $\text{max-rk}^{(b)}(r)$ that equal or exceed the observed statistic $\text{max-rk}(r)$: $\text{bootstrap p-value} = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(\text{max-rk}^{(b)}(r) \geq \text{max-rk}(r))$. We reject the null hypothesis that $\text{rank}(\mathbf{P}_k) \leq r$ for all $k = 1, \dots, T$ at significance level α if the bootstrap p-value is

strictly less than α .

B Additional Tables and Figures

B.1 Controlled-DGP Validation: Size and Power

Table B1 presents size and power for testing $H_0: M_0 = 2$ vs $H_1: M_0 = 3$ under controlled DGPs with conditionally independent errors, based on 500 simulations with $T = 3$. The size DGP uses heterogeneous variances $(\sigma_1, \sigma_2) = (0.8, 1.2)$ to evaluate size under a realistic null where components differ in both location and scale; the power DGP uses homogeneous variances $(\sigma_1, \sigma_2, \sigma_3) = (1, 1, 1)$ to isolate the effect of mean separation on detection power, without confounding from variance differences.

Table B1: Size and Power for testing $H_0: M_0 = 2$ vs $H_1: M_0 = 3$ at the 5% level under controlled DGPs

n	Component means	ave-rk	max-rk	LR	AIC	BIC
<i>Panel A: Size — DGP is two-component, $(\sigma_1, \sigma_2) = (0.8, 1.2)$, $\alpha = 0.5$</i>						
200	$(\mu_1, \mu_2) = (-1, 1)$	3.4	6.0	4.6	9.6	0.0
200	$(\mu_1, \mu_2) = (-0.5, 0.5)$	0.2	0.2	4.6	7.2	0.0
400	$(\mu_1, \mu_2) = (-1, 1)$	5.8	6.4	4.2	7.6	0.0
400	$(\mu_1, \mu_2) = (-0.5, 0.5)$	0.0	0.0	4.6	6.8	0.0
<i>Panel B: Power — DGP is three-component, $\sigma_j = 1$, $\alpha_j = 1/3$</i>						
200	$(\mu_1, \mu_2, \mu_3) = (-0.5, 0, 1.5)$	0.6	0.8	12.0	15.4	0.2
200	$(\mu_1, \mu_2, \mu_3) = (-1.5, 0, 1.5)$	50.8	37.2	99.8	88.6	85.6
400	$(\mu_1, \mu_2, \mu_3) = (-0.5, 0, 1.5)$	3.0	3.2	14.4	23.4	0.0
400	$(\mu_1, \mu_2, \mu_3) = (-1.5, 0, 1.5)$	92.8	81.2	99.8	89.2	100.0

Notes: 500 replications, $T = 3$, conditionally independent normal errors (model 1 with (2)–(3)). Rejection frequencies (%) at 5% level; AIC/BIC report % selecting $M = 3$ over $M = 2$.

Size (Panel A): LRT maintains near-nominal 5% rejection rates (4.2–4.6%) across all scenarios. Rank tests (ave-rk, max-rk) achieve nominal size only with large samples ($n = 400$) and well-separated means, but severely underperform with close means. AIC overestimates (7–10%), while BIC remains conservative (0%).

Power (Panel B): LRT achieves near-100% power with well-separated means ($\mu = (-1.5, 0, 1.5)$) and maintains superior power (12–14%) even with close means, vastly outperforming rank tests (0–3%). BIC matches LRT under clear separation but becomes overly conservative as means converge. AIC consistently overestimates across all scenarios.

B.2 Simulation: Sensitivity to Mixing Proportions

Table B2 examines how model selection accuracy varies with mixing proportions, holding all other DGP parameters at Metal Products (CIIU 381) estimates. Two mixing vectors are considered: the empirical weights $\alpha = (0.352, 0.402, 0.245)$ and an unbalanced design $\alpha = (0.60, 0.30, 0.10)$ in which the smallest component has 10% weight.

Table B2: Model selection accuracy (%) under DGPs with varying mixing weights ($M_0 = 3$, Metal Products, $n = 225$, $T = 3$)

DGP α	Method	Normal estimation ($K_\epsilon = 1$)			Mixture estimation ($K_\epsilon = 2$)		
		<	=	>	<	=	>
<i>Panel A: Normal-error DGP ($K_\epsilon = 1$)</i>							
(0.352, 0.402, 0.245)	BIC	0	100	0	0	100	0
	LR (0.01)	0	99	1	3	97	0
	LR (0.05)	0	95	5	0	93	7
	LR (0.10)	0	93	7	0	91	9
(0.60, 0.30, 0.10)	BIC	11	85	4	46	54	0
	LR (0.01)	5	91	4	18	82	0
	LR (0.05)	5	89	6	6	88	6
	LR (0.10)	5	88	7	3	90	7
<i>Panel B: Mixture-error DGP ($K_\epsilon = 2$)</i>							
(0.352, 0.402, 0.245)	BIC	5	95	0	45	55	0
	LR (0.01)	1	99	0	6	94	0
	LR (0.05)	0	93	7	1	95	4
	LR (0.10)	0	91	9	0	89	11
(0.60, 0.30, 0.10)	BIC	67	33	0	93	7	0
	LR (0.01)	23	75	2	72	28	0
	LR (0.05)	15	79	6	45	53	2
	LR (0.10)	11	80	9	38	58	4

Notes: 100 replications, $M_0 = 3$, $T = 3$, $n = 225$. Only mixing proportions α vary; all other parameters held at Metal (CIIU 381) estimates. Baseline regularisation: variance floor $\sigma_j^2 \geq 0.05\hat{s}^2$, Dirichlet(3) penalty on α_j and τ_{jk} . “</=>”: under/correct/over-selection rates (%).

With empirical mixing weights (near-balanced), BIC and LRT achieve high accuracy under both DGPs. With the unbalanced design ($\alpha_{\min} = 0.10$), performance degrades: under the mixture-error DGP, mixture estimation struggles (BIC correct: 7%), while normal estimation remains substantially better (LRT 0.01: 75%). This reflects a power limitation

from the high parameter count of $K_\epsilon = 2$ at $n = 225$; in our empirical application the sample sizes are larger ($N = 157\text{--}669$, $T = 10$), and the empirical mixing weights are approximately balanced, so this degradation is not a practical concern.

B.3 Realistic-DGP Simulation: Full Results

Table B3: Full selection frequencies (%) under mixture-error DGP ($M_0 = 3$, $K_\epsilon = 2$, $T = 3$, $n = 277$, Metal industry)

Methods (q_n , Error Density)	$M=1$	$M=2$	$M=3$	$M\geq 4$
AIC (Normal)	0	0	1	99
BIC (Normal)	0	0	24	76
LR (0.01, Normal)	0	0	18	82
LR (0.05, Normal)	0	0	12	88
LR (0.10, Normal)	0	0	11	89
AIC (Mixture)	0	0	76	24
BIC (Mixture)	0	0	98	2
LR (0.01, Mixture)	0	13	86	1
LR (0.05, Mixture)	0	1	97	2
LR (0.10, Mixture)	0	1	97	2
ave-rk (0.05)	0	49	51	0
max-rk (0.05)	0	66	34	0

Notes: Based on 100 simulation repetitions with $n = 277$ and $T = 3$. DGP calibrated to Chilean fabricated metal products data with normal mixture errors ($K_\epsilon = 2$). True number of components $M_0 = 3$. The $M\geq 4$ column combines $M = 4, 5, 6$. **LRT**: sequential parametric bootstrap ($B = 199$) at the stated significance level. **ave-rk** / **max-rk**: nonparametric rank test (single step, $H_0 : M \leq 2$); reject $\Rightarrow \hat{M} = 3$. See Table 1 for the normal-error DGP counterpart.

Table B4: Bootstrap LRT size across model specifications ($H_0 : M = 2$ vs $H_1 : M = 3$, $n = 277$, $T = 5$, Metal industry)

Case	Dynamics	K_ϵ	Rej. 1%	Rej. 5%	Rej. 10%	Mean LRT
1	IID	1 (Normal)	0.5%	4.5%	8.5%	15.81
2	IID	2 (Mixture)	1.5%	5.0%	9.0%	16.52

Notes: Empirical rejection rates of the sequential bootstrap LRT under $H_0 : M_0 = 2$ across all four model specifications in Section 2. DGP parameters calibrated to Metal Products (CIU 381) estimates under each specification. $n = 200$ Monte Carlo replications, each with $B = 199$ bootstrap draws. Baseline regularisation: variance floor $\sigma_j^2 \geq 0.05\hat{s}^2$, α -penalty ($a_\alpha = 3$), τ -penalty ($a_\tau = 1.5$). All rejection rates are close to nominal levels, confirming bootstrap validity across specifications.

Table B5: Bootstrap LRT size: effect of hard parameter bounds ($H_0: M = 2$ vs $H_1: M = 3$, $K_\epsilon = 2$)

Setting	Rejection rate at nominal level		
	1%	5%	10%
Panel A: Mixing proportion (α)			
Bound: $\alpha_j \in [0.05, 0.95]$	1%	1%	5%
No bound	2%	4%	10%
Panel B: Variance floor (σ^2)			
Floor: $\sigma_j^2 \geq 0.05 \hat{s}^2$	3%	4%	9%
No floor	3%	10%	13%
Panel C: Sub-component weight (τ)			
<i>DGP $\tau = 0.50$ (interior)</i>			
Bound: $\tau_{jk} \in [0.05, 0.95]$	0%	4%	7%
No bound	1%	6%	9%
<i>DGP $\tau = 0.25$</i>			
Bound: $\tau_{jk} \in [0.05, 0.95]$	1%	3%	5%
No bound	1%	4%	9%
<i>DGP $\tau = 0.10$ (near boundary)</i>			
Bound: $\tau_{jk} \in [0.05, 0.95]$	1%	6%	10%
No bound	1%	7%	14%

Notes: Size of the bootstrap sequential LRT under $H_0: M = 2$ vs $H_1: M = 3$. DGP is a hand-specified $M = 2$ mixture with $K_\epsilon = 2$ and all parameters in the interior of the parameter space. All configurations include the default regularisation: variance floor $\sigma_j^2, \sigma_{1,j}^2 \geq 0.05 \hat{s}^2$ (only the variance is hard-bounded), and the logarithmic penalties α -penalty ($a_\alpha = 3$) and τ -penalty ($a_\tau = 1.5$) that keep α_j and τ_{jk} away from zero without imposing a separate hard bound. Each panel adds or removes one hard bound on top of this baseline. Rejection rates from 100 Monte Carlo replications, each with $B = 199$ bootstrap draws; binomial standard errors for the 5% rejection rate range from ± 0.010 to ± 0.030 . $\hat{s}^2$: sample variance of the dependent variable.

Table B6: Robustness: selected $\hat{M}$ under three-component within-type errors ($K_\epsilon = 3$, no CRE)

Model	Scheme	AIC			BIC			LRT			LRT ($\tau \geq 0.05$)		
		M	F	T	M	F	T	M	F	T	M	F	T
Cobb-Douglas	1	9	11	7	5	8	5	5	8	7	5	5	5
Translog	1	4	8	6	3	3	3	3	3	3	3	2	4

Notes: $\hat{M}$ selected under $K_\epsilon = 3$ (three-component within-type error). M = Metal (381), F = Food (311), T = Textiles (321). Last three columns impose a lower bound $\tau_{jk} \geq 0.05$ to prevent degeneracy of the third sub-component. Without the τ bound, the third sub-component can receive near-zero weight ($\tau \rightarrow 0$), collapsing the three-component error into an effectively two-component specification. BIC selections under $K_\epsilon = 3$ are comparable to the $K_\epsilon = 2$ baseline (Table 3); LRT shows more sensitivity, with the τ bound reducing over-selection for Cobb-Douglas Food (from 8 to 5).

B.4 Additional Empirical Results

This appendix collects supplementary results for the empirical analysis in Section 7 of the main paper, organised into seven margins. Section B.4.1 reports the effect of Mundlak CRE augmentation on model selection. Section B.4.2 documents the input winsorisation robustness check (5/95 percentile vs. raw; and the rationale for not extending to 1/99). Section B.4.3 compares Normal and Mixture error distributions. Section B.4.4 gives short-panel ($T=5$) model selection, per-type Step 1 coefficients, and a cross-panel elasticity comparison. Section B.4.5 gives the full per-type Step 1 nonlinear coefficient table at $T=10$. Section B.4.6 discusses the misclassification bias arising from the two-step classify-then-GNR procedure. Section B.4.7 explains why hard plant-to-type assignment (binarization) is not used.

B.4.1 Correlated Random Effects (Mundlak) Augmentation

The main results treat firm fixed effects as random noise captured by the EM residual structure. The Correlated Random Effects (CRE) extension (Mundlak, 1978) projects each firm's time-average log input $\bar{k}_i$ onto the regression index, adding a common (across-type) coefficient δ on $\bar{k}_i$ to absorb potential correlation between the unobserved type assignment and firm-level capital intensity. This is equivalent to a within-type Mundlak correction and leaves the EM identification strategy unchanged.

Table B7 reports $\hat{M}$ from the sequential bootstrap LRT under the Nonlinear Mixture ($K_\epsilon=2$) + CRE specification at $T=10$, with the no-CRE baseline reproduced for comparison.

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Table B7: Sequential bootstrap LRT $\hat{M}$ under the Nonlinear Mixture ($K_\epsilon=2$) specification with and without Mundlak CRE augmentation, $T=10$, $B=199$, $\alpha=0.05$.

Spec	No CRE (baseline)			+ CRE		
	Metal	Food	Textiles	Metal	Food	Textiles
CD ($q=0$) $\times$ Mixture	3	3	4	2	4	5
Linear ($q=3$) $\times$ Mixture	4	4	5	2	3	2
Nonlinear ($q=3$) $\times$ Mixture	2	6	2	3	4	2

Notes. CRE augmentation adds the firm-mean $\bar{k}_i$ as a shared (across-type) regressor with a common coefficient δ (shared-Mundlak design). Raw (no-winsor) data. Source: [results/empirical_test_2026-06-07_T10_n50_h0free_lightH1/](#).

Adding CRE reduces $\hat{M}$ by 0–2 for the less flexible Cobb-Douglas and Linear specifications (which were absorbing capital-correlated heterogeneity into extra types), while the Nonlinear Mixture specification shifts modestly: Metal increases from 2 to 3, Food decreases from 6 to 4, and Textiles stays at 2. The reduction in Food under CRE (6→4) is consistent with the Mundlak regressor absorbing some correlated capital-technology variation that was driving the high no-CRE selection. The qualitative conclusion — multiple latent types are selected in all three industries — is unchanged.

Phase 2 elasticities under CRE. The main paper’s Phase 2 results use the no-CRE, no-winsor baseline ($\hat{M}$: Metal 2, Food 6, Textiles 2 at $T=10$). Under CRE, the anchors shift to Metal 3, Food 4, Textiles 2 (Table B7). The per-type GNR Step 1 estimates in Table B13 use the v5 winsor+CRE anchors from an earlier run; these will be updated to the no-winsor no-CRE baseline in a future revision. The qualitative patterns are: Step 1 β_0 heterogeneity spans 0.10–0.40 across types, $\beta_{MM} < 0$ and $\beta_{KM} > 0$ everywhere, and Step 2 capital elasticities have wide CIs on thin types.

B.4.2 Input Winsorisation Robustness

The baseline in the main paper uses the *raw* (*no-winsor*) panel. This section documents a robustness check where we winsorise the centred log inputs ($\tilde{k}_{it}, \tilde{\ell}_{it}, \tilde{m}_{it}$) at the 5/95 percentile (across the pooled panel; thresholds fixed pre-bootstrap) and re-run the full pipeline — model selection and Phase 2 GNR — on the winsorised data. A small number of extreme plants exert disproportionate leverage on the per-type GNR Step 2 GMM fit; winsorisation assesses whether these observations drive the baseline $\hat{M}$ selections.

Effect on $\hat{M}$ selection. Table B8 reports $\hat{M}$ side-by-side for the no-winsor and winsor 5/95 baselines across all 16 specifications.

Table B8: Sequential bootstrap LRT $\hat{M}$, no-winsor vs. 5/95 winsor baseline. Same EM pipeline; only data preparation differs. Bold rows: preferred Mixture specification at each flexibility level.

Panel A: $T = 5$

Spec	Error	CRE	No winsor			Winsor 5/95		
			Me	Fo	Tx	Me	Fo	Tx
CD ($q=0$)	Normal	–	8	10	9	3	4	4
CD ($q=0$)	Normal	✓	7	9	7	4	5	4
CD ($q=0$)	Mixture	–	3	4	5	4	5	4
CD ($q=0$)	Mixture	✓	2	4	4	2	5	4
Linear ($q=3$)	Normal	–	6	2	5	2	5	4
Linear ($q=3$)	Normal	✓	1	2	3	3	2	2
Linear ($q=3$)	Mixture	–	3	2	4	3	2	3
Linear ($q=3$)	Mixture	✓	4	1	2	2	2	2
Nonlinear ($q=3$)	Normal	–	3	7	2	4	6	5
Nonlinear ($q=3$)	Normal	✓	2	5	2	4	4	5
Nonlinear ($q=3$)	Mixture	–	2	7	3	4	5	4
Nonlinear ($q=3$)	Mixture	✓	3	4	2	3	4	2

Panel B: $T = 10$

Spec	Error	CRE	No winsor			Winsor 5/95		
			Me	Fo	Tx	Me	Fo	Tx
CD ($q=0$)	Normal	–	9	7	9	8	4	5
CD ($q=0$)	Normal	✓	6	8	6	7	4	3
CD ($q=0$)	Mixture	–	3	3	4	3	3	5
CD ($q=0$)	Mixture	✓	2	4	5	2	3	3
Linear ($q=3$)	Normal	–	6	2	6	5	5	4
Linear ($q=3$)	Normal	✓	3	3	3	3	2	2
Linear ($q=3$)	Mixture	–	4	4	5	5	3	3
Linear ($q=3$)	Mixture	✓	2	3	2	2	3	3
Nonlinear ($q=3$)	Normal	–	4	7	2	4	4	8
Nonlinear ($q=3$)	Normal	✓	5	7	2	8	4	10
Nonlinear ($q=3$)	Mixture	–	2	6	2	3	3	4
Nonlinear ($q=3$)	Mixture	✓	3	4	2	5	4	3

Notes. No-winsor source: `results/empiricaltest_2026-06-07-T{5,10}_n50_h0free_lightH1`. Winsor 5/95 source: `results/empiricaltest_2026-06-11-T{5,10}_winsor_all16_lightH1`. Same EM pipeline throughout (H_0 unconstrained, H_1 light penalty, $B=199$, $\alpha=0.05$). Winsorisation has the largest effects under Cobb-Douglas and Linear Normal, where high-leverage plants were driving implausibly large $\hat{M}$ (≥ 8). The Nonlinear Mixture row (the paper’s baseline) gives $\hat{M} \in \{2, 6, 2\}$ (Metal, Food, Textiles) at $T=10$ and $\{2, 7, 3\}$ at $T=5$ on raw data. Winsorisation shifts these to $\{3, 3, 4\}$ and $\{3, 4, 5\}$ respectively, reducing the Food $\hat{M}$ substantially — consistent with a small number of extreme observations inflating the Food selection.

Effect on Pool $M=1$ elasticity CIs. Winsorisation fixes three borderline zero-coverage cases in the Pool $M=1$ capital elasticity: (i) Metal $T=10$ Cobb-Douglas

($[-0.017, +0.108] \rightarrow [+0.016, +0.120]$); (ii) Textiles $T=5$ Python GNR ($[-0.014, +0.160] \rightarrow [+0.031, +0.163]$); (iii) Metal $T=10$ Python GNR improves from -0.092 to -0.015 on the lower bound ($[-0.092, +0.278] \rightarrow [-0.015, +0.278]$), though still nominally containing zero. Labour elasticity CIs are negligibly affected, consistent with labour being well-identified regardless of input-tail outliers.

Why 1/99 winsorisation was not run. A tighter 1/99 percentile threshold removes fewer observations but might still reduce the residual leverage of the most extreme 2–3% of the distribution. We did not run the 1/99 grid because the full 16-specification grid (3 flexibility levels $\times$ 2 error distributions $\times$ 2 CRE settings $\times$ 2 panel lengths) with $B=199$ bootstrap replications takes approximately 18–24 hours per pass on the available hardware. Running an additional pass at 1/99 would double this cost without a clear prior that the five affected plants (roughly $N \times 0.01=5-8$ per cell) drive the remaining instabilities. Given that the 5/95 winsor produces $\hat{M}$ values within two of the no-winsor baseline in most cells, and that the key robustness conclusion (meaningful latent heterogeneity across all industries) is unchanged, the marginal information from a 1/99 run was judged insufficient to justify the computational cost.

B.4.3 Normal vs. Two-Component Mixture Error Distribution

The main analysis assumes a $K_\epsilon=2$ normal mixture for the within-type residual u_{it} , which allows heteroskedastic residual shapes within each technology type. Table B9 compares $\hat{M}$ selected under Normal ($K_\epsilon=1$, standard Gaussian) and Mixture ($K_\epsilon=2$) residuals at the Nonlinear FOC specification, both with and without CRE, for $T=10$.

Table B9: $\hat{M}$ under Normal vs. Mixture residuals, Nonlinear inside-log specification, $T=10$, raw (no-winsor) data.

CRE	Normal ($K_\epsilon=1$)			Mixture ($K_\epsilon=2$)		
	Metal	Food	Textiles	Metal	Food	Textiles
No CRE	4	7	2	2	6	2
+ CRE	5	7	2	3	4	2

Notes. Nonlinear inside-log specification. Source: `results/empirical_test_2026-06-07.T10_n50_h0free_lightH1`

Under Normal errors, Food persistently selects $\hat{M}=7$ regardless of CRE, while Mixture errors reduce this to $\hat{M}=6$ (no CRE) or 4 (CRE). The Mixture error model is preferred because it separates within-type residual non-Gaussianity from across-type count, avoiding spurious inflation of $\hat{M}$ seen in Normal specifications.

Normal errors substantially over-select $\hat{M}$ in Textiles at $T=10$ (8 without CRE, 10 with

CRE, versus 4 and 3 under Mixture), indicating that the Textiles share residual has a bi-modal or heavy-tailed marginal distribution that a Normal single-type model absorbs by adding spurious types. The Mixture residual structure ($K_\epsilon=2$) absorbs this within-type non-Gaussianity explicitly, preventing it from inflating the type count. Metal and Food are less sensitive: Mixture gives $\hat{M}$ 0–2 below Normal, a more modest correction. The Mixture error specification is therefore the preferred baseline and is more conservative in the sense of selecting fewer latent types.

B.4.4 Short Panel Robustness: $T = 5$ Results

The main results focus on the $T=10$ panel (1987–1996), which provides a longer earnings window for the cubic-Markov GMM Step 2. Here we report model selection, per-type Step 1 coefficients, and a cross-panel aggregate elasticity comparison for the $T=5$ panel (1992–1996), which has larger samples ($N_{\text{Food}}=862$, $N_{\text{Metal}}=242$, $N_{\text{Tex}}=203$) but a shorter time dimension.

Table B10: Sequential bootstrap LRT $\hat{M}$ at $T = 5$, raw (no-winsor) data. Panel A: without CRE; Panel B: with CRE. ($N_{\text{Food}} = 862$, $N_{\text{Metal}} = 242$, $N_{\text{Tex}} = 203$, $B=199$, $\alpha=0.05$.)

Panel A: Without CRE

Spec (per-type q)	Error dist.	Metal	Food	Textiles
CD ($q=0$)	Normal	8	10	9
CD ($q=0$)	Mixture	3	4	5
Linear ($q=3$)	Normal	6	2	5
Linear ($q=3$)	Mixture	3	2	4
Nonlinear [‡] ($q=3$)	Normal	3	7	2
Nonlinear[‡] ($q=3$)	Mixture	2	7	3

Panel B: With Mundlak CRE augmentation

Spec (per-type q)	Error dist.	Metal	Food	Textiles
CD ($q=0$) + CRE	Normal	7	9	7
CD ($q=0$) + CRE	Mixture	2	4	4
Linear ($q=3$) + CRE	Normal	1	2	3
Linear ($q=3$) + CRE	Mixture	4	1	2
Nonlinear [‡] ($q=3$) + CRE	Normal	2	5	2
Nonlinear[‡] ($q=3$) + CRE	Mixture	3	4	2

Notes. [‡]Nonlinear uses the inside-log design. Bold Mixture rows are preferred at each flexibility level.

Source: results/empirical_test_2026-06-07.T5_n50_h0free_lightH1/.

$T = 5$ **Step 1 coefficients.** Table B11 reports the per-type Step 1 nlbeta at $T=5$ under the Nonlinear Mixture ($K_\epsilon=2$) specification, conditioned on the CRE+winsor $\hat{M}$ anchors (Food $T=5$: $\hat{M}=4$; Metal $T=5$: $\hat{M}=3$; Textiles $T=5$: $\hat{M}=2$).

Table B11: Step 1 inside-log nonlinear coefficients per type, $T=5$, with Phase 2 v5 cluster-bootstrap conditional 95% CIs. Type-rank sorted by data $\bar{\epsilon}_M$ ascending.

Source	const (β_0)	β_{MM}	β_{LM}	β_{KM}
Food CIU 311, $T = 5$, $N = 862$, $\hat{M} = 4$				
Type 1 ($\hat{\alpha} = 0.185$, $N = 146$)				
point estimate	+0.567	−0.089	−0.032	+0.181
cond. 95% CI	[+0.266, +0.657]	[−0.100, −0.009]	[−0.075, −0.001]	[+0.113, +0.201]
Type 2 ($\hat{\alpha} = 0.377$, $N = 336$)				
point estimate	+0.652	−0.069	−0.007	+0.168
cond. 95% CI	[+0.519, +0.734]	[−0.101, −0.011]	[−0.074, −0.001]	[+0.120, +0.189]
Type 3 ($\hat{\alpha} = 0.394$, $N = 342$)				
point estimate	+0.769	−0.040	−0.001	+0.175
cond. 95% CI	[+0.557, +0.794]	[−0.067, +0.001]	[−0.067, +0.004]	[+0.123, +0.185]
Type 4 ($\hat{\alpha} = 0.044$, $N = 38$)				
point estimate	+0.689	+0.000	+0.000	+0.120
cond. 95% CI	[+0.619, +0.777]	[−0.047, +0.032]	[−0.061, +0.001]	[+0.050, +0.170]
Metal CIU 381, $T = 5$, $N = 242$, $\hat{M} = 3$				
Type 1 ($\hat{\alpha} = 0.280$, $N = 65$)				
point estimate	+0.369	−0.081	−0.061	+0.147
cond. 95% CI	[+0.366, +0.479]	[−0.238, −0.052]	[−0.098, +0.023]	[+0.125, +0.218]
Type 2 ($\hat{\alpha} = 0.288$, $N = 66$)				
point estimate	+0.562	−0.125	−0.090	+0.214
cond. 95% CI	[+0.417, +0.615]	[−0.257, −0.021]	[−0.114, +0.056]	[+0.103, +0.264]
Type 3 ($\hat{\alpha} = 0.433$, $N = 111$)				
point estimate	+0.469	−0.136	−0.094	+0.227
cond. 95% CI	[+0.449, +0.666]	[−0.230, −0.000]	[−0.149, −0.000]	[+0.073, +0.287]
Textiles CIU 321, $T = 5$, $N = 203$, $\hat{M} = 2$				
Type 1 ($\hat{\alpha} = 0.472$, $N = 96$)				
point estimate	+0.505	−0.109	−0.057	+0.217
cond. 95% CI	[+0.476, +0.538]	[−0.206, −0.079]	[−0.082, −0.007]	[+0.185, +0.249]
Type 2 ($\hat{\alpha} = 0.528$, $N = 107$)				
point estimate	+0.674	−0.125	−0.056	+0.228
cond. 95% CI	[+0.627, +0.708]	[−0.248, −0.041]	[−0.072, +0.003]	[+0.174, +0.245]

Notes. Phase 2 v5 cluster-bootstrap ($B_{\text{out}}=300$ firm-level resamples). Textiles $T=5$ has the sharpest CIs of any cell: the two-type separation at $\beta_0=0.505$ vs. 0.674 is estimated to within ± 0.04 (conditional). Food $T=5$ Types 2–3 dominate by weight (38% and 39%) and have CIs tighter than the comparable $T=10$ types.

$T = 5$ **vs. $T = 10$ model selection.** The selected $\hat{M}$ values under the Nonlinear Mixture (no CRE): Food 7 vs. 6, Metal 2 vs. 2, Textiles 3 vs. 2. The $T=5$ Food selection (7) is higher than $T=10$ (6), consistent with two mechanisms: (i) the larger $T=5$ samples (30–80% more firms) give the LRT more power to detect additional types; and (ii) the $T=5$ sub-panel (1992–1996) covers a narrower and potentially more heterogeneous time window than $T=10$ (1987–1996), which pools across a longer growth trajectory and may merge types

that are distinct only in the early-to-mid 1990s transition period. Under CRE, the Nonlinear Mixture gives Metal $T=5$: $\hat{M}=3$, Food $T=5$: $\hat{M}=4$, Textiles $T=5$: $\hat{M}=2$ — all below the no-CRE $T=5$ values. The convergence of CRE selections relative to no-CRE suggests that some of the no-CRE $\hat{M}$ is driven by unabsorbed capital-productivity correlation.

$T = 5$ vs. $T = 10$ aggregate elasticity comparison. Table B12 reports the $\hat{\alpha}$ -weighted aggregate GNR Step 2 elasticities at $T=5$ and $T=10$ from the Phase 2 v5 cluster-bootstrap. The cross-panel comparison reveals strong stability in materials and returns-to-scale elasticities: Food's $\bar{\varepsilon}_M$ is +0.682 at $T=5$ and +0.698 at $T=10$; Textiles' RTS is +1.061 and +1.037. The capital elasticity aggregate is more variable because a single thin type with an atypical ε_K can shift the weighted average (Metal Type 1 at $T=5$: $\varepsilon_K = -0.439$, $\hat{\alpha}=0.28$, $N=65$). Nevertheless, the sign pattern (Food: positive ε_K ; Metal: near-zero or negative; Textiles: positive) is consistent across panel lengths.

Table B12: $\hat{\alpha}$ -weighted aggregate GNR Step 2 elasticities at $T=5$ and $T=10$, Phase 2 v5 cluster-bootstrap. CRE+winsor $\hat{M}$ anchors: Food $T=5$: 4, $T=10$: 4; Metal $T=5$: 3, $T=10$: 5; Textiles $T=5$: 2, $T=10$: 3.

Industry	Panel	$\bar{\varepsilon}_K$	$\bar{\varepsilon}_L$	$\bar{\varepsilon}_M$	RTS
Food CIIU 311	$T=5$ ($\hat{M}=4$)	+0.082	+0.160	+0.682	+0.924
	$T=10$ ($\hat{M}=4$)	+0.095	+0.153	+0.698	+0.945
Metal CIIU 381	$T=5$ ($\hat{M}=3$)	-0.039	+0.415	+0.463	+0.839
	$T=10$ ($\hat{M}=5$)	-0.007	+0.334	+0.501	+0.828
Textiles CIIU 321	$T=5$ ($\hat{M}=2$)	+0.099	+0.368	+0.593	+1.061
	$T=10$ ($\hat{M}=3$)	+0.103	+0.321	+0.612	+1.037

Notes. $\hat{\alpha}$ -weighted sums $\sum_j \hat{\alpha}_j \hat{\varepsilon}_{k,j}$. Conditional and unconditional CIs for the $T=10$ estimates are in the main paper figure. Metal $T=5$ Type 1 (lowest $\bar{\varepsilon}_M$, $\hat{\alpha}=0.28$, $N=65$) has $\varepsilon_K = -0.439$, pulling the $T=5$ aggregate negative; the $T=10$ aggregate (-0.007) reflects a different five-type structure. Source: results/postsel_phase2_*.winsor.lightH1_v5/ci.summary.csv.

B.4.5 Per-Type Step 1 Intercepts and Elasticities at $T = 10$

Table B13 reports the materials-elasticity intercept $\beta_{0,j}$ and output elasticities ($\varepsilon_{K,j}, \varepsilon_{L,j}, \varepsilon_{M,j}, \text{RTS}_j$) evaluated at the grand-mean centred input bundle ($\tilde{m}=\tilde{\ell}=\tilde{k}=0$), under the Nonlinear Mixture ($K_\epsilon=2$), no-CRE no-winsor baseline at $T=10$ ($\hat{M}=6$, 2, 2 for Food, Metal, Textiles). Types are ranked in ascending order of $\beta_{0,j}$. Unconditional 95% cluster-bootstrap CIs are shown ($B=350$).

Table B13: Per-type materials-elasticity intercept and output elasticities, $T=10$, Nonlinear Mixture ($K_\epsilon=2$), no-CRE no-winsor baseline. Types ranked by $\beta_{0,j}$ ascending. Unconditional 95% cluster-bootstrap CIs [$B=350$, L2-Hungarian type alignment].

Type ($\hat{\alpha}$, N)	$\beta_{0,j}$	$\varepsilon_{K,j}$	$\varepsilon_{L,j}$	$\varepsilon_{M,j}$	RTS _{j}
Food CIIU 311 , $T = 10$, $N = 669$, $\hat{M} = 6$					
T1 ($\hat{\alpha} = 0.068$, $N = 44$)					
point estimate	+0.496	+0.083	+0.300	+0.497	+0.880
95% unc. CI	[+0.409, +0.653]	[−0.596, +0.424]	[−0.096, +0.645]	[+0.377, +0.659]	[+0.028, +1.284]
T2 ($\hat{\alpha} = 0.288$, $N = 200$)					
point estimate	+0.651	+0.095	+0.242	+0.656	+0.993
95% unc. CI	[+0.508, +0.757]	[+0.019, +0.681]	[+0.035, +0.428]	[+0.528, +0.759]	[+0.835, +1.667]
T3 ($\hat{\alpha} = 0.171$, $N = 122$)					
point estimate	+0.720	+0.067	+0.140	+0.705	+0.912
95% unc. CI	[+0.600, +0.796]	[−0.096, +0.205]	[+0.011, +0.217]	[+0.594, +0.790]	[+0.626, +1.012]
T4 ($\hat{\alpha} = 0.287$, $N = 191$)					
point estimate	+0.706	+0.023	+0.033	+0.746	+0.802
95% unc. CI	[+0.566, +0.808]	[−0.551, +0.173]	[−0.253, +0.175]	[+0.567, +0.835]	[+0.145, +0.955]
T5 ($\hat{\alpha} = 0.089$, $N = 51$)					
point estimate	+0.706	+0.057	+0.236	+0.746	+1.039
95% unc. CI	[+0.609, +0.814]	[−0.063, +0.900]	[+0.071, +0.460]	[+0.647, +0.846]	[+0.807, +1.985]
T6 ($\hat{\alpha} = 0.097$, $N = 61$)					
point estimate	+0.739	+0.105	−0.039	+0.882	+0.948
95% unc. CI	[+0.607, +0.836]	[−0.249, +1.140]	[−0.286, +0.212]	[+0.623, +0.871]	[+0.170, +2.163]
Metal CIIU 381 , $T = 10$, $N = 157$, $\hat{M} = 2$					
T1 ($\hat{\alpha} = 0.380$, $N = 56$)					
point estimate	+0.441	+0.077	+0.518	+0.432	+1.026
95% unc. CI	[+0.416, +0.611]	[−0.095, +0.661]	[+0.169, +0.739]	[+0.376, +0.638]	[+0.893, +1.543]
T2 ($\hat{\alpha} = 0.620$, $N = 101$)					
point estimate	+0.579	−0.091	+0.341	+0.584	+0.835
95% unc. CI	[+0.440, +0.642]	[−0.540, +0.312]	[−0.079, +0.643]	[+0.413, +0.679]	[+0.130, +1.251]
Textiles CIIU 321 , $T = 10$, $N = 162$, $\hat{M} = 2$					
T1 ($\hat{\alpha} = 0.555$, $N = 90$)					
point estimate	+0.502	+0.146	+0.352	+0.480	+0.978
95% unc. CI	[+0.447, +0.675]	[−0.093, +0.474]	[−0.013, +0.568]	[+0.387, +0.720]	[+0.613, +1.425]
T2 ($\hat{\alpha} = 0.445$, $N = 72$)					
point estimate	+0.667	+0.018	−0.006	+0.700	+0.713
95% unc. CI	[+0.465, +0.709]	[−0.307, +0.558]	[−0.472, +0.403]	[+0.411, +0.735]	[+0.071, +1.534]

Notes. v8 Phase 2 cluster-bootstrap ($B=350$ firm-level resamples, inner $B_{\text{in}}=19$ parametric LRT for $\hat{M}$ re-selection). CIs are unconditional (all B replicates), with L2-Hungarian alignment of types across bootstrap replicates. The intercept $\beta_{0,j}$ equals each type's materials elasticity at the industry grand-mean centred input bundle ($\tilde{m}=\tilde{\ell}=\tilde{k}=0$). Source: results/postsel_phase2-`{food,metal,textiles}`-T10_nocre_v8/ci-summary.csv.

Type heterogeneity is concentrated in $\beta_{0,j}$: in Food, six types span $\beta_{0,j} \in [0.496, 0.739]$; Metal and Textiles each show a clear low-high contrast ($[0.441, 0.579]$ and $[0.502, 0.667]$ respectively). The materials elasticities $\varepsilon_{M,j}$ closely track the intercepts, confirming that variation in $\beta_{0,j}$ drives type differentiation. Capital elasticities $\varepsilon_{K,j}$ carry wide CIs, consistent with the discussion in Section 7.6 that capital plays a secondary role in type differentiation.

B.4.6 Misclassification Bias and the Two-Step Estimation Procedure

The empirical analysis recovers type-specific production-function parameters by a two-step plug-in: (i) fit the inside-log nonlinear share-equation EM mixture and MAP-assign each plant to $\hat{j}_i \in \{1, \dots, \hat{M}\}$ at the fitted posterior; (ii) run the GNR Step 1 and Step 2 estimators on each type-conditional sub-sample. Because the Chilean panel is short ($T=5$ or $T=10$), MAP classification is imperfect *even in the limit* $n \rightarrow \infty$ — misclassified plants contaminate the per-type sub-samples and the per-type GNR estimator converges to a pseudo-true value $\psi_j^*(T) \neq \psi_j^0$.

Exponential bias bound. Let $P_{kj}(T) = \Pr(\hat{j} = j \mid \text{type} = k)$ be the population MAP confusion matrix. Under regularity (Gaussian errors, bounded inputs, identified per-type moment), the bias satisfies

$$\|\psi_j^*(T) - \psi_j^0\| \leq C_j \cdot e^{-T\rho_j}, \quad \rho_j = \min_{k \neq j} \bar{C}_{kj},$$

with $\bar{C}_{kj}$ the path-averaged Chernoff information between the share-equation densities of types k and j , and C_j scaling with GNR instrument strength and the type-parameter contrast.

Step-1 vs. Step-2 asymmetry. The classifier observes only the materials-share equation. A misclassified plant therefore resembles the target type on Step 1 (materials elasticity) but carries alien output-side structure (capital, labour, persistence) not seen by the classifier. Monte-Carlo simulations show that at moderate separation ($n=277$, $M_0=2$, misclassification rate 18.7% at $T=5$), the Step-2 persistence bias $|\Delta\rho|^{\text{MAP}}=0.042$ exceeds the Step-1 materials bias $|\Delta\varepsilon_M|^{\text{MAP}}=0.0025$ by an order of magnitude. The Step-2 capital elasticity bias ($|\Delta\varepsilon_K|^{\text{MAP}}=0.012$ at $T=5$, falling to 0.002 at $T=20$) is intermediate. This asymmetry sharpens the “classification is exogenous to the production-function parameters” claim: it holds as $T \rightarrow \infty$ but fails at fixed T , with Step-2 parameters carrying substantially more fixed- T bias than Step-1.

Empirical separation. Pairwise path-averaged Chernoff informations estimated from the inside-log nonlinear mixture fits give worst-pair indices $e^{-5\hat{\rho}_j}$ of 17–25% at $T=5$ across the three industries, matched by the Monte-Carlo actual rate of 18.7% at comparable separation. At $T=10$ the index falls to 3–6%, so the bias is substantially reduced in the main results.

Remedies. The cleanest fix is joint estimation: a single EM treating type membership as latent throughout, eliminating the fixed- T misclassification bias by construction. The two-step approach used here is retained for comparability with earlier industry-level applications; the Phase 2 v5 bootstrap CIs capture post-selection uncertainty via conditional/unconditional percentiles. Full derivations, proofs, and simulation evidence are in `paper/memo/misclassification_bias_subsection.2026-06-10.tex`.

B.4.7 Hard Plant-to-Type Assignment (Binarization)

An alternative to the soft-assignment Phase 2 bootstrap is to assign each plant to a single type by MAP classification (“binarization”) and run ordinary GNR on each hard-assigned sub-sample. We do not adopt binarization for three reasons: (i) it discards the signal in plants with non-trivial posterior probability for a second type, conflicting with GNR’s identification from the full conditional distribution of productivity (Gandhi et al., 2020); (ii) it amplifies the fixed- T misclassification bias $C_j e^{-T\rho_j}$ (Section B.4.6) across every Step-1 and Step-2 moment condition rather than smoothing it via posterior weighting; and (iii) with $\hat{M} \geq 3$ types, discrete membership counts make L2-Hungarian label alignment noisier than posterior-weighted centroids, particularly for thin types. The two-step plug-in used here occupies an intermediate position: it uses MAP assignment but applies GNR to the resulting sub-sample as if it were a clean partition, with the resulting bias bounded as in Section B.4.6.

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